

Anatomy Remediation

Excludes Exam 1 content

Review Exam 1 Neuro Content Separately

What's inside

Exam 2 · Eye & Ear, and more

22 pages · p.2

Exam 3 · GI, and more

33 pages · p.24

Exam 4 · Axial Skeleton, and more

42 pages · p.57

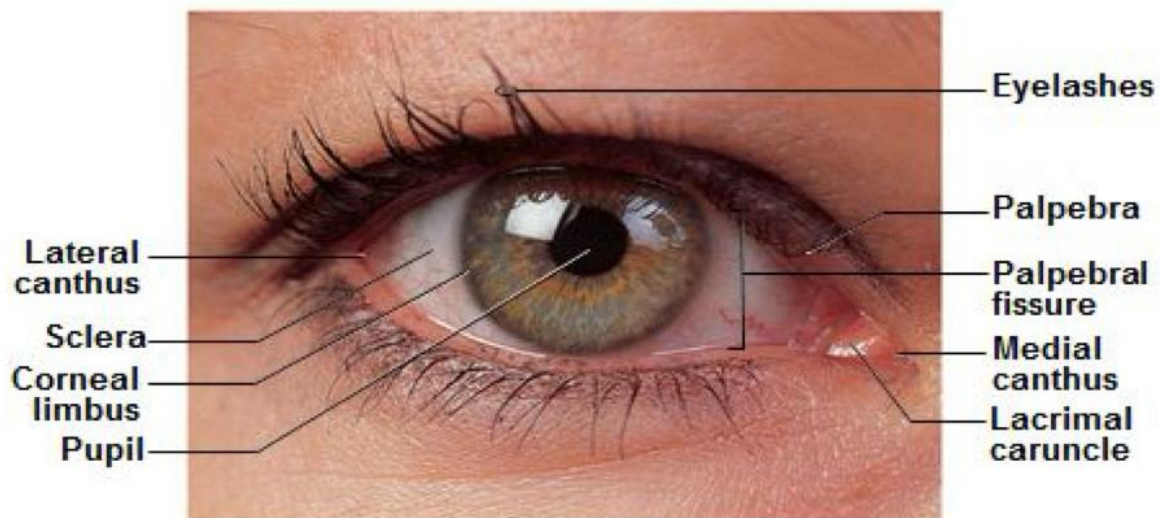
Anatomy of the Eye and Ear

Identify the anatomic structures of the eye including external and internal structures.

External

1. Accessory Structures

- a. **Palpebrae** – eyelids
- b. Medial and Lateral Canthus – connect the eyelids at the corners of the eye
- c. Palpebral Fissure – area between the eyelids
- d. Eyelashes – contain root hair plexus; triggers blinking reflex
- e. Conjunctiva – epithelial lining of the eyelids
 - i. **Palpebral Conjunctiva** – inner lining of the eyelids
 - ii. **Bulbar (Ocular) Conjunctiva** – outer lining of the eye
- f. Glands
 - i. Lacrimal **caruncle** – produce **thick secretions**
 - ii. Lacrimal **glands** – produce **tears**
 - iii. Glands of **Zeis** – **sebaceous glands** *cyst*
 - iv. Tarsal **glands** – secrete a lipid-rich product; keeps eyelids from **sticking to each other** *stye*



a Superficial anatomy of the right eye and its accessory structures

Internal

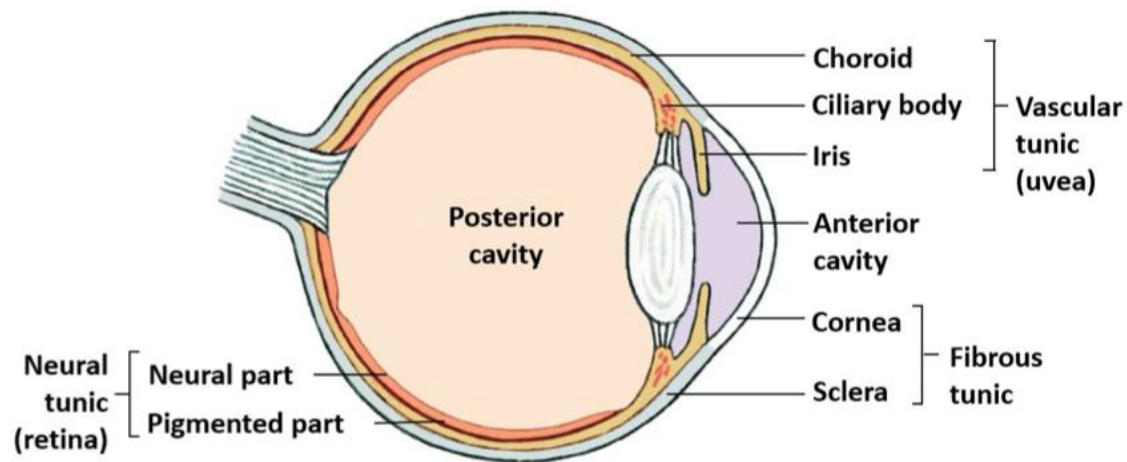
Organs:

- Sclera
- Cornea
- Pupil
- Iris
- Lens
- Anterior Cavity
 - Anterior & Posterior chambers
- Posterior Cavity
- Three tunics:
 - Fibrous tunic - *sclera & cornea*
 - Vascular tunic - *choroid, iris, ciliary*
 - Neural tunic
- Retina

Tunics:

1. The Fibrous Tunic (outer layer)
 - a. Sclera & Cornea
 - b. Provides some degree of protection
 - c. Provides attachment sites for extra-ocular muscles
2. The Vascular Tunic (middle layer)
 - a. Iris
 - i. Consists of blood vessels, pigment, and smooth muscles
 - ii. Pigment = color of eye
 - iii. Smooth muscles contract to change the diameter of the pupil
 - b. Ciliary body
 - i. Consist of ciliary muscles connected to the suspensory ligaments, which are connected to the lens
 - c. Choroid
 - i. Highly vascularized
 - ii. Innermost portion of choroid attaches to the outermost portion of the retina
 - d. Consists of blood vessels, lymphatics, and intrinsic eye muscles
 - e. Regulates the amount of light entering the eye
 - f. Secretes and reabsorbs aqueous fluid
 - g. Controls shape of lens

3. The Neural Tunic (inner layer)
 - a. Retina
 - b. 2 layers:
 - i. Pigmented layer – outer layer
 - ii. Neural layer – inner layer
 - c. Retina cells:
 - i. Rods – night vision
 - ii. Cones – color vision



Cavities and Chambers of the Eye:

1. Anterior Cavity
 - a. Anterior chamber
 - b. Posterior chamber
 - c. Filled with fluid called **aqueous fluid**
2. Posterior Cavity
 - a. Filled with fluid called **vitreous fluid**

Fluids of the Eye:

1. **Aqueous fluid (aqueous humor)**
 - a. Secreted by cells at the **ciliary body area**
 - b. Enters the posterior chamber → flows through the pupil area → enters the anterior chamber → flows through **canal of Schlemm** → **enters venous circulation**
2. Vitreous fluid (Vitreous humor)
 - a. Gelatinous material in the posterior chamber

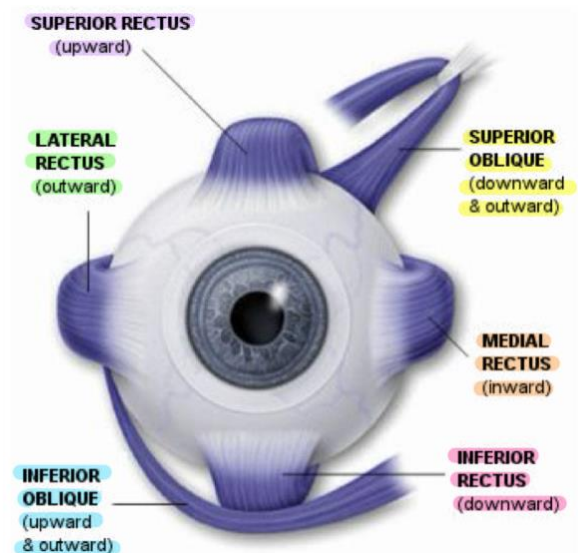
- b. Supports the shape of the eye, position of the lens, and position of the retina
- c. Aqueous humor can flow across the vitreous fluid and over the retina

List the extra-ocular muscles and their corresponding nerve innervations.

1. Extrinsic Muscles

- a. Superior Rectus – CN III, Up
- b. Inferior Rectus – CN III, Down
- c. Medial Rectus – CN III, In
- d. Lateral Rectus – CN VI, Out
- e. Superior Oblique – CN IV, Down and Out
- f. Inferior Oblique – CN III, Up and Out
- g. LR6, SO4, all the rest are 3!

- SO = down and out
- IO = up and out
- SR = up
- IR = down
- LR = out
- MR = in



Describe the structure of the lacrimal system of the eye.

1. Lacrimal Apparatus

- a. Lacrimal glands (produce tears)
 - i. Tears are produced by the lacrimal glands, flow over the ocular surface, flow into the nasolacrimal canal (foramen), the foramen enters into the nasal cavity
- b. Lacrimal canaliculi
- c. Lacrimal sac
- d. Nasolacrimal duct (canal)

Describe the anatomic structures of the external, middle, and inner ear.

1. External Ear

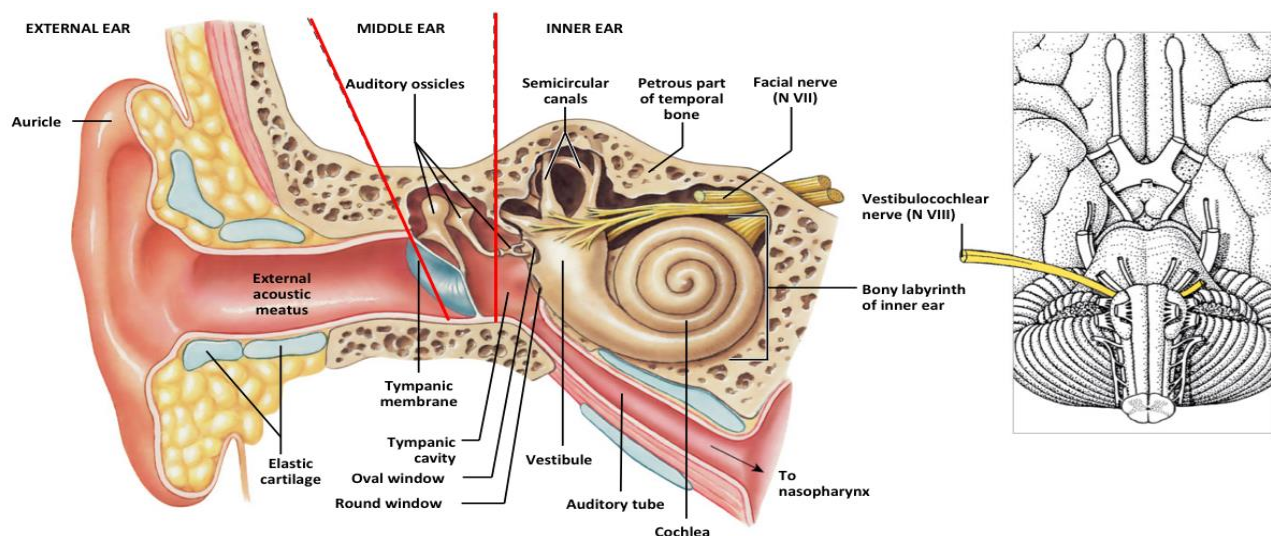
- a. Consists of:
 - i. Auricle
 - ii. External acoustic meatus
 - iii. Tympanic membrane
 - iv. Ceruminous glands *ear wax*

2. Middle Ear

- a. Consists of:
 - i. Tympanic cavity
 - ii. Auditory ossicles
 - 1. Malleus, Incus, Stapes
 - iii. Auditory tube or Eustachian tube
 - 1. Equalizes pressure in the middle ear
 - 2. Allows fluid to drain from the middle ear

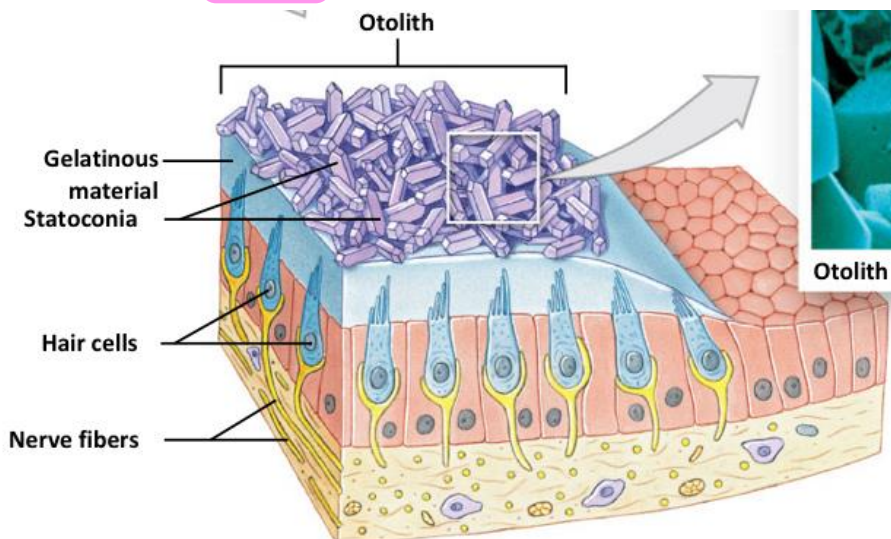
3. Inner Ear

- a. Consists of:
 - i. Receptors
 - ii. Vestibule
 - iii. Semicircular canals
 - iv. Utricle
 - v. Saccule
 - vi. Cochlea
 - vii. Membranous labyrinth
 - viii. Bony labyrinth



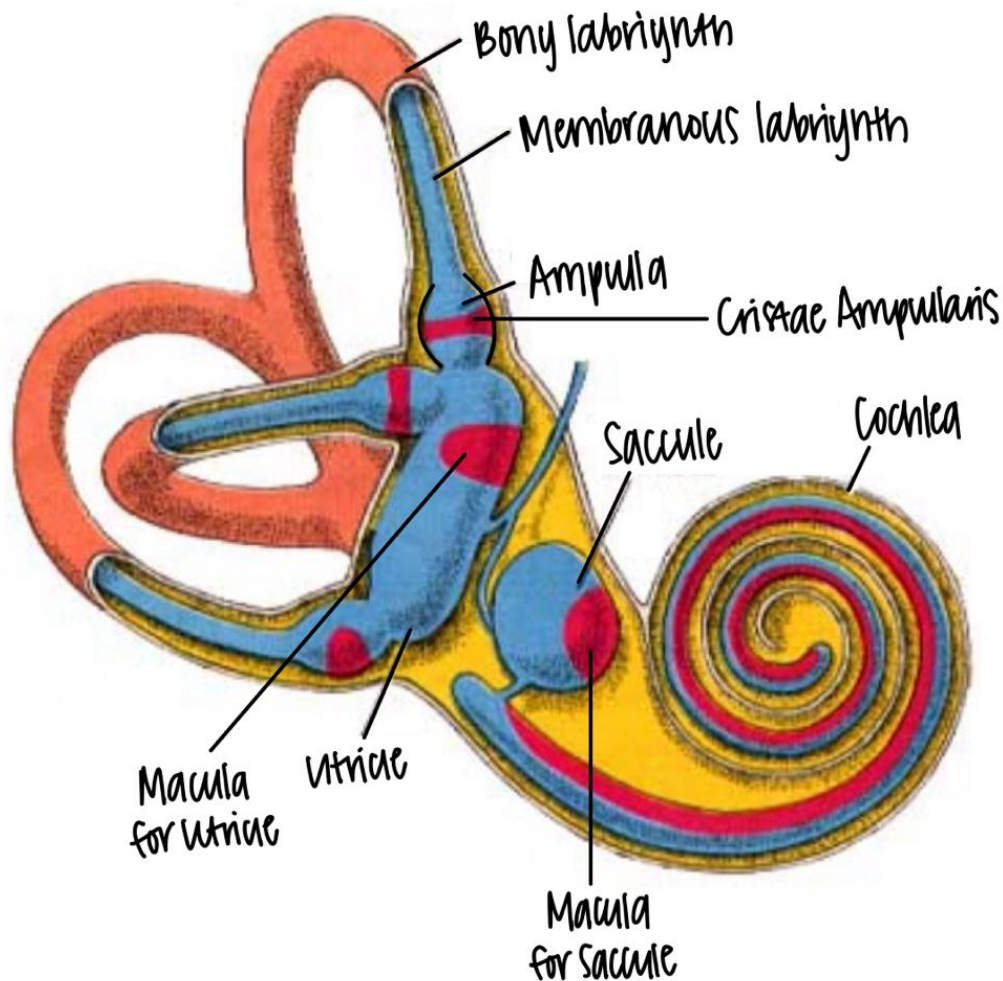
b. Vestibular Complex and Equilibrium

- i. Part of inner ear that provides equilibrium sensations by detecting rotation, gravity, and acceleration
- ii. Vestibule consists of: Semicircular canals, Utricle, Saccule
- iii. Semicircular canals:
 1. Each encases a duct
 2. Beginning of each duct is the ampulla
 3. Within each ampulla is a cristae with hair cells
 4. Each hair cell contains kinocilium and stereocilia
 5. The hair cells are embedded in gelatinous material called the cupula
 6. Movement of the body causes movement of fluid in the canal (endolymph), which in turn causes movement of the cupula and hair cells, which the brain detects
 7. Crista ampullaris detect angular acceleration
- iv. Utricle & Saccule:
 1. Utricle and saccule are connected to the ampulla and to each other and to the fluid within the cochlea
 2. Hair cells of the utricle and saccule are in clusters called maculae
 3. Hair cells are embedded in gelatinous material consisting of statoconia (calcium carbonate crystals)
 4. Gelatinous material and statoconia collectively are called an otolith



a Detailed structure of a sensory macula

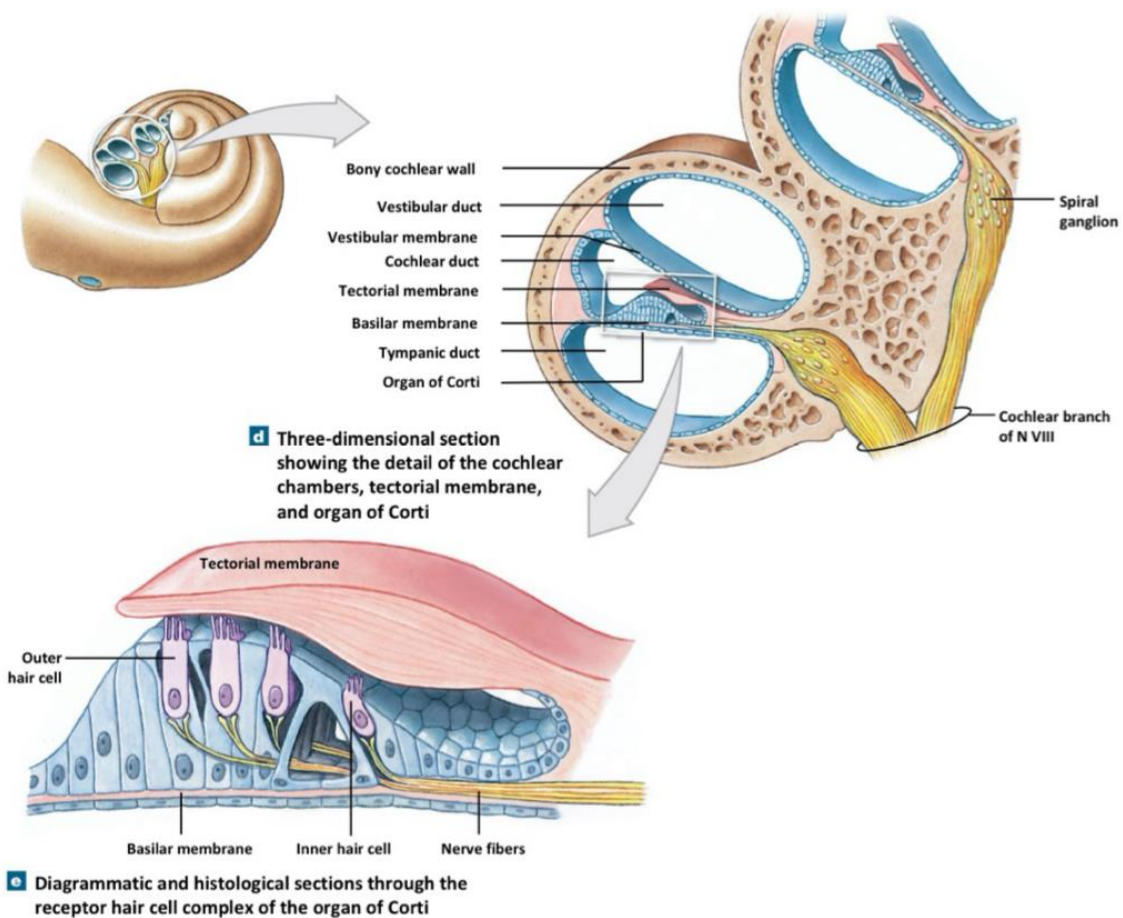
- c. My breakdown of Crista Ampullaris (Cristae of ampulla), Utricle, and Saccule
- i. Crista Ampullaris (Cristae of ampulla)
 - 1. Detects *angular acceleration, rotation*
 - 2. Cristae are enclosed within the ampulla and ampulla is found in each of the 3 semicircular ducts
 - ii. Utricle
 - 1. Detects *linear acceleration, horizontal*
 - 2. Maculae enclosed in utricle sac
 - iii. Saccule
 - 1. Detects *linear acceleration, vertical*
 - 2. Maculae enclosed in saccule sac



d. Hearing

i. Cochlea

1. “Snail-shaped” spirals
2. Spirals coil around a central area called the **modiolus**; within the modiolus are sensory neurons associated with CN VIII
3. Each spiral consists of 3 layers:
 - a. Scala vestibuli (**Vestibular duct**): consists of **perilymph**
 - b. Scala tympani (**Tympanic duct**): consists of **perilymph**
 - c. Scala media (**Cochlear duct**): consists of **endolymph**
4. **Organ of Corti**
 - a. Rests on the basilar membrane between the scala media and scala tympani
 - b. Hair cells are in contact with an overlying tectorial membrane
 - i. Distortion of tectorial membrane stimulates organ of Corti



Integumentary System

Describe the integumentary system.

Cutaneous Membrane:

1. Epidermis (from superficial to deep)

a. Stratum corneum

Come

- i. Most superficial layer
- ii. 15-30 layers of flattened cells
- iii. Dry layer, water resistant but not waterproof, allows perspiration
- iv. Keratinocytes take 15-30 days to travel from basal layer; shed in about 2 weeks

b. Stratum lucidum

lets

- i. Have no nucleus
- ii. Flattened and densely packed

dead cells

c. Stratum granulosum

get
makes
cells
water
resistant

- i. Contains keratinocytes
- ii. Keratohyalin produced here (makes cells water-resistant by coating them with a lipid rich substance)
- iii. Keratin produced here
- iv. Causes top layer of cells to die
- v. Also form calluses

keratin
produced in granulosum

d. Stratum spinosum

sun

- i. Contains mostly keratinocytes
- ii. May also have melanocytes and Langerhans cells (initiate immune response)

Langerhans cell → spinosum

e. Stratum basale

Burned

- i. Cuboidal cells attached to basal lamina
- ii. Separates the dermis from the epidermis
- iii. Composed of basal cells and melanocytes
- iv. Regenerate keratinocytes
- v. Merkel cells (stimulate sensory nerve endings)
- vi. Forms epidermal ridges (ex: fingerprints)

f. Come Let's Get Sun-Burned!

2. Dermis

a. Papillary layer

- i. Contains capillaries and axons

b. Reticular layer

- i. Interwoven collagen fibers bundles

Accessory Structures:

1. Hair follicles and hair

- a. Follicle found in dermis and subcutaneous layer
 - i. Hair papillae – containing capillaries and nerves
 - ii. Hair bulb – epithelial cells
 - iii. Hair matrix – involved in hair production
- b. Follicle organized into 3 layers:
 - i. Internal root sheath – surround hair root and part of shaft
 - ii. External root sheath – from the skin surface to matrix
 - iii. Glassy membrane – thick basal lamina, wrapped by dense connective tissue
- c. Hair composed of
 - i. Medulla (containing soft keratin)
 - ii. Cortex (containing hard keratin)
 - iii. Cuticle (contains hard keratin)

2. Exocrine Glands

- a. Secretions are controlled by the ANS
- b. Sebaceous – produce oily lipid (waxy and oily) that coats the hair and epidermis
- c. Sweat
 - i. Apocrine – axillae, groin, and nipples only
 - ii. Merocrine – everywhere else
 - iii. Clear secretion
 - iv. Functions: Thermoregulation, Excretion, Protection
- d. Mammary – related to apocrine glands
- e. Ceruminous – modified sweat glands located in external ear canals

3. Nails

- a. Nail body/plate
 - i. External component
 - ii. Bounded by nail groves and folds
 - 1. Eponychium – extends over body at nail root
 - 2. Hyponychium – free end of the nail bed
- b. Nail bed
 - i. Internal component
- c. Nail root
 - i. Epithelial fold

- ii. Lies close to periosteum of bones

Skin Color:

1. Variable quantity of carotene and melanin
 - a. Carotene:
 - i. Orange-yellow pigment
 - ii. Converted to vitamin A
 - iii. Found in keratinocytes
 - b. Melanin:
 - i. Produced and stored in melanocytes
 - ii. Prevents skin damage by UV light

Discuss the various cell types found in the skin.

1. Melanocytes – produce pigmentation of skin; found in stratum basale
2. Keratinocytes – produce keratin (fibrous protein); found in stratum spinosum
3. Langerhans cells – initiate immune response; stratum spinosum
4. Merkel cell – stimulate sensory nerve endings; found in stratum basale

Describe and contrast the various components of the skin appendages.

Refer to Accessory Structures in IO 1 😊

Compare and contrast the sensory receptors of the skin.

Putnam said not to worry about them!

Pulmonary System

Compare and contrast the organs that compromise the respiratory system.

Upper Respiratory Organs:

- Nose
- Nasal cavity
- Sinuses
- Pharynx
- Larynx

Lower Respiratory Organs:

- Trachea

- Lungs
- Bronchi
- Bronchioles
- Alveoli

Describe the anatomical structures of the organs of the respiratory system.

1. Nose

- Nasal bones
 - Vomer bone and perpendicular plate of ethmoid
- Nasal septum
- Cartilage
- External nares
- Alar cartilage
- Dorsum and apex of the nose
- Nasal conchae

2. Pharynx

- Nasopharynx
 - Where the nose meets the pharynx (internal nares to soft palate)
- Oropharynx
 - Oral cavity meets the pharynx (soft palate to base of tongue at hyoid bone)
 - Consists of pharyngeal arch and uvula
- Laryngopharynx
 - Area that has the entrance to the trachea and esophagus (hyoid bone to entrance of esophagus)

3. Larynx

- Cylinder whose cartilaginous walls are stabilized by ligaments or skeletal muscles or both
- Begins at C3 or C4, Ends at C7
- Unpaired Cartilages:
 - Thyroid cartilage
 - Cricoid cartilage
 - Epiglottis
 - Closes over the glottis during swallowing of food
- Paired Cartilages
 - Arytenoid cartilages
 - Corniculate cartilages
 - Cuneiform cartilages

A
C
C

- e. **Vocal ligaments form the vocal cords (elastic)**
 - i. **True vocal cords (made from vocal ligaments)**
 - ii. **Pitch depends on the diameter, length, and tension in the vocal cords**

4. **Trachea**

- a. Lining consists of:
 - i. Respiratory **epithelia**
 - ii. Lamina propria – loose connective tissue
 - iii. Submucosa – contains **Goblet cells**
- b. Bifurcates at the carina into the right and left primary bronchi at T5
 - i. Right primary bronchus is steeper, shorter, and larger in diameter than the left

5. **Lungs**

- a. **Right lung has 3 lobes!**
 - i. Superior, Middle, and Inferior
 - ii. Groove for esophagus
- b. **Left lung has 2 lobes!**
 - i. Superior and Inferior
 - ii. Cardiac notch and groove for aorta
 - iii. Lingula – tongue-shaped region of left lung
- c. **Primary bronchi → secondary bronchi → tertiary bronchi → bronchioles**
- d. Right Primary Bronchus
 - i. Divides into 3 secondary bronchi. One for each lobe
 - 1. Superior, Middle, and Inferior lobar bronchus
- e. Left Primary Bronchus
 - i. Divides into 2 secondary bronchi. One for each lobe
 - 1. Superior and Inferior lobar bronchus
- f. **Hilum** – each primary bronchus (right & left bronchus) enters the lung at this point; also the point of entrance and exit of the pulmonary vessels
- g. Secondary bronchi → tertiary bronchi
 - i. Right and left lungs have 10 tertiary bronchi and therefore 10 bronchopulmonary segments
- h. Tertiary bronchi → bronchioles
 - i. **Consist of smooth muscle for bronchodilation (sympathetic) & bronchoconstriction (parasympathetic)**
 - ii. Bronchioles terminate with clusters of **alveolar sacs**
 - iii. Terminal bronchioles: final branches of the conducting zone; **NOT** involved in gas exchange

- iv. Respiratory bronchioles: lead to alveoli and ARE involved with gas exchange; self-sufficient at oxygenation

6. Alveoli

- a. Cells associated with alveoli:
 - i. Lining of alveoli consists of a single layer of **squamous cells**
 - 1. These cells are called **type I pneumocytes or alveolar type I cell; where gas exchange takes place**
 - 2. Scattered among the type I pneumocytes are the **type II pneumocytes; secrete surfactant (prevents alveolar collapse)**
 - 3. **Alveolar macrophages (Dust cell) – phagocytize matter**
- b. Extensive network of **capillaries surrounds each alveolus**
- c. Elastic tissue surrounds each alveolus
- d. **Self-sufficient at oxygenation (due to capillaries)**

Describe the anatomic relations of each organ of the respiratory system and the surrounding anatomy.

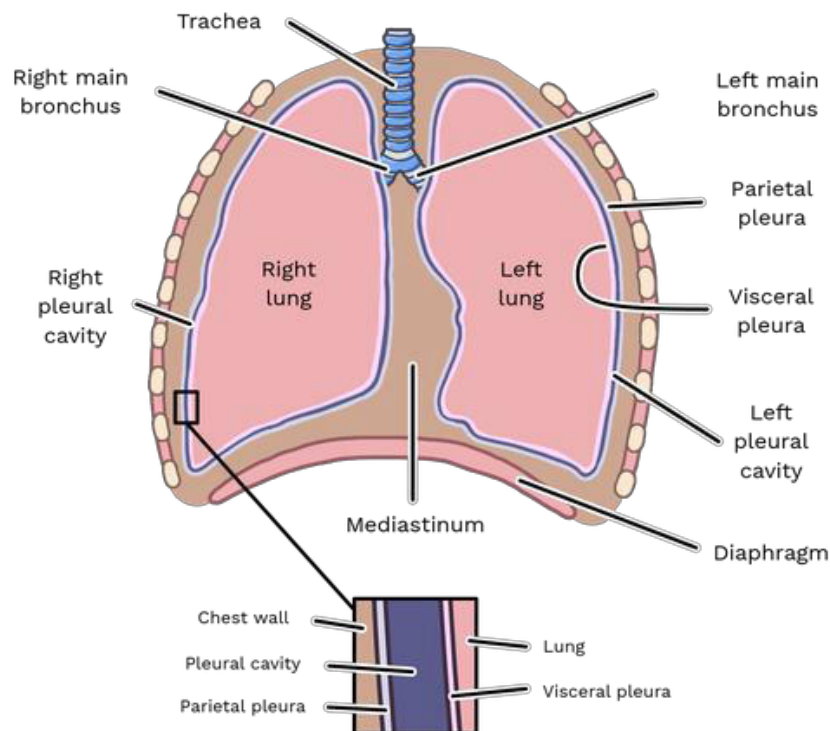
See above IO.

Compare and contrast the segments of the trachea and bronchi.

- 1. **Trachea**
 - a. Bifurcates at the carina into the right and left bronchi at T5
 - b. Contains **15-20 tracheal cartilages**
 - i. **Annular ligaments** – connect 1 cartilage ring to another

Discuss segments of the pleura and pleural cavity.

1. Mediastinum – separates the right and left pleural cavities
2. Each lung is lined by a serous membrane which is made up of 2 layers:
 - a. Visceral pleura – covers the outer surface of the lung
 - b. Parietal pleura – covers the inside lining of the thoracic wall
3. Pleural cavity – space created between the visceral and parietal pleura
 - a. Contains pleural fluid – reduces friction



Identify the blood, nerve, and lymphatic supply of the thorax.

Refer to Professor Shah's lecture!

Identify the major sections of the diaphragm including the hiati of the diaphragm.

1. Diaphragm
 - a. Involved in pulmonary ventilation
 - b. Innervated by phrenic nerve (C3, 4, 5!)
 - c. Contracts (moves inferiorly) to cause inhalation
 - d. Relaxes (moves superiorly) to cause exhalation
 - e. Hiatus – opening in the diaphragm which the esophagus passes through to get to the abdominal cavity (clinical: hiatal hernia)

Pulmonary Side Notes!

Brain Respiratory Centers:

1. Respiratory rhythmicity center – located in medulla; respiratory pace
2. Apneustic center – located in pons; adjusts respiratory intensity by inhalation
3. Pneumotaxic center – located in pons; adjusts the respiratory rate and pattern

Anatomy of the Heart and Great Vessels

Compare and contrast the external and internal anatomical structures of the heart.

Refer to 1st IO in Arteries, Veins, and Lymphatics Lecture

Identify the origin and course and the main branches of the main coronary arteries.

Repeated/More in depth in Arteries, Veins, and Lymphatic System lecture

Describe the cardiac nerve supply and distribution.

1. Innervation of the Heart
 - a. Innervated by the ANS!
 - b. Rich innervation to SA and AV nodes, but also to cardiomyocytes
 - i. Vagal nerve (parasympathetic) – only innervates the SA and AV nodes
 - ii. Right vagus nerve – innervates the SA node
 - iii. Left vagus nerve – innervates the AV node
 - iv. Sympathetic stimulation – directly to cardiomyocytes and nodal structures
 - c. Parasympathetic NS dominates the autonomic stimulation
 - d. Sympathetic innervation increases the rate and force of heart contractions

Explain the anatomical components of the cardiac conduction system.

Sinoatrial Node (SA node) - aka cardiac pacemaker



Internodal pathways



Atrioventricular node (AV node)



AV bundle (Bundle of His)



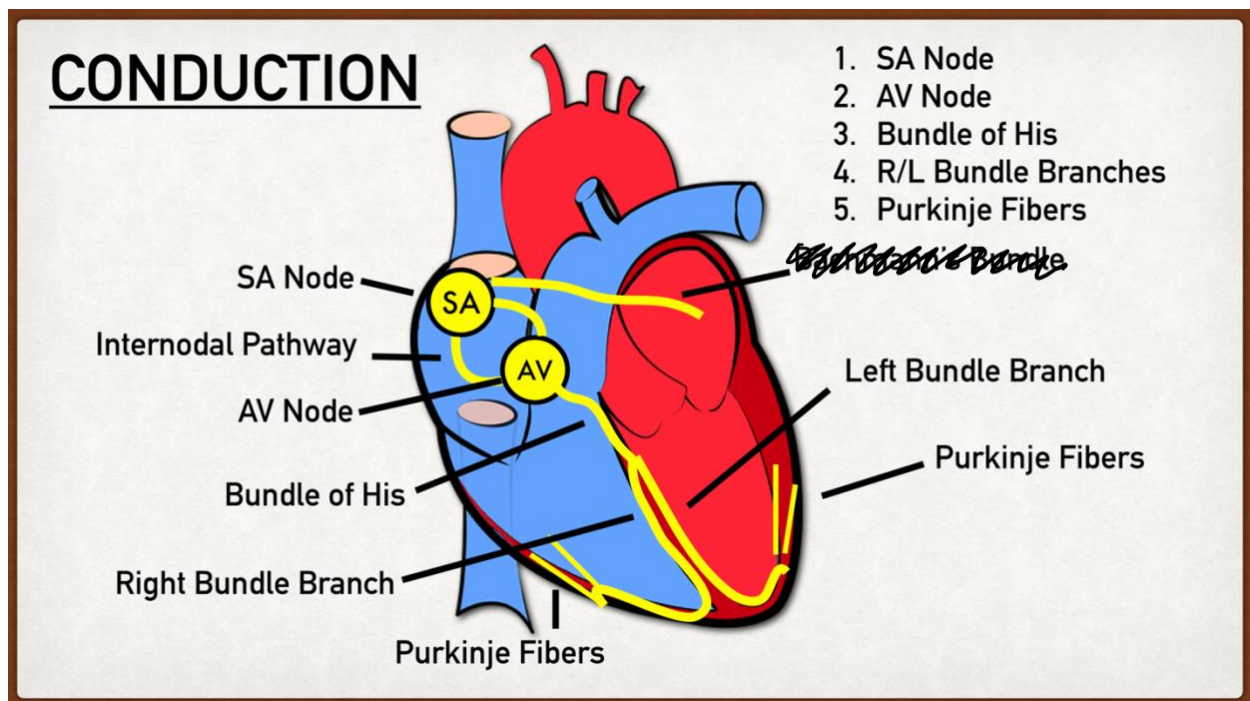
Travels along the interventricular septum and then divides to form the right and left bundle branches



Purkinje fibers



Ventricular myocardium to contract!



Describe the entire course of the aorta and name the origin of the major branches of each division.

1. Ascending Aorta

a. Branches into:

- i. Left & Right Coronary Arteries – the ONLY branches of the ascending aorta; found at the trunk of the ascending aorta

2. Aortic Arch

a. Branches into the following vessels:

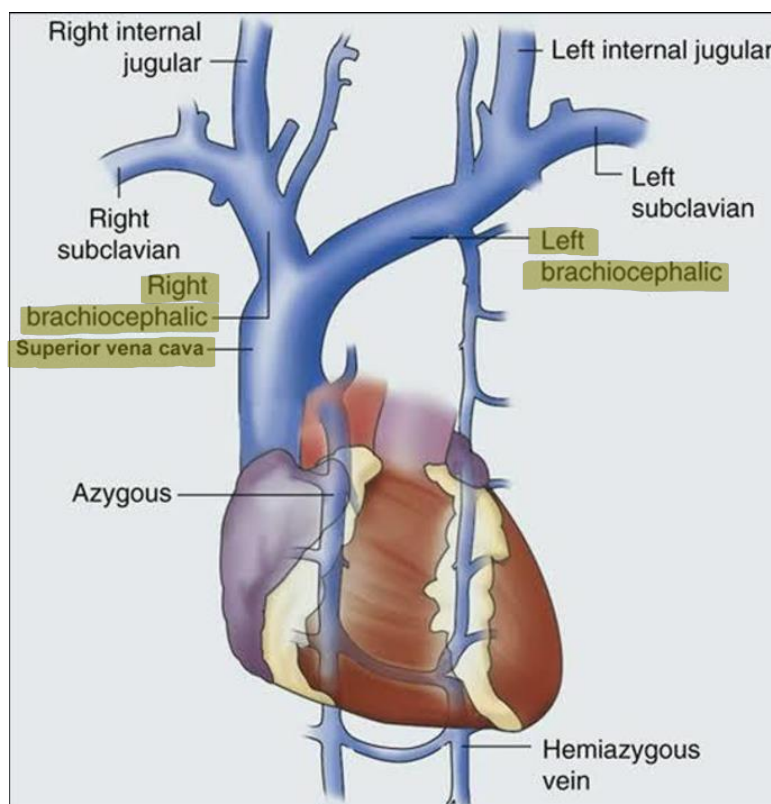
- i. Brachiocephalic Trunk
 - 1. Right common carotid
 - 2. Right subclavian arteries
- ii. Left common carotid Artery
- iii. Left Subclavian Artery

iv. Descending Aorta

- 1. Divides into the thoracic aorta and abdominal aorta

Describe the origins, course and relationships of the brachiocephalic veins and superior vena cava.

Brachiocephalic veins (innominate vein) join to form the superior vena cava.



Arteries, Veins, and Lymphatics System

Compare and contrast the external and internal anatomical structures of the heart.

Structure of the Heart:

1. Pericardium – serous membrane lining the pericardial cavity; has 2 layers
 - a. Visceral pericardium – aka epicardium
 - b. Parietal pericardium – reinforced by a layer called fibrous pericardium
 - c. Parietal fluid is between the 2 layers
2. Heart Wall
 - a. Epicardium – external surface; visceral layer of serous pericardium
 - b. Myocardium – consists of cardiomyocytes (functional cells of heart)
 - c. Endocardium – internal surface
3. Cardiac Muscle
 - a. Intercalated discs – bind the actin of adjacent cardiac cells together and allow synchronized contraction
 - b. Gap junctions – cardiac cells are bound together by these junctions and allows ions to move from one cell to another
4. Fibrous skeleton – provides structure and support
5. Orientation
 - a. Heart lies slightly to the left of midline
 - b. Base is the superior portion
 - c. Apex is the inferior portion
6. Internal anatomy
 - a. Right atrium
 - i. Receives deoxygenated blood via the superior vena cava, inferior vena cava, and coronary sinus
 - ii. Consists of pectinate muscles – “comb-like”; push blood unidirectionally
 - iii. Crista terminalis – origin of pectinate muscles
 - b. Right ventricle
 - i. Receives deoxygenated blood from right atrium
 - ii. Passes through the tricuspid valve
 - iii. Valve is connected to papillary muscles via chordae tendineae
 1. (prevent valve inversion; keeps valves shut against pressure)
 - iv. Trabeculae carneae – “bands of meat”; contract the ventricle from all directions



- v. Moderator band – found ONLY in RV; muscular band with conduction tissue called Purkinje fibers; serves as a conductor!
- vi. Pulmonary valve – leads to pulmonary trunk into L and R pulmonary arteries
- c. Left atrium
 - i. Receives oxygenated blood from lungs via the L and R pulmonary veins
 - ii. Blood passes through bicuspid valve
- d. Left ventricle
 - i. Thickest myocardial tissue; needed for strong contractions to pump blood to the whole body (systemic)
 - ii. Bicuspid valve (Mitral valve) has chordae tendineae connecting to the 2 cusps and 2 papillary muscles
 - iii. Blood leaves by passing through the aortic valve to the aorta then to the rest of the body
- e. Valves prevent backflow of blood!

Body -> Lungs
“RIGHT” to the Lungs

Right

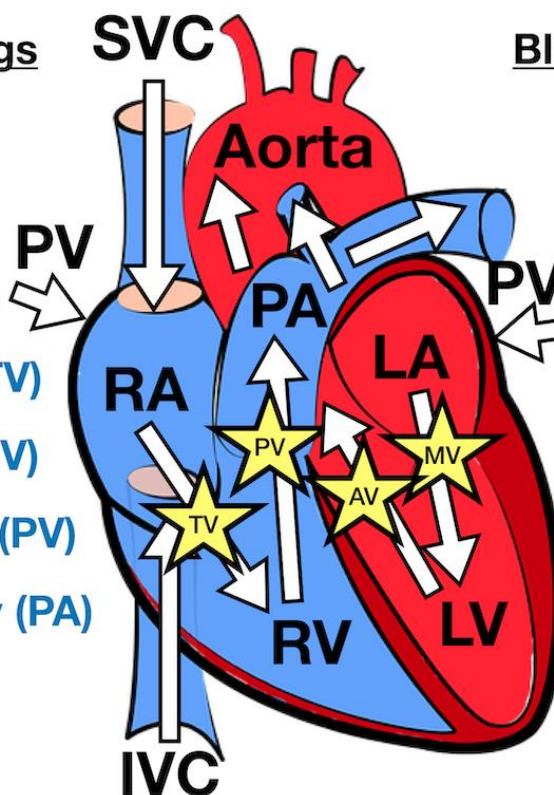
1. SVC/IVC
2. Right Atrium (RA)
3. Tricuspid Valve (TV)
4. Right Ventricle (RV)
5. Pulmonary Valve (PV)
6. Pulmonary Artery (PA)



Lungs -> Body
Blood “LEFT” the Lungs

Left

1. Pulmonary Veins (PV)
2. Left Atrium (LA)
3. Mitral Valve (MV)
4. Left Ventricle (LV)
5. Aortic Valve (AV)
6. Aorta



Identify the origin and course and the main branches of the main coronary arteries.

1. Right Coronary Artery
 - a. Passes between the right auricle and right ventricle
 - b. Major branches:
 - i. Atrial branches
 - ii. Right marginal branch
 - iii. Posterior interventricular branch
 - iv. Conducting system branches
2. Left Coronary Artery
 - a. Major branches:
 - i. Circumflex branch
 - ii. Anterior interventricular branch

Identify the surface markings of the major vessels of the thorax, abdomen, and lower and upper extremities.

1. Thorax
 - a. Aorta
 - b. Pulmonary trunk & arteries
 - c. Superior vena cava
2. Abdomen
 - a. Abdominal aorta
 - b. Inferior vena cava
3. Lower extremities
 - a. Femoral artery
 - b. Popliteal artery
 - c. Great saphenous vein
4. Upper extremities
 - a. Subclavian
 - b. Axillary
 - c. Brachial
 - d. Radial
 - e. Ulnar
5. Pulse Points!
 - a. Radial
 - b. Carotid
 - c. Brachial
 - d. Femoral

- e. Popliteal
- f. Posterior tibial
- g. Dorsalis pedis
- h. Temporal
- i. Abdominal aortic

Identify the azygos system of veins.

1. Azygos veins
 - a. Drain into the superior vena cava then drain into the right atrium
 - b. Includes:
 - i. Azygos
 - ii. Hemi-azygos
 - iii. Accessory Azygos
 - c. Drains blood from thoracic and posterior abdominal walls
 - d. Acts as a backup pathway if the inferior vena cava is obstructed

Compare and contrast the cell types within blood and lymphatics.

1. Blood Cell Types
 - a. Erythrocytes (Red Blood Cells) – transport O₂ and CO₂
 - b. Leukocytes (White Blood Cells) – function in the immune system
 - i. Granulocytes
 1. Neutrophils – kill bacteria; phagocytic
 2. Eosinophils – reduce inflammation; allergic reaction, parasites
 3. Basophils – release histamine and heparin
 - ii. Agranulocytes
 1. Monocytes – large phagocytic cells
 2. Lymphocytes – primary cells of lymphatic system; responsible specific immunity
 - c. Platelets – involved in blood clotting

Compare and contrast the organs that comprise the lymphatic system.

Compare and contrast the anatomical structures of the lymphatic system.

- **Lecture 12 - GI digestive organs**

- **GI tract composed of two organs**

principle ○ Digestive organs collectively make up the GI tract → AKA **digestive tract** AKA **alimentary tract**

*digestive
&
Accessory*

- Includes:

- Oral cavity
- pharynx
- esophagus (no serosa membrane located in thorax, has adventitia)
- Stomach
- small intestine
- large intestine

*peristalsis
segmentation*

- Smooth muscle in the wall - visceral muscle *motility*
 - Pushes material from one end to the other
- Continuous tube about 30 ft
 - From mouth to anus

- **Accessory digestive organs**

- Connected to the GI tract via some ducts
- Does not form the GI tube
- Assist GI tract in digestion of foods
- Includes:
 - Teeth
 - Tongue (big muscle)
 - Salivary glands
 - Liver
 - Gall bladder
 - pancreas

- **GI tract layers (tunics)**

- From inside to outside "My Special Muscle Sandwich"

- M* ■ **Mucosa** → Epithelium → no blood vessels *inner*
- S* ■ **Submucosa** → vascular → submucosal nerve plexus *2nd*
- 3rd M* ■ **Muscularis externa** → two layers separated by myenteric nerve plexus:
outer longitudinal smooth muscle & inner circular smooth muscle
- outer S* ■ **Serosa** → epithelium, connective tissue
 - Covers all GI organs
 - 2 layers: **Visceral peritoneum** (covers external surfaces of most abdominal organs) & **parietal peritoneum** (lines inner surface of abdominal wall & underside of diaphragm)
 - Parietal & visceral in between peritoneal cavity
 - Peritoneal cavity contains small amt of fluid; is divided into
 - Greater & lesser sac connect via epiploic foramen of Winslow
 - Depending on organ serosa changes name
 - **Mesentery**: holds small intestines

continuous

- Suspends jejunum and ileum
 - Greater omentum
 - Attaches to greater curvature of stomach
 - Covers transverse colon and small bowel
 - 1st thing you see in abdominal cavity
 - Lesser omentum
 - Between liver & lesser curvature
 - Extends from porta hepatis to lesser curvature
 - Hepatogastric & hepatoduodenal ligaments
- Apron*
- extends from lesser portion of liver*
- **Special features of GI tract**
 - Most digestion occurs in duodenum
 - Folding increases surface area for absorption
 - rugae → mucosa longitudinal folds of stomach; disappear as digestive tract fills
 - Plicae → in intestinal lining, most abundant in jejunum, permanent features
 - Innervated primarily by parasympathetic nervous system
 - Enteric nervous system
 - Intrinsic innervation
 - Distributed in 2 intramural plexi that extend along the entire length of the gut *control coordinated smooth muscle contractions and other functions*
 - **Oral cavity**
 - Entrance to GI tract
 - Initial site of digestion
 - Mechanical digestion (via mastication)
 - Chemical digestion (enzymes in saliva)
 - **Salivary glands**
 - Three pairs of salivary glands that function to wash the palate & secrete saliva
 - Salivary glands consist of 99% H₂O & amylase (first step of chemical digestion, breaks down carbs to disaccharides)
 - Parotid glands
 - Largest salivary glands *25-30% of saliva*
 - Anterior & inferior to ear
 - Partially overlying masseter muscle
 - Close relationship with facial nerve
 - Submandibular glands *inferior to mandible 60-70% saliva opens through a papilla in floor of mouth*
 - Sublingual glands *inferior to tongue - has many sublingual ducts 3-5% saliva*
 - **Esophagus**
 - Not covered by peritoneum, instead covered by adventitia (loose connective tissue)
 - Three portions: cervical, mediastinal, abdominal
 - Histology layers of esophagus
 - Mucosa (**stratified squamous**)
 - Submucosa: thick elastic fibers, mucous glands
- pierces diaphragm T10 T11 joins Stomach*
- pharynx ↓ Stomach*
- left gastric artery*

art
-in
-f
-l
-

- Muscularis: inner circular, outer longitudinal
- Adventitia (no serosa) *loose CT; not covered by peritoneum*
- *○ Supplied by celiac trunk w/ left gastric artery, hepatic artery, & splenic artery
- Constrictions: beginning (C6), arch of aorta, left bronchus
- Diaphragm (T10)
- Termination (T11) by joining the stomach, inferior to diaphragm
- Venous drainage to portal vein *& azygous vein*
- Superior esophageal sphincter AKA cricopharyngeus muscle
 - Skeletal muscle
 - Where pharynx & esophagus meet
- Inferior esophageal sphincter (circular smooth muscle) AKA cardiac sphincter
 - NOT A TRUE SPHINCTER!!
 - Circular smooth muscle
- Tubular passageway
 - Pharynx to stomach
 - Bolus of food swallowed
 - 25 cm in adult
 - Esophageal hiatus: through diaphragm *2 leaves → R & L*

• GI viscera

• Stomach

regions
cardia
fundus
body
pyloric
antrum
canal
sphincter
curvatures
-lesser
-greater

- Gastroesophageal junction
 - Between mucosa of esophagus and mucosa of stomach
 - Crucial role in preventing stomach acid from flowing back into esophagus
- Regions:
 - Cardiac notch
 - Fundus (most superior region)
 - Body
 - Greater and lesser curvature
 - Pyloric antrum, pyloric canal, pyloric sphincter
 - Pyloric sphincter regulates intake into duodenum as duodenum cannot handle acidity
- Layers:
 - Now has an oblique layer in muscularis externa
 - 1) longitudinal layer (superficial)
 - 2) circular layer
 - 3) oblique layer (deep)
- Stomach interior:
 - Intraperitoneal: whole stomach covered by peritoneum
 - Mucosal specializations
 - Rugae: more within pylorus and greater curvature
 - Gastric canal *→ disappears when filled*
 - Pits

Chyme

finishes
Chemical
Digestion

- glands

• Small intestine

- Extends from pylorus of stomach to cecum of large intestine
- Thin-walled tube → 6 meters, 20ft in length - really long! Small b/c of diameter
- Coiled
- Intestinal wall:
 - Mucosa
 - Muscularis mucosae
 - Submucosa
 - Muscularis externa
 - Serosa → known as mesentery
- Includes:

• absorbs most
nutrients

■ Duodenum

• closest to
Stomach
• "mixing Bowl"
• chyme from
Stomach
• digestive secretions
from pancreas &
liver

- First segment of small intestine
- 25 cm long 10 inches
- Originates at pyloric sphincter
- Major duodenal papilla (aka popular vater)
- Shortest & widest
- Retroperitoneal
- Duodenal cap/bulb/ampulla - 1st portion (intraperitoneal)
- Descending part
 - Greater and lesser papillae
- Transverse part
 - Longest part
- Ascending part
- Duodenojejunal flexure

• celiac trunk
• SMA

■ Jejunum

- Middle segment of small intestine
- Has the most plicae → absorption
- Microvilli cells make up the villi
- Primary region for chemical digestion and nutrient absorption
- Proximal 40% 2/5 of length
- Mostly in left upper quadrant
- Larger diameter, thicker walls
- More vascular → lots of absorption

■ Ileum

Last

- Terminates at ileocecal valve (controlling mechanism)
- Sphincter that controls entry of materials into large intestine
- Distal 60% 3/5th
- Right lower quadrant
- Thinner walls
- Peyer's patches → lymphoid tissue
- Terminates at ~~ileocecal junction~~ → acts as sphincter - protects against reflux ileocolic junction

Ileum empties into cecum (LI)
Ileocecal valve

plicae:
- Transverse folds in
Intestinal lining
- permanent features

- Majority of H₂O absorbed here

- **Large intestine**

- Absorption of fluid/electrolytes H₂O
- Taenia coli is functional muscle (3) → longitudinal muscle condense into three thin muscle strips that pull large intestines to form haustra
 - Absent in rectum & appendix
- Haustra are formed from the condensing of the longitudinal muscular layer (contractions)
- Horseshoe-shaped
- Extends from end of ileum to anus
- Lies inferior to stomach & liver
- Watery material that first enters the large intestine soon solidifies and becomes feces
- Flora importance - produce vitamin K - involved in coagulation
- Store fecal matter
- Lack of villi
- Less than 10% absorption of GI tract
 - **Histology**
 - Goblet cells for lubrication for defecation
 - **Four sections include**
 - cecum (w/ vermiform appendix)
 - Colon
 - Rectum
 - anal canal
 - **Cecum & appendix**
 - Intraperitoneal Blind pouch
 - Narrow tube
 - Fixed by mesoappendix
 - Prevents reflux into ileum
 - **Vermiform appendix** a lymphoid organ
 - Slender, hollow appendage
 - Dominated by lymphoid nodules

Colon
cecum

(4)

Rectum
anal canal
(Sacculations)

Colon

right

Lack villi (no absorption)

less than 10% absorption

goblet cells

No enzymes

feces

large Diameter

Thinner walls than SI

- **Ascending colon** → fluid contents

- **Retroperitoneal**
- Begins @ ileocecal valve
- Terminates at right colic flexure (hepatic)
- Large diameter

- **Transverse colon**

- **Intraperitoneal** most mobile portion
longest
- Hangs by transverse mesocolon

- Left ○ **Descending colon** → diameter is smaller, less fluid (present w/ constipation)

- **Retroperitoneal**
- Narrowest diameter

- **Sigmoid colon**

- **Intraperitoneal**
- Extends into pelvis right into rectum
- Storage of fecal matter until time is right
- Distal to descending colon

- **Rectum**

- Forms last 15cm of digestive tract
- Expandable organ for temporary storage of feces
- **Retroperitoneal**

anal
sphincters }

- Internal anal sphincter: smooth muscle cells; involuntary control
- External anal sphincter: skeletal muscle fibers; voluntary control
- Part of levator ani

Distal portion

- **Lecture 13 - Accessory digestive organs**

- **The pancreas**

- Retroperitoneal
- Regions:
 - Head: in the C of the duodenum
 - Uncinate process: endocrine & exocrine
 - Neck, body, tail
 - Pancreatic ducts
 - Main and accessory
 - Pancreatic islets aka islets of langerhan
 - Endocrine tissues of pancreas
 - Scattered 1% of pancreatic cells

2 glands in 1

- **The liver**

- Largest visceral organ & largest gland
- Completely covered by glisson's capsule: very shiny
- Incomplete covering by peritoneum: bare area is not covered
- Triple blood supply
 - Portal vein: inflow of blood needing metabolic processing
 - Brings most GI venous blood to the liver first
 - Portal vein is made up of the splenic vein and superior mesentary vein
 - Hepatic veins: outflow to right atrium Drains into IVC
 - Hepatic artery
 - Supplies bile ducts
 - Gives about 15% of flow to liver tissue
 - Comes off celiac trunk off of the descending abdominal aorta
 - Celiac trunk
 - Left gastric
 - Splenic artery
 - Common hepatic artery
 - Proper hepatic artery brings arterial blood flow to liver

*

- Anatomy of liver → peritoneal ligaments: reflect onto liver from diaphragm
 - Surround bare area
 - Falciform ligament: separates right and left lobes
 - Ligamentum teres: holding the duodenal flexure? *round ligament*
 - Coronary
 - triangular

- Lobes of liver

- Right: largest
- Left: separated from right by falciform ligament
- Quadrate: between gallbladder & ligamentum venosum
- Caudate: between inferior vena cava & ligamentum venosum
 - Ligamentum venosum: was ductus venosus in the fetus where blood bypassed the liver

- Lesser omentum

- Sleeve-like structure
- Connects lesser curvature of stomach and bulb of duodenum to inferior surface of liver
 - AKA gastrohepatic or hepatoduodenal ligament

- Histology of liver

- Hepatic lobule
 - In the center is central vein drains from sinusoids
 - Hepatic portal triad → portal vein, hepatic artery, bile duct
 - Hepatic sinusoids: capillaries between hepatic artery and central vein
 - Hepatic veins: drain into inferior vena cava

- Blood flow of veins

- Artery
- Sinusoids (capillary)
- Central vein
- Hepatic vein
- Bile canaliculi send blood out of liver

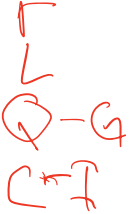
- Gallbladder

- Contains

- Fundus
- Body
- Neck
- Cystic duct
- Between right and quadrate hepatic lobes

- Blood flow

- Right & left hepatic duct → common hepatic duct → merge with pancreatic duct in duodenum and form → major papilla of Vater
- Common hepatic duct joins cystic duct to form the common bile duct which enters the duodenum & is met by main pancreatic duct



(4)

- portal vein
- hep artery
- bile duct



X

vascular
supply

- **Spleen**
 - **Intraperitoneal**
 - **Anterior to ribs 9-11**
 - Hematopoietic
 - Lymphatic: major lymphatic organ & taking old RBCs out of circulation due to vasculature in spleen being tortuous and the break up/die

- **Celiac trunk**
 - Comes off abdominal descending aorta
 - Aortic hiatus between T12-L1
 - Branches:
 - **Left gastric artery**
 - Lesser curvature
 - Esophageal branches
 - **Splenic artery**
 - Pancreatic branches
 - **Common hepatic artery**
 - Right hepatic
 - Cystic
 - Left and right hepatic
 - Right gastric
 - Gastroduodenal
 - supraduodenal

IVC

- common iliac veins
- enters
 - Right gonadal
 - Suprarenal
 - Inferior phrenic
 - Renal
 - Hepatic veins
 - Azygous

- **Superior mesenteric artery**
 - Feeds right side of body
- **Inferior mesenteric artery**
 - Left side of intestines to the left side of body
- **Lecture 14 endocrine - always remember major modulating system - takes longer to put into effect, but effect lasts longer**
- **All endocrine glands have ample blood supply!!!!**
- **Endocrine glands secrete hormones right into the bloodstream**
- **Main endocrine glands are**
 - Pituitary gland
 - Hypothalamus
 - Anatomical & physiological connection between pituitary gland & hypothalamus
 - Thyroid gland
 - Thymus gland (present in childhood - undergoes atrophy in adulthood)
 - Suprarenal gland
 - Pineal gland
 - Parathyroid glands posterior to thyroid → 4 of them
 - Pancreas
 - Reproductive glands
- **Other endocrine tissues are**
 - Heart

- Kidney
- Adipose cells
- Digestive tract

- **Hormones & receptors**

- Hormones are chemical messengers released by endocrine glands
 - Acts as “key & lock” ~ fits into receptor perfectly
 - Receptor causes second messenger to react intracellularly → to modulate function of cell

- **Hypothalamus & pituitary gland**

- **Adenohypophysis**

- “Anterior pituitary”

- **Neurohypophysis**

- “Posterior pituitary”

- **Pituitary gland (hypophysis cerebri)**

- “Hypophysis”
- Attached to hypothalamus via the infundibulum
- Sits in hypophyseal fossa of the sella turcica in sphenoid bone
- reddish-grey, ovoid body
- Covered superiorly by diaphragma sellae (saddle) of dura mater which is pierced centrally by the infundibulum and separates the gland from the optic chiasm
- Inferiorly & in front it is related to sphenoid air sinus
- It is related on either side of the cavernous sinus
- It is related to behind the dorsum sellae
- **Consists of 2 lobes:**
 - **Adenohypophysis** - anterior lobe (synthesize & secrete 9 peptide hormones)
 - Consists of **Pars tuberalis, Pars distalis, Pars intermedia**
 - **Pars distalis** is formed of thick irregular cords of cells & blood sinusoids - takes up majority of adenohypophysis
 - Two cell types: chromophobes, chromophils (acidophils & basophils) 50/50 Granules; larger
 - Receives portal blood (vein → capillary → vein)
 - **Neurohypophysis** - posterior lobe (stores 2 hormones & released when needed)
 - Close relationship with hypothalamic nuclei (paraventricular nucleus & supraoptic nucleus) via neuronal axons
 - Consists of **Pars nervosa** which is connected to the hypothalamus via infundibular stalk - takes up majority of neurohypophysis
 - Pars nervosa contains bundles of nerve fibers, cells called pituicytes, homogenous intercellular substance → herring body

The Master Gland

know

T
D
I

portal Blood

Arterial Blood

- Bundles of nerve fibers arise in hypothalamus and descend to the posterior lobe forming the hypothalamo-hypophyseal tract (HHT)
 - Storage site for ADH & oxytocin
 - Receives arterial blood (artery → capillary → vein)
- **Hypophyseal portal system**
 - Fenestrated capillary between two veins
 - Within infundibulum is a plexus of fenestrated capillaries
 - Regulatory hormones (inhibiting or releasing hormones) leave the hypothalamus and pass through the portal vessels to the adenohypophysis
 - Does not involve neurohypophysis
- **Adenohypophysis of pituitary gland**
 - Consists of five different cell types

Ant.
pituitary

- **Thyrotropes** → release TSH
- **Corticotropes** → release ACTH & melanocyte-stimulating hormone
- **Gonadotropes** → release FSH & LH
- **Lactotropes** → release prolactin
- **Somatotropes** → release GH *soma-self Growth*

● **Thyroid gland**

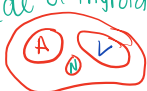
little lungs



- Lies in the lower part of the neck at the level of the 5th cervical to the 1st thoracic vertebrae
- It is formed of two lobes connected by an isthmus
- Each lobe is pyramidal in shape with an apex, a base, & 3 surfaces (**medial, lateral, & posterior**)
- The apex is @ the level of oblique line of thyroid cartilage of the larynx
- The base is at level of the 4th or 5th tracheal ring
- **Anterior border** is thin and is fed by the **anterior branch of the superior thyroid artery**
- **Posterior border** is fed by the **inferior thyroid artery** (which comes off thyrocervical trunk)
- **Medial surface** is related to:
 - Larynx, pharynx, & external laryngeal nerve
 - trachea, esophagus, & recurrent laryngeal nerve
- **Lateral surface** is related to skin, & fascia
- **Posterior surface** is related to **carotid sheath - surrounded by light connective tissue** (overlapping common carotid artery)
 - ****Structures of carotid sheath (posterior surface):**
 - **Jugular vein, carotid artery, vagus nerve**
 - **Jugular vein lateral to carotid artery**
 - **Vein has greater diameter**
- Thyroid gland is found within the **pretracheal fascia** – move up and down when swallow
- Thyroid isthmus: 2nd & 3rd tracheal rings
 - A conical pyramidal lobe is often present along its upper border

Thyroid

1 on each side of thyroid



jugular
carotid
vagus

Thyroid made up of follicles central mass colloid

Histology of thyroid gland

- Thin capsule of connective tissue
- The thyroid follicle forms the structural unit of the gland
- Each follicle consists of follicular epithelium (simple cuboidal) and a central mass of colloid which contains thyroid hormones
 - Iodine needed to make proper thyroid hormones
- Each follicle is surrounded by a delicate layer of loose connective tissue, containing a capillary network
- Thyroid medulla is made of parafollicular cells AKA c cells between follicles
 - Parafollicular or c cells are near the follicular basement lamina but separated from the lumen of the follicle by follicular epithelium. These cells secrete calcitonin

Arterial supply of thyroid gland

- Superior thyroid artery → arises from external carotid artery
- Inferior thyroid artery → arises from thyrocervical trunk of subclavian artery

Venous drainage - thyroid gland is drained by...

- Right & left Superior and middle thyroid veins
- Right & left Inferior thyroid veins

Lymphatic drainage

- Drained by lymph nodes beside the trachea (paratracheal nodes) and to deep cervical nodes

Innervation

- Autonomic innervation via sympathetic trunk & parasympathetic fibers from vagus nerve

Parathyroid gland

- Parathyroid glands are located on the posterior aspect of the thyroid → usually the tissue is embedded within thyroid tissue
 - Lie between posteromedial thyroid lobes and carotid sheath
- Small, yellow-brown ovoid in shape, not very large, usually lying between posterior borders of thyroid gland and its capsule
- Usually there are two on each side, one superior (more constant in location) and one inferior (can vary in location) - **total of four parathyroid glands**
 - Inferior pair may be above or below inferior thyroid arteries or in the thyroid gland near its base
- Has a capsule & is well vascularized
- **Principal (chief) cells:** produce PTH
- **Oxyphil cells:** larger; unknown function but they may also secrete PTH
- Adipose tissue is frequently found in this gland in older individuals
- Aberrant glands may lie between trachea and thyroid

Pancreas

- Two glands in one

Know <

S, I, M

Blood
lymph
nerve

principle → PTH
Abberant *

*

A - glucagon
 B - insulin
 F - polypeptides
 D - somatostatin

LPA

- Exocrine → (majority) secretes product in duct system (main pancreatic duct or accessory pancreatic duct) L, P, A
 - Pancreatic acini (exocrine cells)
- Endocrine → located in pancreatic islets (islets of langerhan)
 - Pancreatic Islets composed of dif cells
 - Alpha cells (glucagon)
 - Beta cells (insulin)
 - F cells (pancreatic polypeptide)
 - Delta cells (somatostatin)
 - One of secretions is insulin → allows glucose into cells → stimulated by hyperglycemia which is sensed by pancreatic cells
- Pancreas is highly vascularized!!
 - Fenestrated capillaries
 - 2 sources of arterial blood bringing blood to pancreas: pancreaticoduodenal and pancreatic arteries
 - Anterior branch of pancreaticoduodenal artery
 - Superior and inferior
 - Posterior branch of pancreaticoduodenal artery
 - Superior and inferior
 - Body and tail of pancreas
 - Supplied by splenic artery - branch of celiac trunk
 - Has three surfaces

2

Head

- Large rounded end of pancreas connects to the c-shaped part of duodenum of the small intestine
 - Pointed tail extends toward the spleen
- Located in epigastric & left hypochondriac regions
- Lies retroperitoneally at T-12/L1
- Head fills concavity of duodenum
- Body crosses left kidney
- Tail reaches hilus of spleen
- Related anteriorly to transverse colon
- Aorta & IVC lie posteriorly
- Uncinate process
 - Lies posterior to SMA and SMV
 - Lies anterior to aorta
- Pylorus is above neck of pancreas
- Adrenal glands aka suprarenal glands
 - Superior border of kidneys
 - Highly vascularized!!
 - Supplied by renal artery, inferior phrenic artery, middle suprarenal artery
 - Made up of two parts:
 - Suprarenal cortex

Venous drainage

R
 right suprarenal
 into IVC

L
 Left suprarenal
 into renal V.

- Zona **glomerulosa** (superficial - **secretes mineralocorticoids - aldosterone**), zona **fasciculata** (**secretes glucocorticoids - cortisol**), zona **reticularis** (deep - **secretes androgens**)
- **Suprarenal medulla** deep to zona reticularis (**modified sympathetic ganglion**)
- Two small flat yellowish bodies, situated on the upper renal ends
 - They are **enclosed in renal fascia but separated from the kidneys**
- **Histology**
 - Triangular in cross section
- **Principal cells of the medulla** are the **chromaffin cells** - considered modified postganglionic sympathetic neurons that lack dendrites and axons. **They secrete adrenaline/epinephrine (85%) and noradrenaline/norepinephrine (15%)**

• LECTURE 15 MALE REPRODUCTIVE

- The reproductive system produces, stores, nourishes, and transports **gametes**
- Male produces gametes called **sperm**
- Female produces gametes called **oocytes**
- The joining of a sperm cell and an oocyte is **fertilization** → results in formation of a **zygote**

• Organization of the reproductive system

- The reproductive system includes:
 - **Gonads**
 - **Testes/ovaries** that produce gametes and hormones
 - **Reproductive tract**
 - Ducts that **receive, store, and transport gametes**
 - **Accessory glands**
 - **Secrete fluids**
 - **External genitalia**
 - **Perineal** structures (region between genitals and anus)

○ Anatomy of the male reproductive system

- Principal structures:
 - **Scrotum** (1)
 - **Testis** (2)
 - **Epididymis**
 - **Ductus deferens / vas deferens**
 - **Urethra** → main highway for urine & sperm
 - Changes names depending on location
 - **Accessory Glands**
 - **Seminal gland** (seminal vesicles)
 - Produce **60% of semen volume**
 - Seminal fluid consists of **fructose, prostaglandins, clotting factors**
 - Contents empty into the ductus deferens during **ejaculation**
 - **Seminal fluids enable sperm to become motile**

g
↓
O S
↓
fert.
↓
zygote

- **Prostate gland**
 - Produces 20-30% of semen volume
 - Prostatic fluids secreted into prostatic urethra
 - Weakly acidic
 - Secretions contain enzymes that prevent sperm coagulation in vagina and seminalplasmin → antibiotic belived to prevent UTIs in men
- **Bulbo-urethral gland (cowper's glands)**
 - Produces 5% of semen volume
 - Paired glands at base of penis
 - Contents empty into spongy urethra, immediately distal to body wall (end of membranous urethra)
 - Secretes thick, sticky alkaline mucus
 - Neutralize urinary acids that are remaining in urethra
 - Provide lubrication for tip of penis
- **Penis**
- **The testes (2)**
 - **Descent of the testes**
 - During development, the testes form inside the abdominal cavity near the kidneys
 - **Gubernaculum testis** holds testes in place
 - As body growth proceeds, gubernaculum testis does not elongate
 - Gubernaculum testis pulls testes inferiorly
 - Testes descend toward the scrotum by passing through the inguinal canals
 - The ductus deferens, blood vessels, lymphatics, and nerves remain bundled to the testes in the **spermatic cords**
 - Testicle development requires a temp of 98.6 degrees fahrenheit
 - Hence the development within the abdominal cavity
 - Heat needed during development
 - Sperm development requires temp of 96.6 degrees fahrenheit
 - Hence the development of sperm cells outside the abdominal cavity
 - **Spermatic cords (2)**
 - Consist of:
 - **Ductus deferens**
 - **Blood vessels**
 - Deferential artery
 - Testicular artery
 - Pampiniform plexus of the testicular vein
 - **Nerves**
 - Ilioinguinal nerve

Veins
nerves
Ductus
arteries

- Genitofemoral nerve

- Pass through narrow **inguinal canals** that connect scrotal chambers to peritoneal cavity
- The scrotum & position of testes:
 - Divided into **two chambers (scrotal cavities)** which are separated by the **raphe of the scrotum** → seen as thickened tissue that extends from the anus, across the scrotum, and along the anterior surface of the penis
 - **Tunica vaginalis** lines each cavity → serous membrane that lines the cavity and reduces friction between opposing outer and inner serous layers
 - Muscles of the scrotum → **dartos muscle**
 - Superficial smooth muscle
 - Contraction causes the wrinkling of the scrotal surface
 - "Pulling scrotum up"
 - Muscles of the scrotum → **cremaster muscle**
 - Contraction tenses the scrotal sac and pulls it closer to body
 - Temp for sperm development is 2 degrees cooler than body temp
 - Therefore, if temp is cold outside, the testes move closer to the abdomen to maintain proper temp
 - "Pulling testes up"
 - The **scrotum is highly vascularized** by
 - **Internal pudendal arteries** from internal iliac arteries
 - **External pudendal arteries** from femoral artery
 - **Inferior epigastric arteries** from external iliac artery

reflex arc

X

Ilioinguinal
pudendal
Hypogastric
Genitofemoral

Scrotum is highly innervated by

- Hypogastric plexus
- Ilioinguinal nerves
- Genitofemoral nerves
- Pudendal nerves

- **Structure of the testes**

- **Tunica albuginea** → **septa**
 - Fibrous, connective tissue lining the testes
 - Extends deep into the testes and forms **septa**
 - **Septa** converge toward the mediastinum of the testis
 - **Mediastinum** supports ducts that move material toward the epididymis
- **Tunica vaginalis**

Vaginalis - out
albuginea - inner

inner

outer

semi
* Tubules → straight → rete → efferent → epididymis → ductus deferens

SSKEDD

- Serous tissue that surrounds tunica albuginea and is continuous with the surrounding tissue of the epididymis
- **Sperm transport**
 - Seminiferous tubules connect through straight tubules into rete testis which is inside the mediastinum of the testis → then to efferent ductules → then to epididymis → ductus deferens → ejaculatory ducts → urethra (prostatic → membranous → penile) → ejaculation!
- **Histology of testes**
 - Septa testes separates the testes into lobules
 - Each lobule contains tightly coiled seminiferous tubules
 - Each tubule is approx. 80cm long
 - There is a space between the coils that contain interstitial endocrine cells (Leydig cells) which produce and release testosterone
 - Seminiferous tubules begin to uncoil leading to straight tubules in the mediastinum area
 - The straight tubules interconnect with each other forming the retes testes within the mediastinum
 - Rete testis connect to the efferent ductules
 - Efferent ductules lead to the epididymis
- **Testosterone**
 - Stimulates spermatogenesis *creation of sperm*
 - Promotes sperm maturation
 - Maintains accessory organs
 - develops secondary sex characteristics (facial, axillary, pubic hair, lengthens penis)
 - Stimulates growth & metabolism
 - Stimulates sexual behavior and sex drive
- **Spermatogenesis & meiosis**
 - **Spermatogenesis**
 - Formation of sperm cells via meiosis
 - Begins in outer layer of seminiferous tubules
 - **Spermatogonia**
 - Stem cells that will give rise to sperm cells
 - Formed during the embryonic development but dormant until puberty
 - At puberty, spermatogonia continuously divide throughout reproductive years
 - As each division occurs
 - One of the cells produced by this division remains in the outer layer of the seminiferous tubules as a stem cell, as a spermatogonium
 - The other cell produced by this division differentiates to become a primary spermatocyte

- The primary spermatocyte begins to undergo meiosis

■ Meiosis

- Creation of sex cell/gametes *spermatogenesis*
- Cell division that produces haploid (half amt of chromosomes) gametes
- The primary spermatocyte undergoes meiosis 1 to produce two secondary spermatocytes
- Each diploid (has all the chromosomes) secondary spermatocyte undergoes meiosis 2 to produce two haploid spermatids
- Each spermatid matures to become a haploid sperm cell
 - Haploid cells contain 23 chromosomes
 - A single primary spermatocyte gives rise to a total of four sperm cells

(4)

○ Spermiogenesis

- Spermatids mature to form physically mature sperm cells
- This maturation process is called **spermiogenesis** *sperm puberty*
 - During spermiogenesis, the spermatids are embedded in **nurse cells** (sertoli cells)
 - Nurse cells function to maintain blood testis barrier → isolates seminiferous tubules from circulation to prevent autoimmune destruction of sperm cells
 - Nurse cells also function to..
 - Support spermatogenesis & spermiogenesis
 - Secretion of **inhibin** (hormone that controls rate of sperm production)
 - Secretion of androgen binding protein (ABP) → binds testosterone within seminiferous tubules so testosterone will continue to have an effect on spermiogenesis
 - Upon maturation, the spermatids (now mature sperm cells) enter into the lumen of the seminiferous tubules

*inhibin-rate
ABP - Testosterone
will have
effect*

○ Anatomy of a sperm

- Each spermatozoon has three areas..
 - **Head** → contains chromosomes, acrosomes consisting of enzymes (most important)!!!
 - Enzyme, hyaluronidase, required for fertilization of egg
 - **Middle piece (with neck)** → contains mitochondria (energy so tails can flip around) and centrioles
 - **Tail** → only flagellum found in the human body
 - Enables mobility of the sperm cell

○ Male reproductive tract

- Involved in the functional maturation, nourishment, storage, and transport of sperm

■ Consists of...

● **Epididymis**

- Fluid currents in the straight tubules move sperm cells into epididymis (sperm cells are physically but not functionally mature - cant move independently yet)
- Monitors composition of fluid from seminiferous tubules
- Recycling center for damaged spermatozoa
- Stores spermatozoa for further maturation

- Spermatozoa take 2 weeks to pass through epididymis
- Mature sperm cells are not motile until they undergo capacitation which happens outside the epididymis

● 2 steps:

- Sperm cells become motile when mixed with secretions from the seminal glands
- Become capable of fertilizing an egg when exposed to the female reproductive tract

○ **Three parts..**

- **Head** → receives spermatozoa via efferent ducts
- **Body** → coiled tubes pass through the body
- **Tail** → tubes begin to uncoil to become single tube called the ductus deferens

● **Ductus deferens** aka vas deferens

- Begins at the tail of the epididymis
 - Ascends into abdominal cavity through inguinal canal
- Curves around urinary bladder and the ureter, then descends back toward and through the prostate gland
 - Before entering the prostate gland, the ductus deferens expands into the ampulla
 - The ampulla joins the excretory duct of the seminal gland to become the ejaculatory duct
 - The ejaculatory duct empties into the prostatic urethra

● **Urethra**

- Divided to form three regions
 - **Prostatic urethra** → extends from urinary bladder through the prostate gland
 - **Membranous urethra** → passes through muscular wall of pelvic cavity

- **Spongy (penile) urethra** → passes through the penis

○ **Semen**

- Each ejaculation releases 2 to 5 mL of **semen**
- Ejaculate material consists of
 - **Sperm cells**
 - Normal **sperm count** from 20-100 million per mL
 - **Seminal fluid**
 - 60% from the seminal vesicles
 - 30% from prostate
 - 5% from bulbo-urethral glands
 - 5% from the epididymis
 - Enzymes
 - Dissolve vaginal mucus & acts as antibiotic

○ **The Penis**

- Divided into three regions
 - **Root** → attaches base of penis to **rami of the ischia**
 - **Body (shaft)** → consists of **erectile tissue** needed for penetration
 - **Glans Penis** → expanded distal end of the penis; surrounds external urethral orifice

- Consists of three cylindrical columns of **erectile tissue** and blood vessels

• **Two posterior corpora cavernosa**

- **Deep artery of the penis** is in the center of this tissue
- Extends posteriorly forming a **crus of the penis**
- Each crus is bound to the **ramus of the ischium** (otherwise penis would fall off)!

2 Cave

• **One anterior corpus spongiosum**

- Surrounds the spongy penile urethra
- Thickened, proximal end forms the **bulb of the penis**

1 Sponge

- The **prepuce (foreskin)** surrounds the tip of the penis

- Attaches to the **neck of the glans**
- **Preputial glands** in the skin of the neck and inner surface of foreskin secrete waxy **smegma**
 - Smegma encourages bacterial growth → infection & inflammation
 - Surgical removal of prepuce (**circumcision**) avoids this problem

Parasym.

■ **Erection of the penis**

- **Parasympathetic nerves** are activated
- Smooth muscles in **arterial walls relax**
- Arterial vessels dilate
- Arterial vessels and vascular channels within the **corpora cavernosa** and **corpus spongiosum** become **engorged with blood**
- Erection occurs

Sympathetic

■ Ejaculation

- Sympathetic activity
- Emission
 - Sperm mixed with semen by peristaltic actions of the smooth muscle in the ductus deferens, seminal glands, prostate, & bulbo-urethral glands
 - **Ejaculation:** powerful contractions of ischiocavernosus and bulbospongiosus push semen toward external urethral orifice

■ Male climacteric

- Period of decrease of reproductive capacity
- Testosterone levels begin to decline
- Occurs gradually between ages 50 & 60
- Reduction in sexual activity
 - Get testosterone injections!

● LECTURE 16 FEMALE

● Tangent facts:

- Uterus is anteflexed and anteverted
- Ovaries have no physical connection with fallopian tubes
- Fertilization occurs in the fallopian tubes

● Perineum

- Diamond shaped area between the thighs this is circumscribed anteriorly by the pubic symphysis, laterally by the ischial tuberosities, and posteriorly by the coccyx
- **2 distinct triangle bases** - formed by imaginary horizontal line extending between ischial tuberosities of the ossa coxae
 - **Anterior triangle, or urogenital triangle**
 - Contain urethral and vaginal orifices in females
 - Contains base of the penis and scrotum in males
 - **Posterior triangle, or anal triangle**
 - Location of anus in both sexes
- Boundaries of perineum...
 - **Mons pubis:** medial aspects of the thigh
 - **Gluteal folds:** the superior end of the intergluteal cleft - the cleft between the buttocks - where the tip of the coccyx is palpable
 - **Prepuce:** covers clitoris
 - **Labia majora:** scrotum in male
 - **Labia minora**
 - **Vestibule:** consists of vaginal orifice, external urethral meatus, clitoris

● Pelvic diaphragm

- **Levator ani and coccygeus muscles**
- **Levator ani** made up of...
 - Puborectalis
 - Pubococcygeus

- Iliococcygeus
 - Supports pelvic organs and lifts to help release feces during defecation
- **Anatomy of female reproductive system**
 - **Peritoneum** folds around the various pelvic organs and creates two major dead-end recesses, or pouches
 - **Anterior vesicouterine pouch** forms the space between the uterus and the urinary bladder
 - **Posterior rectouterine pouch**, AKA “**pouch of douglas**” forms the space between the uterus anteriorly and the rectum posteriorly
 - Primary sex organs, or gonads, of the female are the ovaries, which produce gametes
 - Accessory sex organs include ..
 - Uterine tubes
 - Uterus
 - Vagina
 - Clitoris
 - Mammary glands (modified sudoriferous glands - excrete milk)
- **Internal genitalia / peritoneal folds / ligaments**
 - **Mesovarium**
 - Double folds of peritoneum
 - Attaches ovaries to broad ligament
 - **Broad ligament**
 - Peritoneum
 - ✗ ■ Drapes over the uterus
 - Serosa membrane
 - Supports ovaries
 - ✗ ○ **Ovarian ligament**
 - Ovary to uterus
 - Supports ovaries
 - ✗ ○ **Suspensory ligament**
 - Ovary to lateral pelvic wall
 - Supports ovaries

extends from the mesovarium to the body
 - ✗ ○ **Round ligament**
 - Uterus through the inguinal canal to labia majora of external genitalia
- **Ovary**
 - Primary sex organ, or gonads, of the female
 - Suspensory and ovarian ligaments anchor
 - Blood supplied via the ovarian artery from the abdominal aorta
 - Venous drainage to right inferior vena cava and left renal vein
 - Composed of ovarian follicles
 - Structure of follicle include oocyte and follicular cells
 - Homologous to testes
- **Uterus - FBI CU BLADDER AND RECTUM**
 - Hollow organ, located between the urinary bladder and rectum

uterine
artery

- Regions include..
 - Fundus
 - Body
 - Isthmus - Narrow portion leading to cervix
 - Cervix
 - Cervical canal - between the two os
 - Internal os - opening to isthmus
 - external os - opening to vagina
- Uterine layers..
 - **Perimetrium**: outer tunic, serosa of the uterus
 - Peritoneum of uterine tissue
 - **Myometrium**: thick smooth muscle
 - Becomes thicker & distended during pregnancy
 - **Endometrium**: inner tunic, hormone sensitive
 - **Two layers**: stratum basalis (proliferates every month) and stratum functionalis
 - Basalis layer gives way to stratum functionale → which is what sloughs off during period
 - **Blood supply**: uterine, vaginal, and ovarian arteries
 - **Lumen**: inside/opening - not large when not pregnant
- ○ Intraperitoneal
- Covered by broad ligament
- **Uterus is anteflexed** at cervix and body and **anteverted** at junction of cervix and vagina
- Supported by levator ani muscles, and urogenital diaphragm
- **Uterine blood supply**
 - Uterine arteries from internal iliac and ~~ovarian arteries~~ from abdominal aorta
 - Very vascular!
- **Uterine tubes**
 - AKA **fallopian tubes or oviducts**, extend laterally from both sides of the uterus toward the ovaries
 - Secondary oocyte from meiosis 2 is fertilized here
 - **Parts** lined with **simple ciliated columnar** and serosa (Cilia helps to move oocyte from tube to uterus)
 - **Infundibulum** - flared end of tube
 - **Ampulla** - long, wide portion
 - **Isthmus** - narrow, attaches to uterus
 - **Interstitial segment "intramural"** - through uterine wall
 - **Fimbriae** - finger like projections at distal end
 - Blood supply via ovarian and uterine arteries
 - Nerve supply from ovarian and uterine plexi
 - Attaches to the uterus
 - ○ Does not physically attach to ovary

- Supported by broad ligament
- Homologous to vas deferens
- **Vagina**
 - Extends from cervix to exterior of body
 - Posterior to urethra, anterior to rectum
 - Birthing canal
 - Supplied by uterine and vaginal arteries
 - **Hymen**: mucous membrane over entry
 - **Vestibule** of vagina is region between labia minora
 - Thick-walled, fibromuscular tube
 - Forms the inferior most region of the reproductive tract
 - **Copulatory organ** of the female
 - Relatively thin, distensible wall consists of **three tunics**...
 - An inner mucosa, a middle muscularis, and outer adventitia (outside pelvic cavity)
- **Cervix and vagina**
 - Cervix attaches to vagina at ~90 degree angle
 - Cervix creates the **fornix** → pocket surrounding uterine cervix
- **Lymphatic drainage**
 - Superior internal and external iliac nodes
- **External genitalia**
 - The external sex organs of the female are collectively called the **vulva**
 - **Mons pubis** is expanse of skin and subcutaneous connective tissue anterior to the pubic symphysis
 - Covered with pubic hair in post-pubescent females
 - **Labia majora**
 - **Labia minora**
 - Contains the **vestibule**: space between labia minora
 - Urethral orifice "meatus"
 - Vaginal orifice "introitus"
 - **Clitoris** located at anterior regions of labia minora
 - **Glans**
 - Glans and crus of clitoris is how it attaches to pelvis
 - **Prepuce**: external fold of labia minora that forms a hoodlike covering over the clitoris
 - Contains erectile tissue
 - Homologous to penis
 - **Labia Majora**
 - Longitudinal folds of skin
 - Derived from same embryonic structures as scrotum
 - **Labia Minora**
 - No hair or fat
 - Contains erectile tissue
 - Anteriorly forms **prepuce** of clitoris

- Prepuce of clitoris contains two crura (each is a crus) "wing" & two corpora cavernosa

○ Vestibule

- Contains openings..
 - Urethra
 - Vagina
 - Vestibular glands

○ Greater vestibular glands aka bartholin's glands

• Anatomy of the breast

○ Location

- Lies between the 2nd & 6th rib
- Between the sternal edge & midaxillary
- $\frac{2}{3}$ superficial to pectoral (major) muscle
- $\frac{1}{3}$ superficial to serratus anterior
- Nipple and areola lateral midline of breast

○ Composition

- Glandular tissue / lactiferous glands - modified sudoriferous glands
- Organized in lobes around the nipple
- Duct drains each lobe via pore on the nipple
- Ampulla holds milk

○ Fibrous tissue

- Includes suspensory ligament (cooper's ligament) that connects skin to breast fascia

○ Adipose tissue

- Surrounds glandular tissue / lactiferous glands
- Predominates superficially and deeply

○ Vascular supply

- To medial quadrant from anterior intercostal artery
- To lateral quadrants from axillary artery

○ Lymphatics

- Extensive lymph system b/c nipple is port of entry
 - Drainage is towards axilla and to collateral side
- Group of nodes
 - Pectoral (anterior) nodes
 - Lower border of pectoralis major
 - Inside axillary fold
 - Drains most of breast tissue
 - Subscapular (posterior) nodes
 - Lateral border of scapula
 - Deep in posterior axillary fold
 - Lateral nodes
 - Along upper humerus
 - Central nodes
 - High in axilla

- Most frequently palpable
 - Infra & supra clavicular nodes
 - Receives from central nodes
- **Surface anatomy**
 - **Nipple**
 - Surface elevation
 - Lactiferous ducts open up into nipple
 - Smooth muscle
 - Usually everted, inverted normal variant
 - **Areola**
 - Pigmented area around nipple
 - Sometimes forms a secondary mound above the level of the breast

● **LECTURE 17 NEPHROLOGY / UROLOGY**

● **Organs involved**

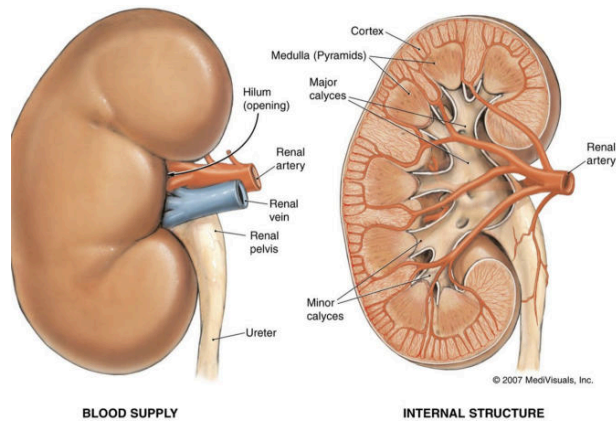
- **Primary organs: Kidneys**
 - Filter waste products from bloodstream
 - Convert filtrate to urine
- **Ureters**
- **Urinary bladder**
 - Smooth muscle, appears smaller when relaxed
- **Urethra**

● **Systems involved**

- **Urinary tract (urology)**
 - Ureters
 - Urinary bladder - storage organ for urine
 - Urethra
 - Transport urine out of body
- **Micturition:** medical term for urination

● **Kidney location**

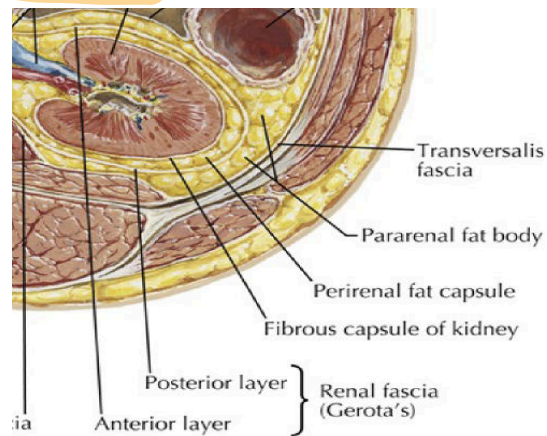
- **Retroperitoneal organs**
 - T12-L3 on the right kidney
 - T10-L2 on the left kidney
- Average size is 11 cm in length
- **NAVs enter at hilum**
 - Nerves - all autonomies - renal plexus
 - Renal artery off abdominal aorta
 - Renal vein drains to IVC
 - Left receives gonadal vein
- **Renal pelvis** exits at **hilum** (concave medial border)
- **Renal sinus:** internal space
 - Houses blood & lymphatic vessels, nerves
 - Houses renal pelvis, renal calyces
 - fat



- **Surrounding tissue**

- From **deep to superficial**

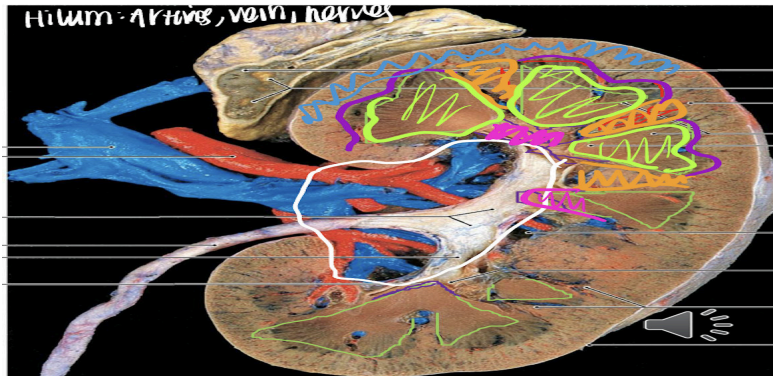
- **Fibrous capsule (renal capsule) - deepest**
 - Dense, irregular, connective tissue
- **Perinephric fat (adipose capsule)**
 - Aka **perirenal fat capsule**
 - Cushioning and insulation
- **Renal fascia**
 - Dense, irregular connective tissue
 - Anchors kidneys to posterior wall & peritoneum
- **Paranephric fat**
- **Peritoneum - most superficial**



- **Coronal plane sections**

- **Renal cortex**
 - **Renal arches**
 - **Renal columns (14)**
- **Renal medulla (10)**
 - Divided into renal pyramids (8 to 15 per kidney)
- Base against cortex (9)

- Apex called renal papilla (11)



• Blood supply

- Aorta → renal artery → segmental arteries → lobar arteries → interlobar arteries → arcuate artery → interlobular artery → afferent arteriole → glomerulus
- Blood plasma is filtered across the glomerulus into the glomerular space
- From the glomerulus → efferent arterioles → venous system
- The efferent arterioles branch into peritubular capillaries and vasa recta
 - **Vasa recta** - capillary networks responsible for exchange of solutes / molecules
- Vasa recta → interlobular vein → arcuate vein → interlobar vein → renal vein → IVC

• Urinary filtrate/flow

- **Nephrologic system:**
 - Glomerulus/Bowman's capsule → proximal convoluted tube → descending limb (thick to thin limb) → nephron loop (loop of henle) → ascending limb (thin to thick limb) → distal convoluted tube → collecting duct → papillary duct → renal papilla (last filtrate stop station) → minor calyx (finally becomes urine) → major calyx
- **Urology system**
 - Renal sinus/pelvis → ureter → bladder → urethra → external urethral orifice/meatus

• Innervation

- Autonomic nervous system fibers called renal plexus
 - Accompanies each renal artery
 - Enters kidney via hilum
- Receptor terms
 - Adrenergic = sympathetic
 - Cholinergic = parasympathetic
 - Muscarinic = parasympathetic (minus sweat glands)
- Sympathetic nerves comes from splanchnic nerves which come from thoracic and abdominal sympathetic trunks (T5-L2 nerve roots)
 - Greater splanchnic nerves (T5-T9)
 - Lesser splanchnic nerves (T10-T11)
 - Least splanchnic nerve (T12) - mainly sensory for kidney

A
Really
Smart
Lady
Is
Always
In
A
Good
Environment

Very
Intrest
Art
Is
Really
Impressive

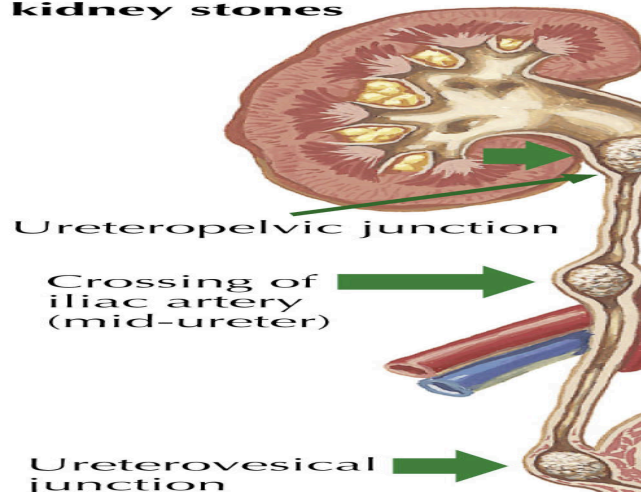
- Sympathetic nervous system related to kidney function
 - Vasoconstrict the afferent and efferent arteriole → decreases blood flow to the glomerulus → decreasing urine production
- Vagus nerve (CN X)
 - Role is unclear
- Glomerular filtration increases with a **decrease in sympathetic tone** - maybe due to vagus nerve?
- Most other controls of the kidney result from **hormonal** stimuli
- **The nephron - functional unit of kidney**
 - **Nephron structures**
 - Renal corpuscle
 - Glomerulus
 - Glomerular capsule (bowman's capsule)
 - Renal tubules
 - Proximal convoluted tubule (PCT)
 - Nephron loop (loop of henle - LOH)
 - Descending limb - thick to thin
 - Ascending limb - thin to thick
 - Distal convoluted tubule (DCT)
 - Collecting tube
- **Type of nephrons**
 - **Medullary (juxtamedullary) type**
 - 15% of nephrons
 - Extends into medulla
 - Long nephron loops
 - Primary purpose: recovery of H₂O, Na⁺, and Cl⁻
 - Have vasa recta (branching bed of capillaries) → gives more surface area for H₂O, Na⁺ and Cl⁻ to come back into bloodstream
 - Helps concentrate urine
 - **Cortical type**
 - 85% of nephrons
 - Near peripheral edge of cortex
 - Short nephron loops
 - Have peritubular capillaries
 - Produce “standard” urine
- **The glomerulus - beginning of the bloods filter**
 - Filtration membrane - filters blood
 - Fluid from blood (plasma) collected here before entering nephron system, going to PCT
 - Large filtration pores (fenestrations)
 - **Endothelial fenestrations** to allow filtration of substances to cross through into capsular space, where the filtrate collects as it begins to flow through the nephron to become urine
 - Created by **podocytes**

- **Glomerular basement membrane** → provides initial structure and is wrapped by **podocytes** with foot-processes (pedicles) which are coated in **Glycocalyx**, which has negative charge which aids in allowing for filtration to occur across membrane
 - negative charge affects absorption
 - Negatively charged filtration slits prevents large anions and proteins from absorption
 - Larger molecules, i.e albumin, are not able to pass through → so we shouldn't see much in urine → should stay in the blood and not within the capsular space
- **Filtration membranes** allow for H₂O, electrolytes, glucose, fatty acids, amino acids, urea, and vitamins to pass
-
- **Renal corpuscle**
 - Parietal layer: simple squamous epithelium
 - Visceral layer: podocytes (pedicles & filtration slits)
 - Capsular space (bowman's capsule)
 - Filtration membrane: fenestrations and filtration slits
 - Tubular pole: connects to PCT
- **Juxtaglomerular apparatus - BP, renin release, & GFR!!!**
 - Senses changes in BP, Na, and tubular flow rate → triggers renin release to control BP & Na
 - **Macula densa**
 - Cluster of specialized cells in the DCT
 - **Stimulates the secretion of renin** from juxtaglomerular cells
 - Senses NaCl and volume
 - Regulates flow to kidneys and the GFR
 - Juxtaglomerular cells or granular cells (smooth muscle fibers from **afferent arteriole**)
 - synthesize, store, and **secrete renin**
 - Released in response to decrease in BP via mechanoreceptors in the vascular walls or when stimulated by the macula densa → increases BP to normal
 - Epo-producing & oxygen sensing (REPOS) cells
 - Sense oxygenation levels and secrete **erythropoietin** (stimulates bone marrow to create more RBC)
 - Kidney function can impact RBC production
- **Proximal convoluted tubule**
 - Begins at tubular pole of the renal corpuscle (glomerulus & bowmans capsule)
 - Cells: simple cuboidal epithelium
 - Ends where the thick descending limb of LOH begins
- **Loop of henle**
 - Originates at end of PCT
 - Projects toward and/or into the medulla

- Each loop has two limbs
 - Descending limb: thick to thin
 - From cortex and PCT toward medulla
 - Ascending limb: thin to thick
 - Returns to renal cortex and leads into DCT
- **Distal convoluted tubule**
 - Contains macula densa
 - Begins at end of ascending limb of LOH
 - Adjacent to the afferent arteriole (juxtaglomerular apparatus)
- **Collecting tubule and ducts**
 - One collecting tubule receives filtrate (urine) from multiple DCTs
 - Duct is where many collecting tubules start to come together, and final filtrate is now urine as it gets to papillary duct
 - No other modifications can be made to the filtrate once it is in the collecting ducts
 - Multiple collecting ducts come together to form a renal papilla
 - The renal papilla drain the urine into the minor calyces
- **Ureters**
 - Long, fibromuscular tubule → bit of peristalsis because of muscular action
 - Conduct urine from kidneys to bladder
 - Retroperitoneal organ
 - Originate at renal pelvis
 - Posterior structure & travels inferiorly to enter at the posterolateral wall at base of urinary bladder
- **Ureter walls**
 - Adventitia (outer) → external connective tissue
 - Muscularis (middle) → smooth muscle
 - Creates peristaltic movement that allows urine to travel to bladder regardless of gravity
 - Innervated by splanchnic nerves for parasympathetic stimulation
 - Mucosa (inner) → transitional epithelium: gives stretch
 - Sensory innervation by T12-L2 vertebral roots
 - Extremely sensitive to pain
 - parasympathetic : pelvic splanchnic nerves from the lumbar spine
- **Zones related to ureters**
 - **Ureteropelvic junction (closest to kidney)** → common site for kidney stone to get stuck
 - **Crossing of iliac artery (mid-ureter)** → also common site for stone to get stuck

- **Ureterovesical junction** (distal ureter) → also common site for stone to get

Common sites of obstruction for kidney stones



- **Water under the bridge**

- The **ureter** travels **deep** to or inferior to the **testicular or ovarian artery and vein bilaterally**
 - This is important for orientation while performing intra-abdominal or genitourinary procedures intraoperatively

- **Bladder**

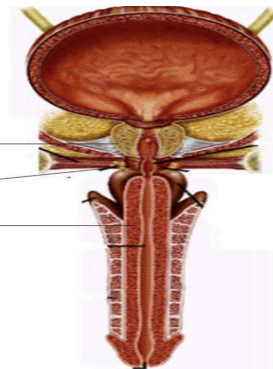
- Serves as reservoir for urine
- **Detrusor muscle** within bladder → expandable
 - What will squeeze to empty the bladder
 - Innervated by hypogastric nerve
- **Trigone**: 2 ureteral openings/orifices & neck of urinary bladder
- Superior and posterior to pubic symphysis
- **Median umbilical ligament** = remnant of the urachus (tube returning fetal urine to the mother via the umbilical cord)
- non-peritoneal organ except for the superior wall → changes where the pain is sensed
- Under both autonomic and voluntary control
 - Will relax to allow storage of urine - parasympathetic
 - Time to urinate → voluntary control where internal & external urethral sphincter is relaxed and can squeeze/control bladder to empty
- **External urethral sphincter** is skeletal muscle that we can voluntarily control
 - In urogenital diaphragm of the pelvis
- Inner bladder is transitional epithelium - can expand 15 times its empty volume to hold ~ 500 mL of urine (average adult)
 - Distal to that is lamina propria, then the submucosa before detrusor muscle and then the outer surface with NAVs in the adventitia
- Superior aspect is peritoneum, innervation is slightly different than the bladder itself
- In females..

- The uterus is posterosuperior to the bladder and the vagina is posteroinferior to bladder
 - In males..
 - The rectum is posterosuperior to the bladder
 - The prostate gland is inferior to the bladder
 - Outer layer is adventitia, except for superior wall which is peritoneal
 - Middle muscular layer = detrusor muscle
 - Inner layer is transitional stretchy epithelium
- **Blood supply to bladder**
 - Aorta → common iliac → internal iliac → anterior division → superior and inferior vesical arteries
 - Bladder is referred to as vesicular
 - Venous drainage via the vesical venous plexus → internal iliac veins → inferior vena cava
- **Urethra**
 - Drains urine from bladder to outside body
 - Last portion of the urethra: urethral meatus, external urethral meatus OR external urethral orifice
 - Female: 3-4 cm long
 - Short urethra and proximity to anus predisposes females to UTIs
 - Male: 18-20 cm long
 - Female urethra is just simply the urethra
 - Male urethra name changes depending on location
 - **Prostatic urethra:** urethra passes through prostate
 - Multiple small prostatic ducts enter here during sexual stimulation
 - **Membranous urethra:** end of the prostatic urethra to beginning of the penile urethra; pierces the urogenital diaphragm
 - **Penile (spongy) urethra:** passes within the penile shaft ending at the external urethral orifice

The male urethra has three regions:

- 1) prostatic urethra
- 2) membranous urethra
- 3) penile urethra.

Difficulty in voiding urine with enlarged prostate



- **Innervation**
 - **Internal urethral sphincter = autonomic control**
 - **Parasympathetic** via the **pelvic nerve** = relaxes the sphincter in order to urinate and help to contract the bladder via muscarinic receptors

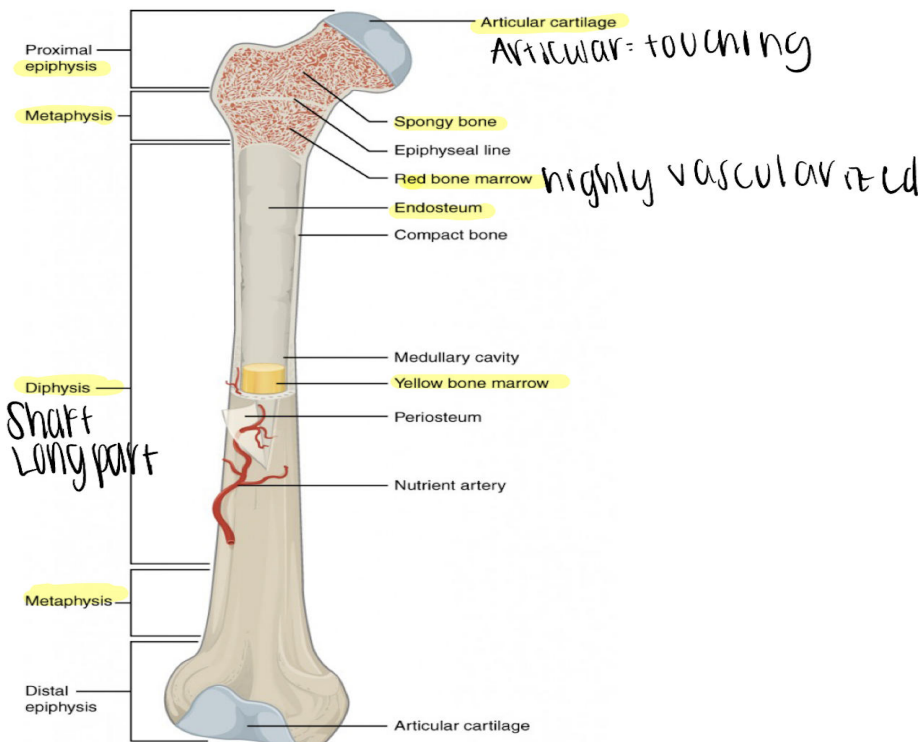
- **Sympathetic** via the **hypogastric nerve** = contracts the sphincter in order to stop urination via adrenergic receptors
- **External urethral sphincter** = **pudendal nerve** - **voluntary control**

ANATOMY EXAM 4

LECTURE 18 - AXIAL SKELETON

Describe the osteon and why it is the structural unit of bone

- **Osteon** is the basic structural unit of skeletal bones
 - Contain **osteocytes**
 - Consists of **central/haversian canal** (blood supply/nerves in center of osteon), **canaliculi** (passage of substances b/w osteocytes & blood vessels), **lacunae** (small cavities that contain osteocytes), & **lamellae** (concentric rings around central canal)
- Osteocytes are mature bone cells that live in the lacunae and communicate via canaliculi



Describe the area of ossification and growth of long bones

- **Primary ossification center** (where bones grow): located in the **diaphysis** (shaft) of the long bone
 - **Function:** 1st area where bone tissue replaces cartilage during fetal development
- **Secondary ossification center:** located in the **epiphyses** (ends) of the long bone
 - **Function:** appears after birth and contributes to the growth of the bones ends
 - **Epiphyseal plate** (growth plate): located between diaphysis and epiphysis
 - **Function:** main site of **longitudinal growth** in children/adolescents
 - Not a plate, just a space where bone continues to grow until skeletal maturity
 - **Structure:** made of hyaline cartilage & organized into zones for completeness

Compare and contrast trabecular and compact bone

"spongy"

Comparison Table: Trabecular vs. Compact Bone		
	<i>trabecular</i>	<i>compact</i>
Structure	Porous, lattice-like network (trabeculae)	Dense, solid, tightly packed <i>no bone marrow</i>
Location	Found at the ends of long bones, inside vertebrae, pelvis, ribs, skull	Forms the outer layer of all bones, especially the diaphysis of long bones
Function	Supports bone marrow, reduces bone weight, absorbs shock	Provides strength, protection, and structural support
Bone Marrow	Contains red bone marrow (hematopoiesis)	Usually does not contain marrow
Vascularization	Highly vascularized	Less vascularized compared to trabecular bone
Remodeling Rate	High – responds quickly to metabolic changes	Slower remodeling rate
Mechanical Role	Distributes forces and stress	Bears weight and resists bending and torsion
Density	Low density	High density

Compare and contrast the various types of marrow



Feature	Red Marrow	Yellow Marrow
Primary Function	Hematopoiesis (production of red blood cells, white blood cells, and platelets)	Fat storage (mainly adipocytes)
Color	Red, due to abundance of blood and hematopoietic cells	Yellow, due to high fat content
Location (in adults)	Flat bones (pelvis, sternum, ribs, vertebrae), ends of long bones (epiphyses)	Medullary cavity of long bones (diaphysis)
Location (in children)	Found in nearly all bones	Minimal; increases with age
Cell Types Present	Hematopoietic stem cells, developing blood cells	Adipocytes, mesenchymal stem cells
Conversion	Can convert to yellow marrow with age	Can revert to red marrow under extreme conditions (e.g., severe blood loss)
Vascularization	Highly vascularized	Less vascularized

Identify the bones of the skull

- **The skull** consists of the **cranium** → **cranial cavity**
 - 8 individual bones
- The skull consists of the **face** → **facial bones**
 - 14 individual bones
- The skull consists of **associated bones**
 - 7 individual bones
- **Sutures of the skull:** made of dense fibrous connective tissue that connect the bones of the skull
 - **Lambdoid suture**
 - b/w occipital bone and 2 parietal bones
 - **Sagittal suture**
 - b/w two parietal bones
 - **Coronal suture**
 - b/w frontal bone & two parietal bones
 - Squamous suture
 - b/w temporal & parietal bone
 - Frontonasal suture
 - b/w nasal bones & frontal bone

occipital
parietal
frontal
temporal
sphenoid
ethmoid

maxillae
palatine
nasal
interior nasal
zygomatic
lacrimal
vomer
mandible

hyoid
& auditory ossicles

Discuss the differences between the cranial bones and the facial bones

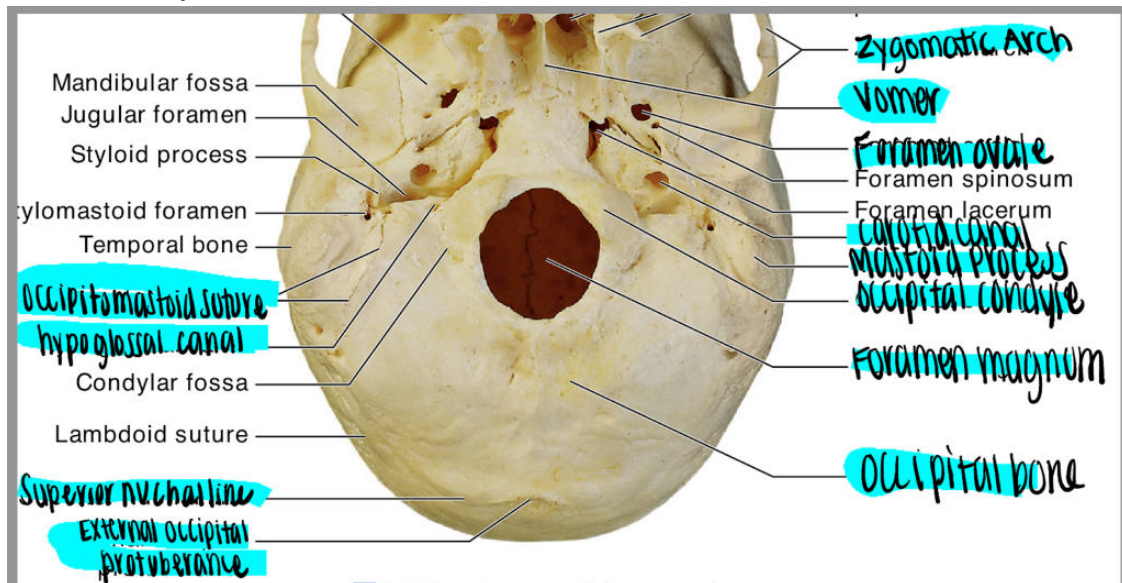
- **Cranial bones:** occipital, parietal, frontal, temporal, sphenoid, and ethmoid

29
Bones

- **Facial bones:** maxillae, palatine, nasal, inferior nasal conchae, zygomatic bones, lacrimal bones, vomer, mandible
- **Associated bones:** hyoid bone, auditory ossicles in temporal bones
- **BONES OF THE CRANIUM**

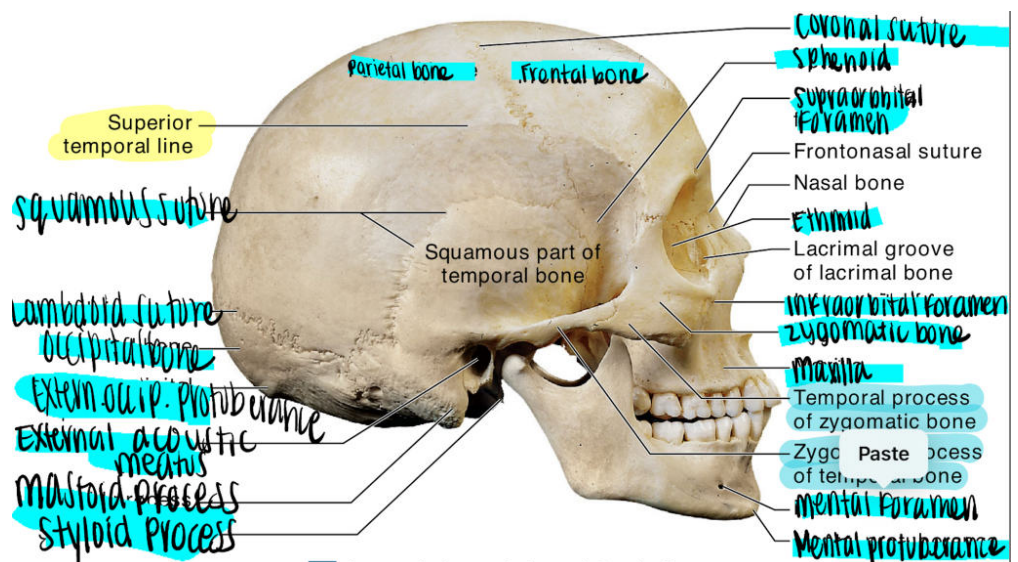
- **Occipital bone** consists of..

- Foramen magnum - spinal canal & vertebral artery travels via here
- Occipital condyles (bony protrusion)
- External occipital crest & protuberance
- Inferior & superior nuchal lines
- Hypoglossal canal
- Condylar fossa



- **Parietal bones** consist of...

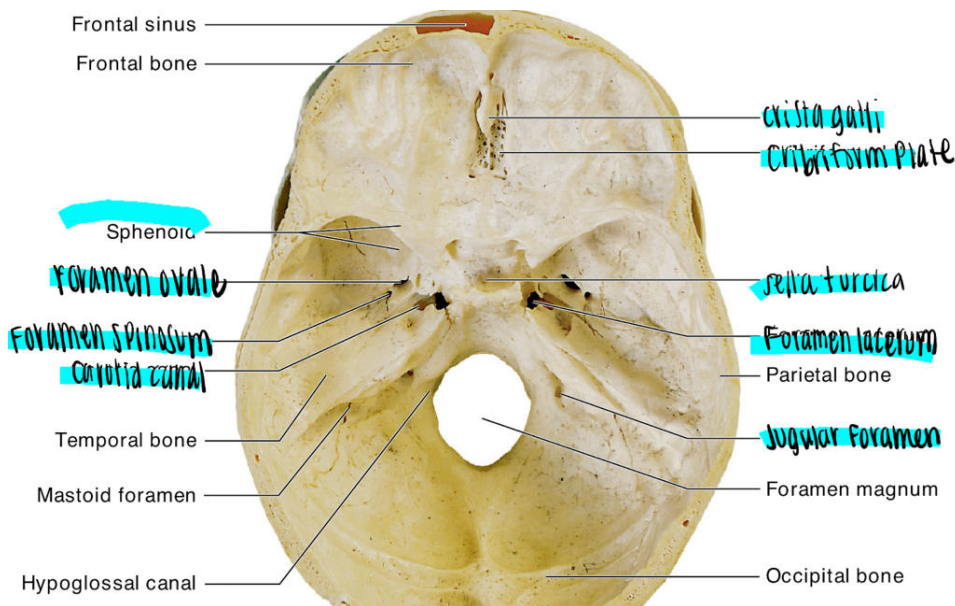
- Superior temporal line
- Inferior temporal line
- Parietal eminence



- **Frontal bones**

- **External view:**

- Frontal suture
 - Frontal eminence
 - Squamous part
 - Superior temporal line
 - Supra-orbital margins
 - Superciliary arches
 - Supra-orbital foramen
 - Orbital part
 - Lacrimal fossa
- **Internal view:**
 - Frontal crest
 - Frontal sinuses
- **Temporal bones:**
 - **External view**
 - Squamous part
 - Zygomatic process
 - Temporal process
 - Zygomatic arch
 - Temporal part
 - External acoustic meatus
 - Tympanic membrane
 - Stylomastoid foramen
 - Styloid process
 - Mastoid process
 - Mastoid foramen
 - Jugular fossa
 - Jugular foramen
 - **Internal view**
 - Carotid canal
 - Tympanic cavity
 - Auditory ossicles
- **Sphenoid bone**
 - Body
 - **Sella turcica** - pituitary gland sits here
 - Optic canals
 - Optic groove
 - **Greater and lesser wings of the sphenoid**
 - **Foramen ovale** (mandibular nerve V3 passes thru here)
 - **Foramen spinosum**
 - **Foramen rotundum** (maxillary nerve V2 passes thru here)



- **Ethmoid bone**

- Crista galli
- Cribriform plate - olfactory nerve passes here
- Superior and middle nasal conchae

- **Cranial fossae**

- Anterior cranial fossa
- Middle cranial fossa

- **BONES OF THE FACE**

- **The maxillae**

- Maxillary sinuses
- Palatine processes
- Infra-orbital groove

- **Palatine bones**

- **Nasal bones**

- **Inferior nasal conchae**

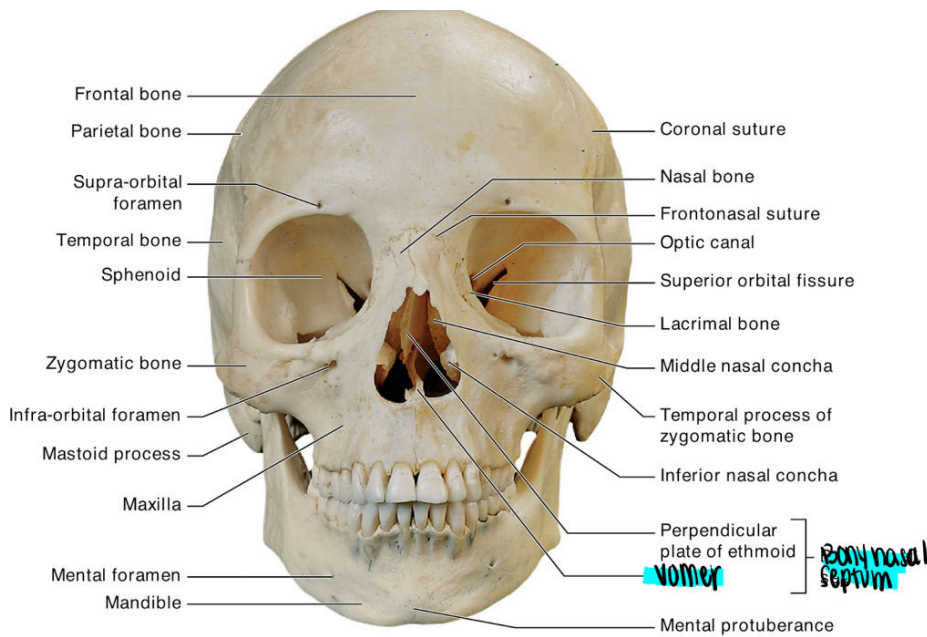
- **Zygomatic bones**

- Articulate with the zygomatic process of the temporal bone. Forms the zygomatic arch

- **Lacrimal bones**

- **The vomer**

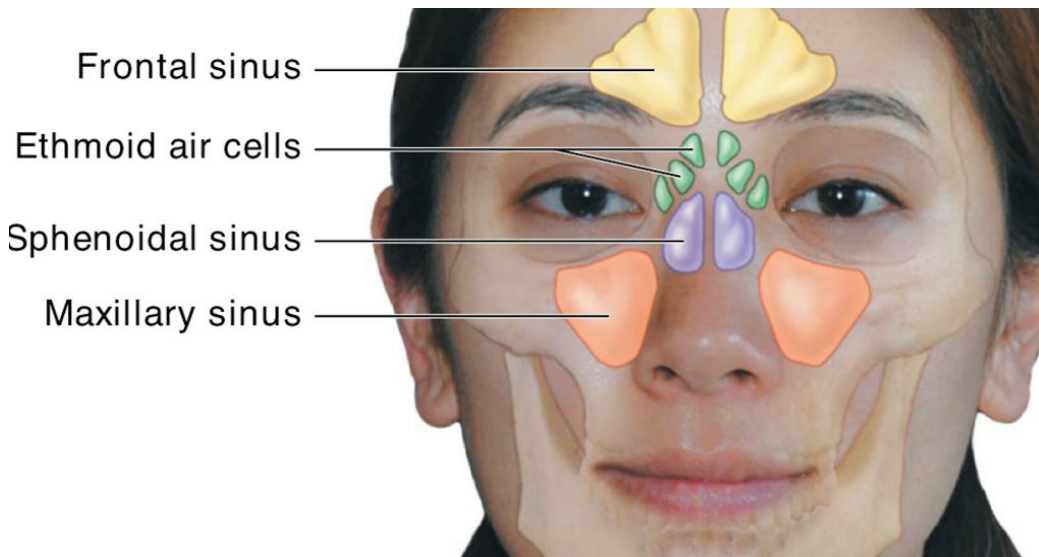
- **Forms inferior portion of nasal septum**



- **The mandible**
 - Body
 - Angle
 - Condylar processes
 - Head
 - Mental foramina
 - Submandibular fossa

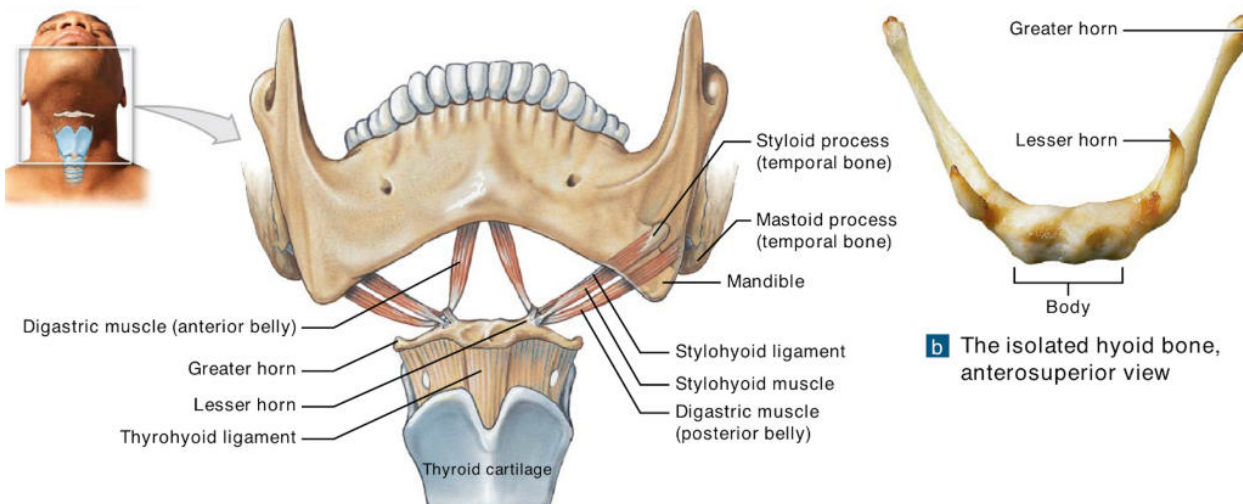
Discuss the orbits, nasal complex, and hyoid bone

- **The orbits**
 - Protects the eyes
 - **Formed from..**
 - **Frontal bone** → makes the roof of the orbit
 - **Maxilla bone** → forms the floor of the orbit
 - **Lacrimal and ethmoid** → form the medial wall of the orbit
 - **Sphenoid bone** → forms the posterior wall of the orbit
 - **Zygomatic bone** → forms the lateral wall of the orbit
- **The nasal complex**
 - **Consists of..**
 - Nostrils
 - Nasal apertures
 - Nasal septum: vomer and perpendicular plate of the ethmoid
 - **Paranasal sinuses**
 - Frontal sinuses
 - Sphenoidal sinuses
 - Maxillary sinuses
 - Ethmoid air cells



- **Hyoid bone**

- Suspended by stylohyoid ligaments
- Does not articulate with any bones and is therefore extremely mobile → moves a lot, connected via ligaments
- Consists of..
 - Greater and lesser horns which serve as points for muscle and ligament attachments



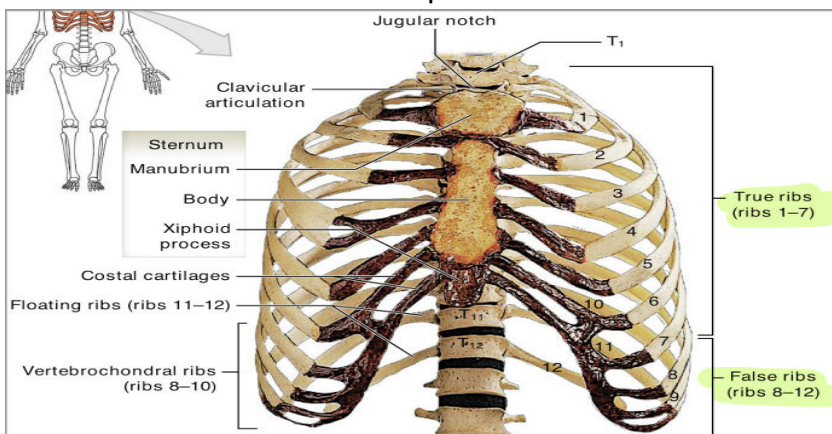
Compare and contrast the bones that make up the axial skeletal system and important landmarks

- 206 separate bones in the body → Axial consists of 80 bones
- Axial skeleton consists of skull bones, auditory ossicles, hyoid (throat), vertebral column, thoracic cage
- Functions..
 - Framework that supports & protects organs in the dorsal & ventral body cavities
 - Protects special sense organs for taste, smell, hearing, balance & vision
 - Adjusts posture of head, neck & trunk

- Moves the thoracic cage for respiration
- Stabilize the appendicular skeleton

Describe the anatomical components of the thoracic cage

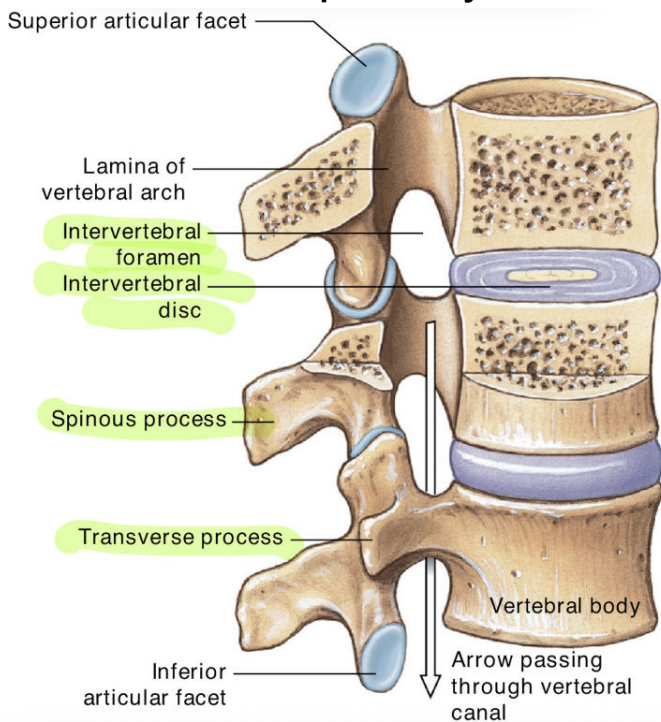
- **The thoracic cage** has two structural functions:
 - protects the heart, lungs, thymus, etc
 - Involved in respiration and positioning of the vertebral column
- **Sternum:**
 - Consists of..
 - Manubrium
 - Body
 - Xiphoid process
 - Jugular notch
- **The ribs:**
 - **True ribs aka vertebrosteral ribs: 1-7** → connects directly to sternum via costal cartilages
 - **False ribs: 8-12**
 - **Vertebrochondral ribs: 8-10** → attach together before sternum
 - **Vertebral ribs are floating ribs: 11-12** → no anterior connection
 - 12 pairs of ribs
 - Each ribs articulates with a thoracic vertebrae at costal facets
 - Structures of a rib:
 - Head (attaches to vertebra)/neck/body (on anterior side)
 - Groove
 - Superior and inferior articular facets



Compare and contrast the different vertebrae

- **Vertebral column** is made of 26 bones (24 vertebrae, 1 sacrum, 1 coccyx)
 - **24 vertebrae**
 - 7 cervical vertebrae
 - 12 thoracic vertebrae
 - 5 lumbar vertebrae
 - **Spinal curves:** help us to maintain balance and stay upright (along with muscle attachment to various vertebral processes) → developing infant lacks balance, lack proper curvature, and less muscle coordination
 - Cervical curve

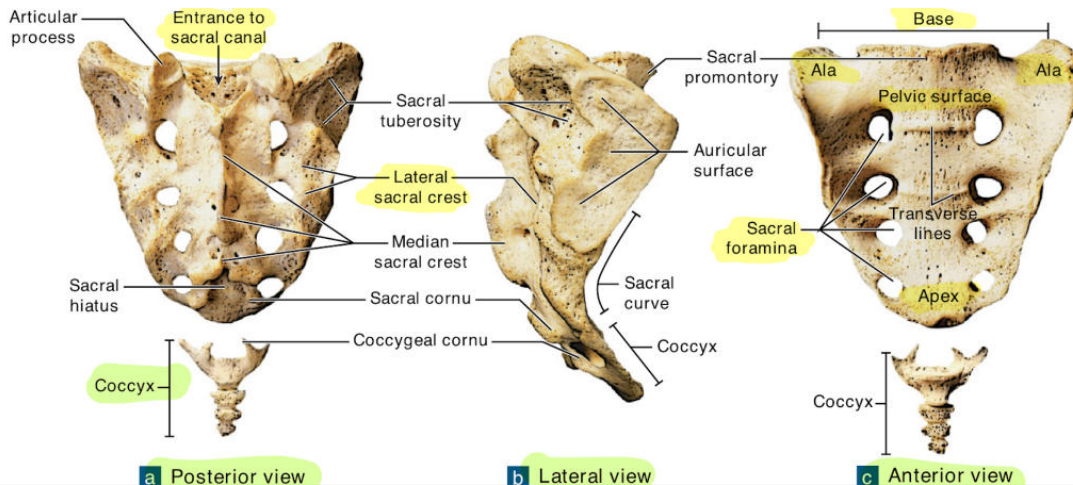
- Thoracic curve
- Lumbar curve
- Sacral curve
- **Vertebral body**
 - Supports weight along axis of body
 - Anterior structure
 - Vertebral body separated from another vertebral body by a pad of cartilage called the intervertebral disc
- **Vertebral arch**
 - Forms vertebral foramen
 - Made of pedicle and laminae
 - Spinous process projects posteriorly
 - Transverse processes project laterally
- **Articular processes**
 - Superior and inferior
- **Vertebral articulation**
 - Articulate at articular facets and a vertebral canal
 - Separated by intervertebral discs



Compare and contrast the different vertebrae

- **The coccyx**
 - Adult male coccyx points anteriorly
 - Adult female coccyx points inferiorly b/c of childbirth
- **The sacrum**
 - One sacrum, but consists of five fused vertebrae
 - Consists of..
 - Apex
 - Base

- Sacral canal
- Sacral foramina
- Ala
- Lateral sacral crest
- Sacro-iliac joint
- Pelvic surface

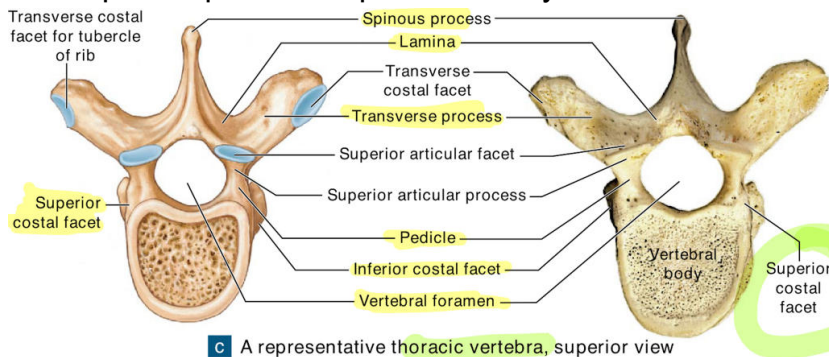


- **Lumbar vertebrae**

- Have the largest vertebral bodies
- Spinous processes point inferiorly

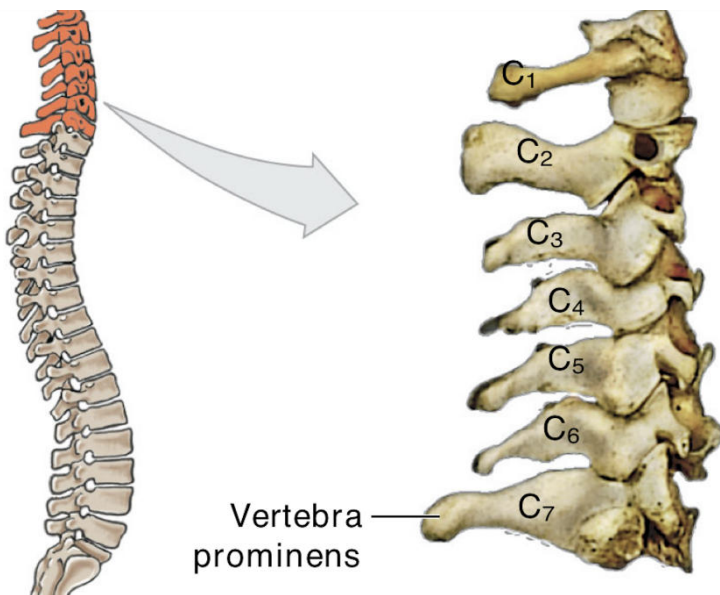
- **Thoracic vertebrae**

- **All have rib articulation points: superior and inferior costal facets**
- Support the ribs
- Most spinous processes point inferiorly

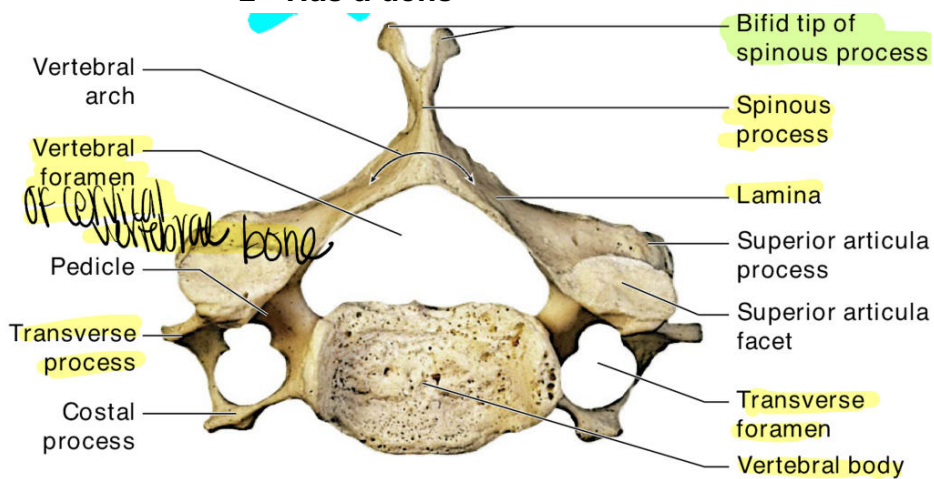


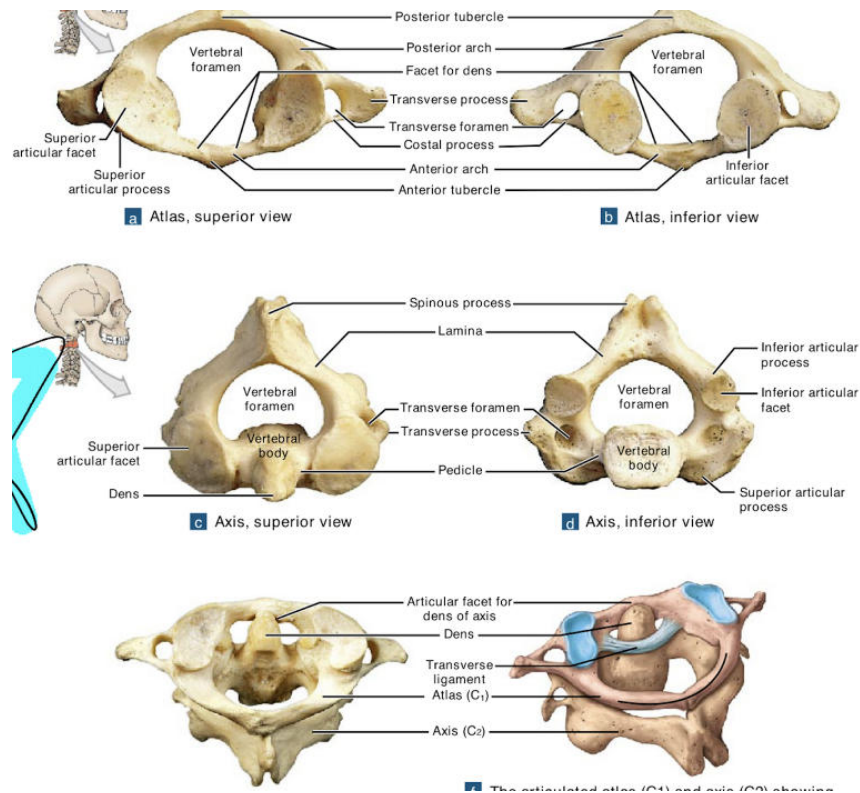
- **Cervical vertebrae**

- Support the weight of the head
- Spinous processes are **bifid** except for C7 → unique here
 - C7 has long prominent spinous process
 - Ligamentum nuchae



- All have transverse foramina
 - Transverse processes are fused to the costal processes
- **C1 is the atlas**
 - **Articulates w/ occipital condyles**
- **C2 is the axis**
 - **Has a dens**





LECTURE 19

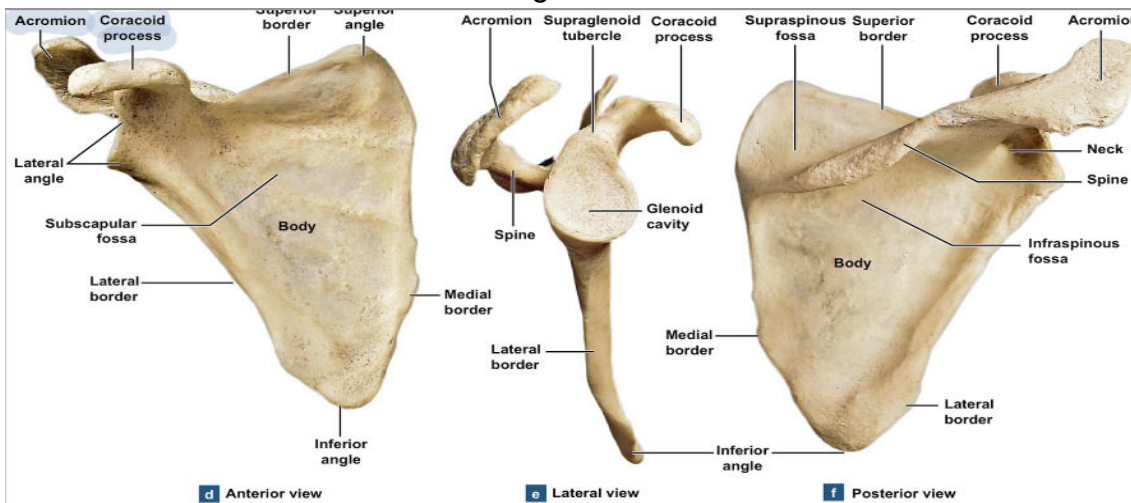
Compare & contrast bones that make up appendicular skeleton and their important landmarks

- **Appendicular skeleton includes..**
 - **Pectoral girdle**
 - Shoulder bones
 - **Upper limbs**
 - **Pelvic girdle**
 - Hip bones
 - **Lower limbs**

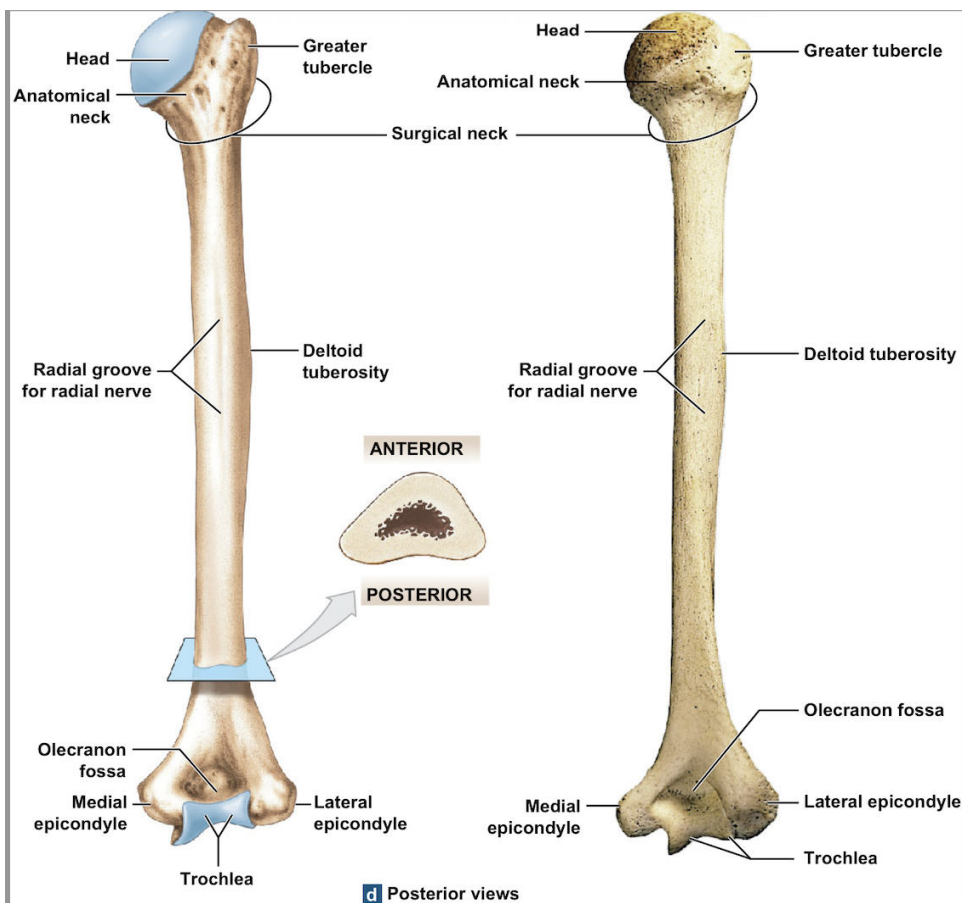
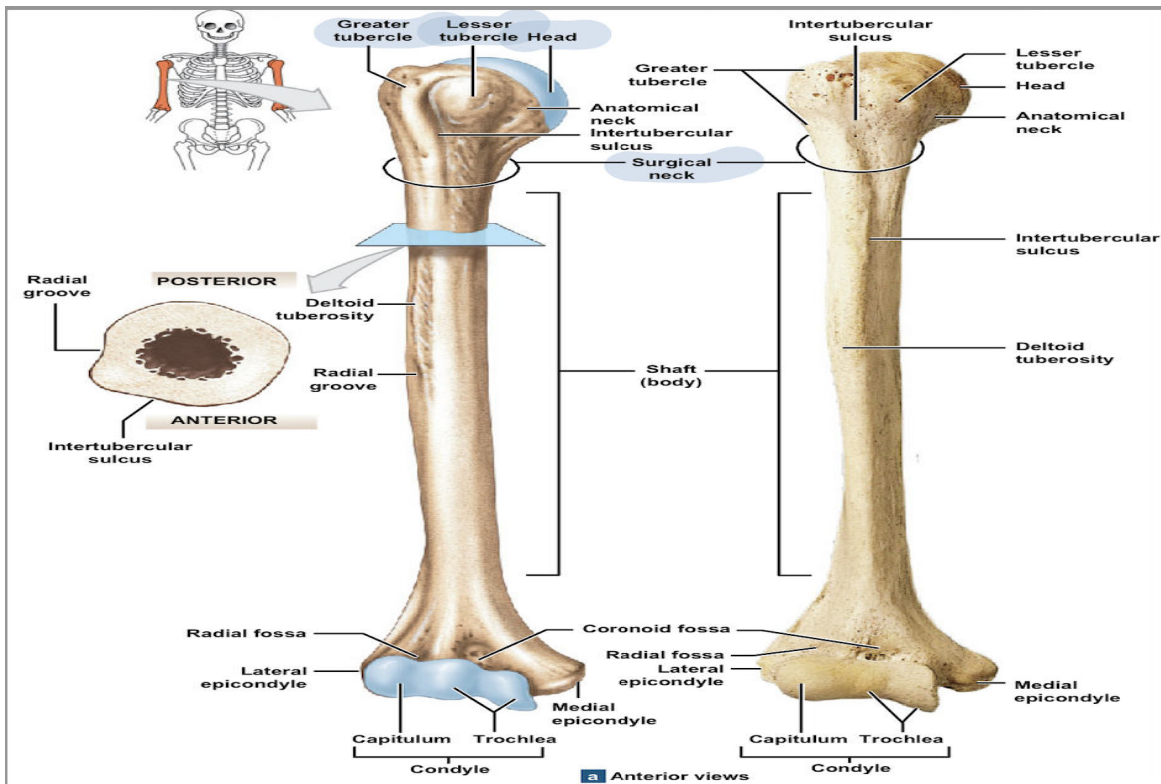
Compare & contrast bones that make up the pectoral girdle & upper limb

- **Pectoral girdle** consists of..
 - **Clavicle**
 - Sternal end is medial end → touches sternum
 - Acromial end is lateral end → touches acromion process of scapula
 - **Scapula** → posterior on rib cage
 - **Anterior view**
 - Body
 - Coracoid process
 - Subscapular fossa
 - **Posterior view**
 - Scapular spine
 - Supraspinous fossa
 - Infraspinous fossa

- Acromion
- **Medial view**
 - Medial border
- **Lateral view**
 - Glenoid cavity
 - Lateral border
 - Lateral angle
 - Glenohumeral joint
 - Neck
 - Acromioclavicular joint
- **Superior view**
 - Suprascapular notch
 - Superior border
 - Superior angle
- **Inferior view**
 - Inferior angle



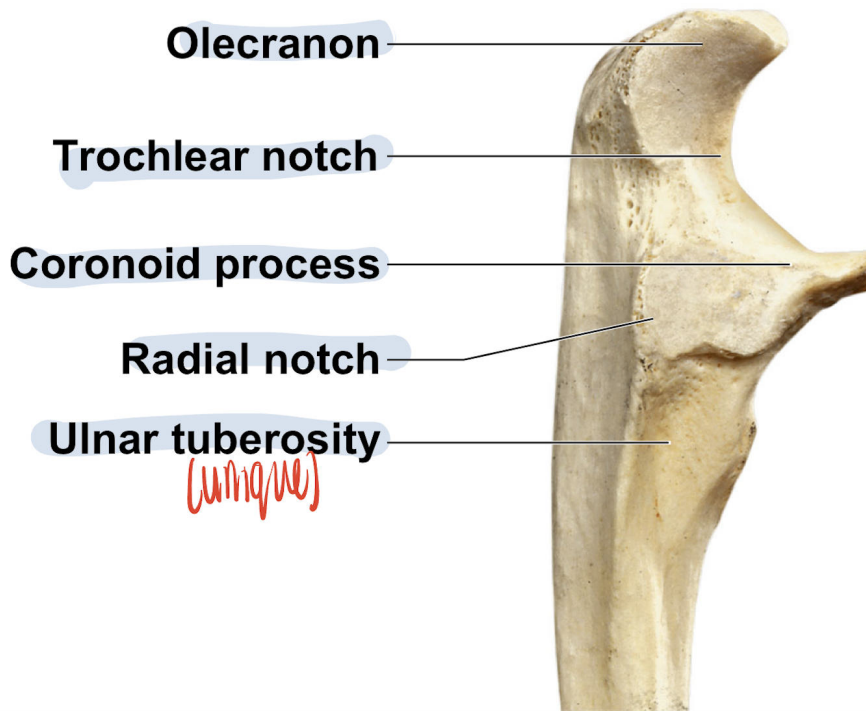
- **Upper limb**
 - Consist of
 - **Humerus (Ball & socket joint)**
 - Proximal structures
 - Head
 - Greater tubercle
 - Lesser tubercle
 - Anatomical neck
 - Surgical neck
 - shaft
 - Distal structures
 - Capitulum
 - Radial fossa
 - Radial groove
 - Trochlea
 - Lateral epicondyle
 - Medial epicondyle



■ Ulna

- Proximal structures
 - Olecranon
 - Trochlear notch

- Radial notch
- Coronoid process
- Ulnar tuberosity
- Distal structures
 - Ulnar styloid process
 - Ulnar head
- The trochlear notch pivots on the trochlea of the humerus
- The head of the radius pivots in the radial notch of the ulna
- Ulna is medial to radius
- **Interosseous membrane:** fibrous sheet between the ulna and radius



■ Radius

- Proximal structures
 - Head
 - Radial tuberosity
 - neck
- Distal structures
 - Radial styloid process
 - Ulnar notch of the radius bone
- The radius is lateral to the ulna
- The head of radius pivots on the capitulum of the humerus
- These bones are parallel to each other in the anatomical position
- **Pronation:** the radius crosses over the ulna
- **Supination:** rotating the bones so they are back in the anatomical position

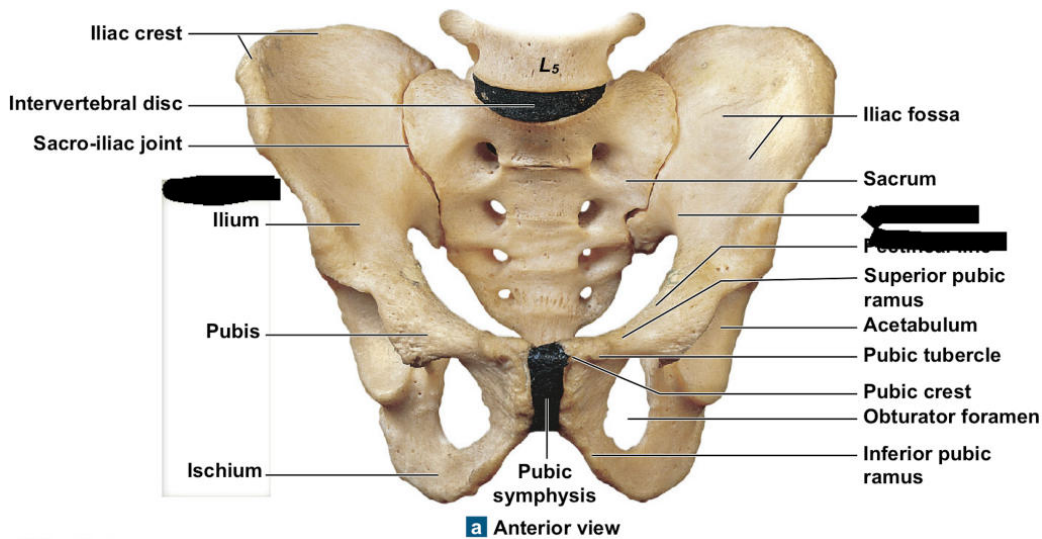
■ Carpal bones

- 8 bones of the wrist
- Consists of two rows..
 - Proximal carpal bones

- Scaphoid
 - Lunate
 - Triquetrum
 - pisiform
- Distal carpal bones
 - Trapezium
 - Trapezoid
 - Capitate
 - Hamate
- **Metacarpal bones**
 - There are 5 metacarpals (numbered I → V)
 - The metacarpal associated with the **pollex** (thumb) is number I
- **Phalanges**
 - 14 phalanges in each hand (proximal, middle, distal)
 - Pollex only has two phalanges (proximal & distal)

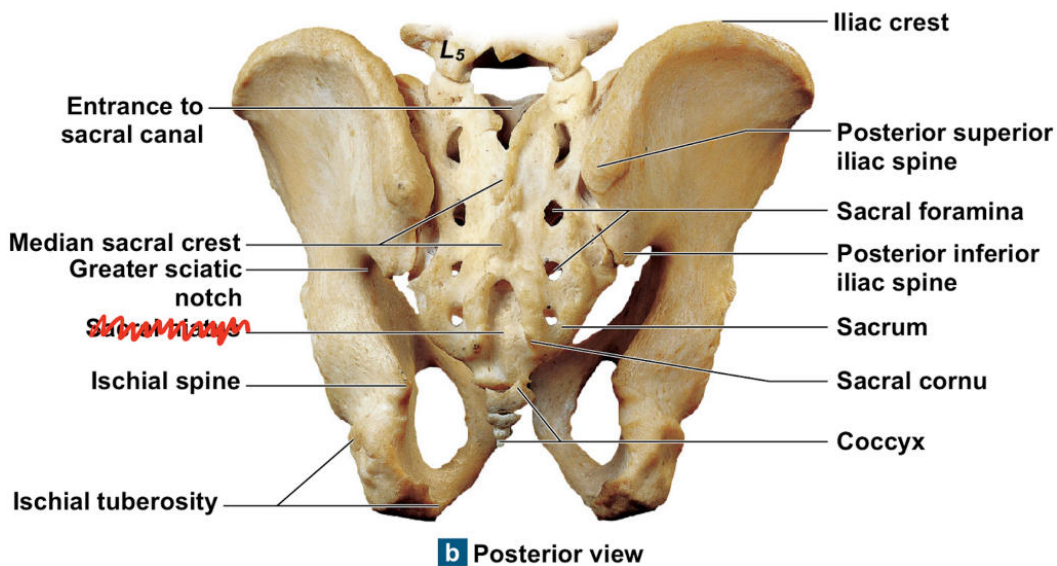
Compare & contrast bones that make up the pelvic girdle & lower limb

- **Pelvic girdle** consists of..
 - **2 pelvic bones** → each called a **coxal bone**
 - **One sacrum and one coccyx**
 - Each coxal bone is made of..
 - **Ilium, ischium, & pubis**
 - **Acetabulum** → **articulating surface b/w pelvic bone & femur**
 - Acetabular notch / acetabular fossa / lunate surface
 - **Pelvic bones**
 - **Anterior structures**
 - Anterior superior iliac spine
 - Anterior inferior iliac spine
 - Pubic tubercle
 - Pubic crest
 - Pubic symphysis
 - Inferior iliac notch
 - Pubis



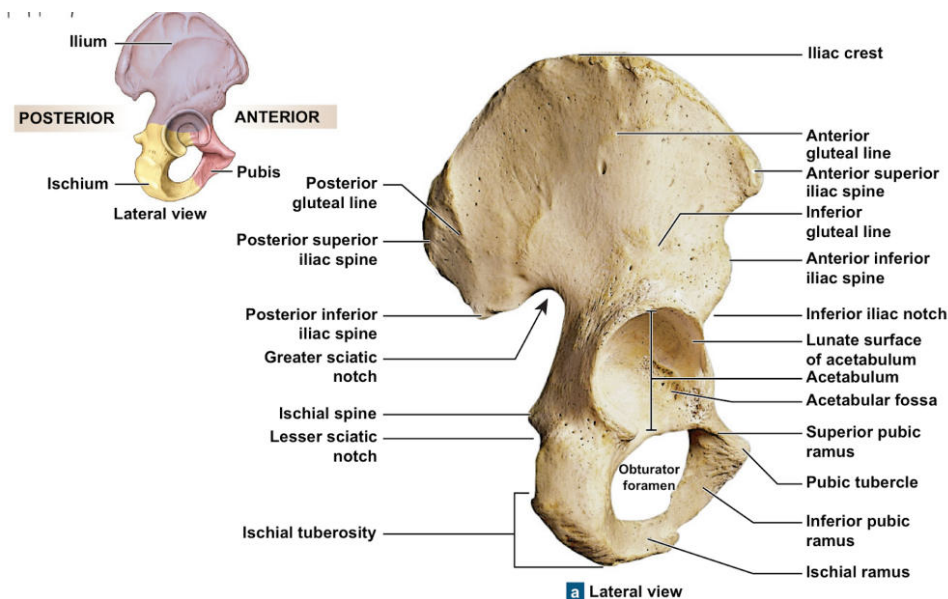
■ Posterior structures

- Posterior superior iliac spine
- Posterior inferior iliac spine
- Ischial tuberosity
- Ischial spine
- Greater sciatic notch
- Lesser sciatic notch
- ischium



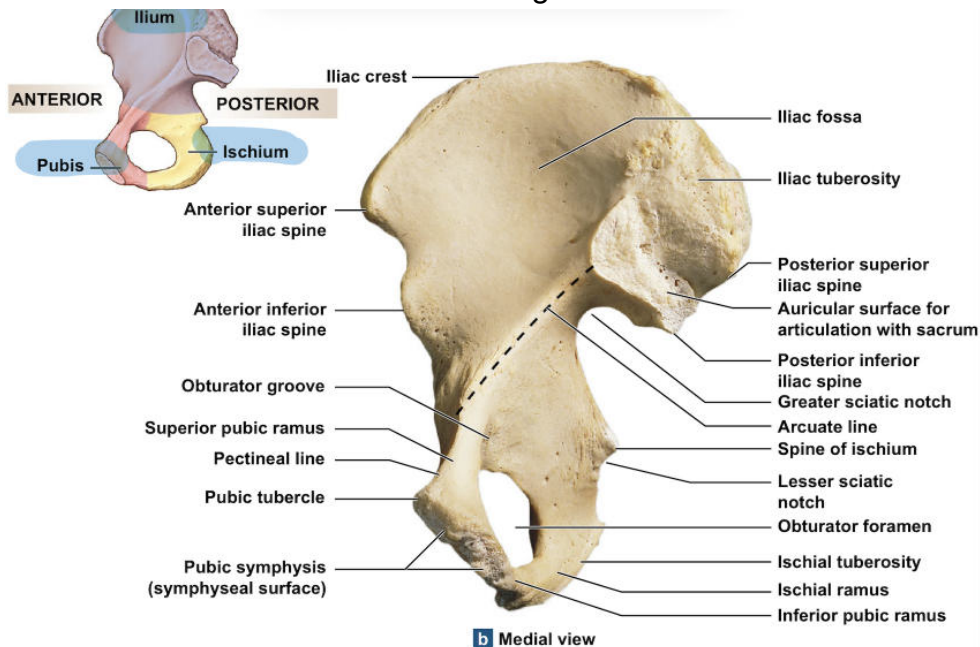
■ Lateral view

- Acetabulum
- Anterior, posterior, & inferior gluteal lines
- Obturator foramen
- Iliac crest



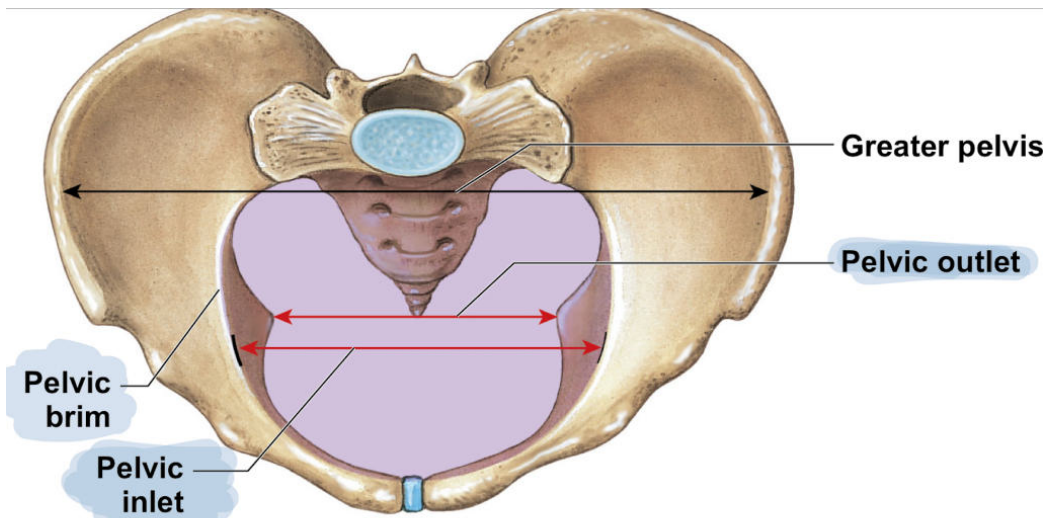
■ Medial view

- Superior pubic ramis
- Obturator groove



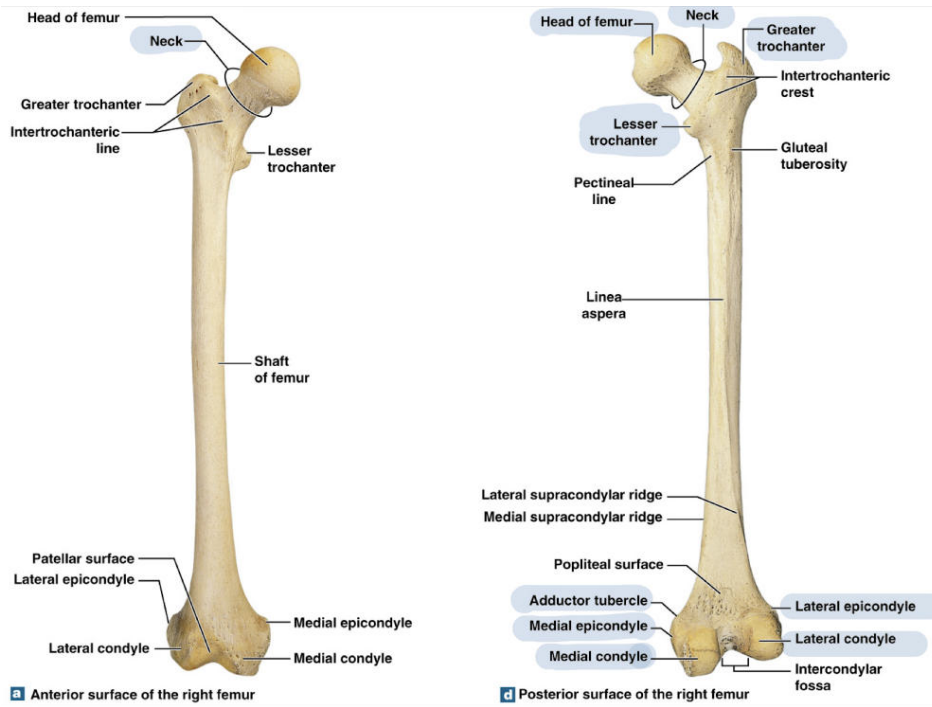
○ The pelvis consists of..

- Two coxal bones, one sacrum, one coccyx
- Two pelvic spaces
 - **Pelvic inlet:** superior
 - **Pelvic outlet:** inferior



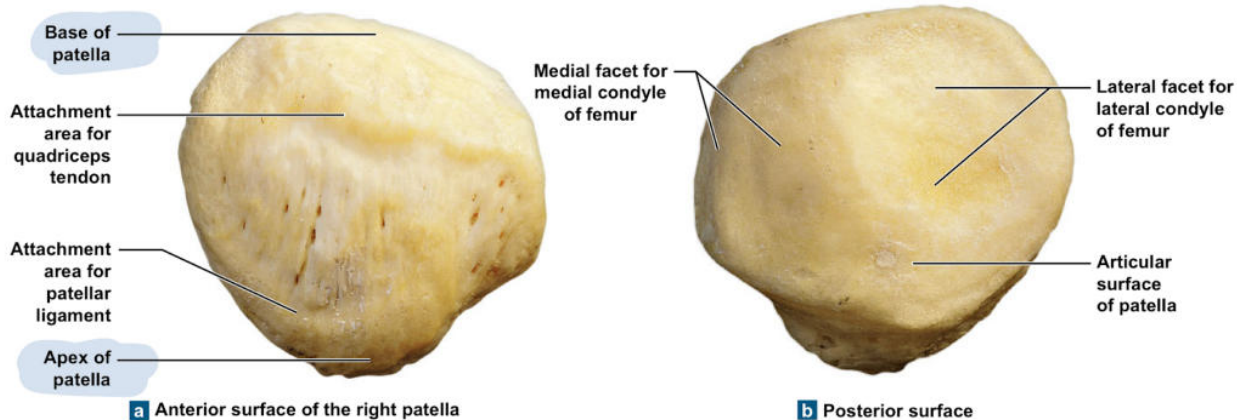
a Superior view of the pelvic brim and pelvic inlet of a male.

- Male and female pelvis differ due to childbearing female anatomy
 - Pelvic outlet → larger in females
 - Sacrum curve → less curvature in females
 - Pelvic inlet → wider in females
 - Pubic angle → greater than 100° in females and less than 90° in males
- **Lower limb** consists of..
 - Femur / patella / tibia / fibula / tarsals / metatarsals / phalanges
 - **Femur:**
 - Proximal structures
 - Head
 - Neck
 - Greater and lesser trochanter
 - Distal structures
 - Medial and lateral condyles
 - Medial and lateral epicondyles
 - Patellar surface
 - Posterior view
 - Linea aspera
 - Lateral supracondylar ridge
 - Pectineal line
 - Gluteal tuberosity
 - Adductor tubercle



○ Patella

- Large sesamoid bone (sesamoid → own little bone)
- Protects knee joint
- Anterior surface is rough for strong tendon attachment
- Posterior surface is smooth & has concave facets for femoral condyles
- Structures ..
 - Apex
 - Base
 - Medial facet
 - Lateral facet

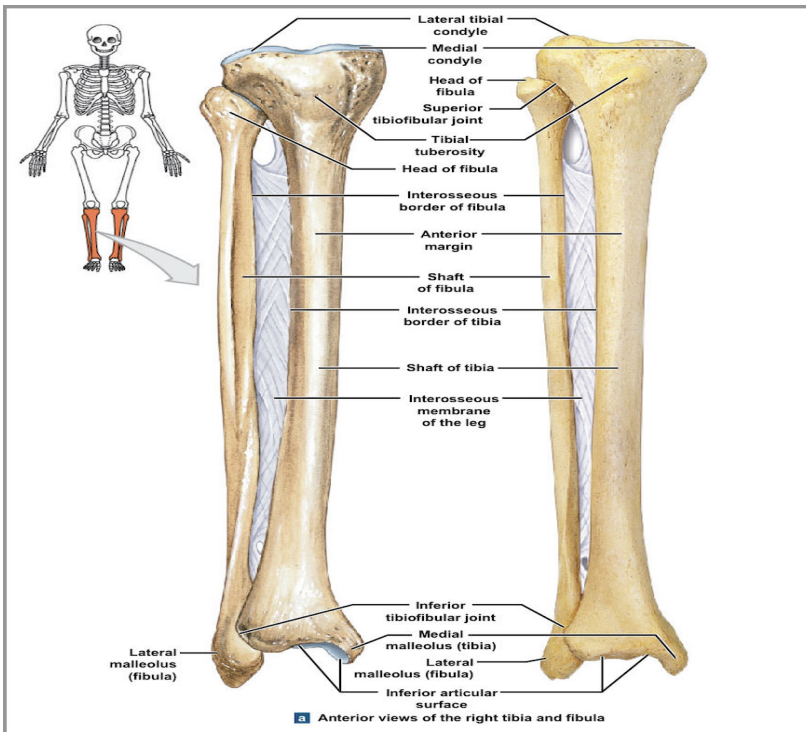


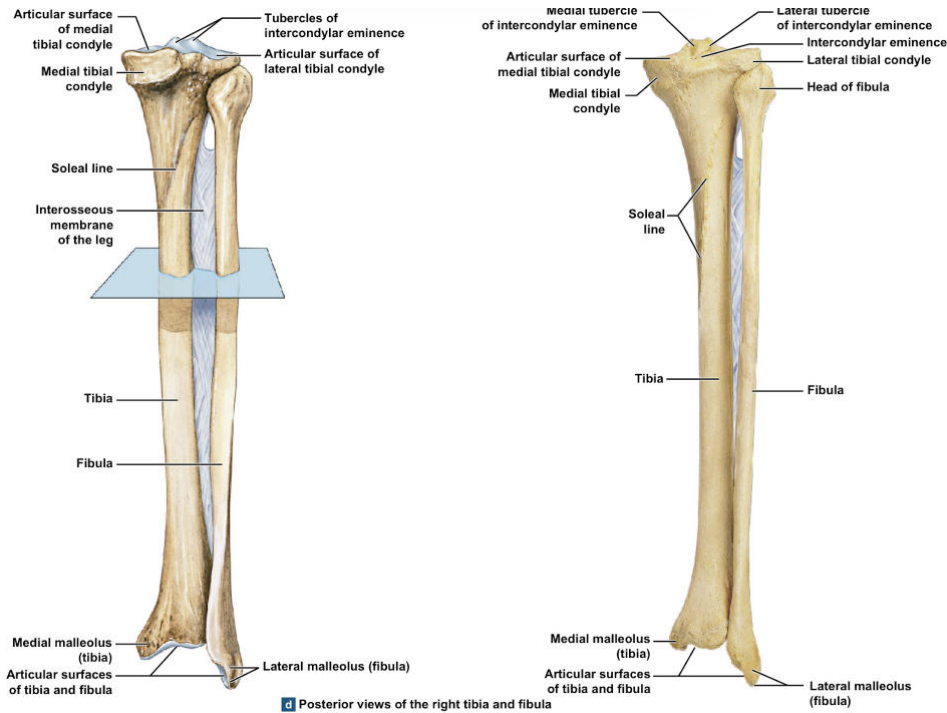
Smooth!

○ Tibia

- Medial to fibula
- Proximal structures
 - Tibial tuberosity
 - Intercondylar eminence
 - Medial and lateral tubercles

- Soleal line
- Medial and lateral condyles
- **Distal structures**
 - **Medial malleolus**
- **Interosseous membrane of the leg**
 - Membrane between the fibula and tibia - entire length
- **Fibula**
 - Lateral to tibia
 - **Proximal structures**
 - Head
 - **Distal structures**
 - **Lateral malleolus**





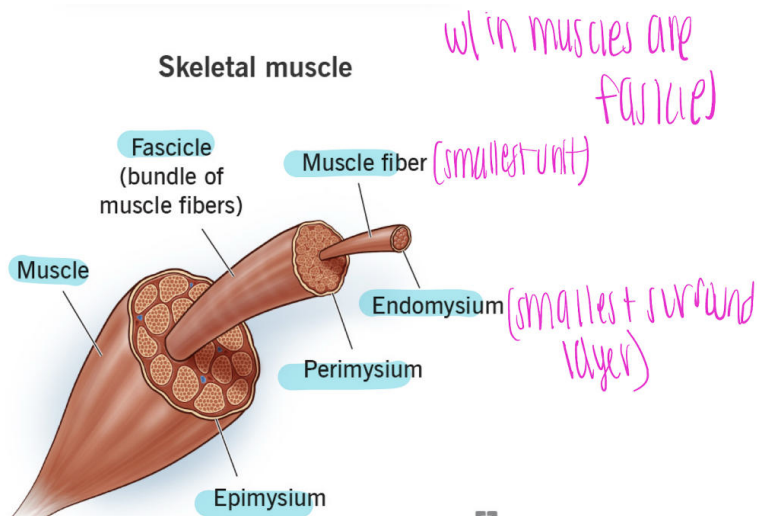
- **Tarsal bones**
 - 7 bones of the ankle (**tarsus**)
 - Calcaneus
 - Talus (contains trochlea)
 - Navicular
 - Medial, intermediate, lateral cuneiform
 - Cuboid
- **Metatarsals**
 - Five metatarsals (numbered I → V)
 - Metatarsal associated with **hallux** (big toe) is number I
- **Phalanges**
 - 14 phalanges in each foot (proximal, middle, & distal)
 - **Hallux** only has two phalanges (proximal & distal)
 - There are two arches of the foot
 - Longitudinal arch
 - Transverse arch

LECTURE 20 AXIAL MUSCLES

Compare & contrast connective tissue covering of muscles

- **Epimysium**
 - Surrounds entire muscle
- **Perimysium**
 - Surrounds bundles of muscle fibers (fascicles)
- **Endomysium**
 - Surround individual muscle fibers
- **Sarcolemma**

- Muscle cell membrane



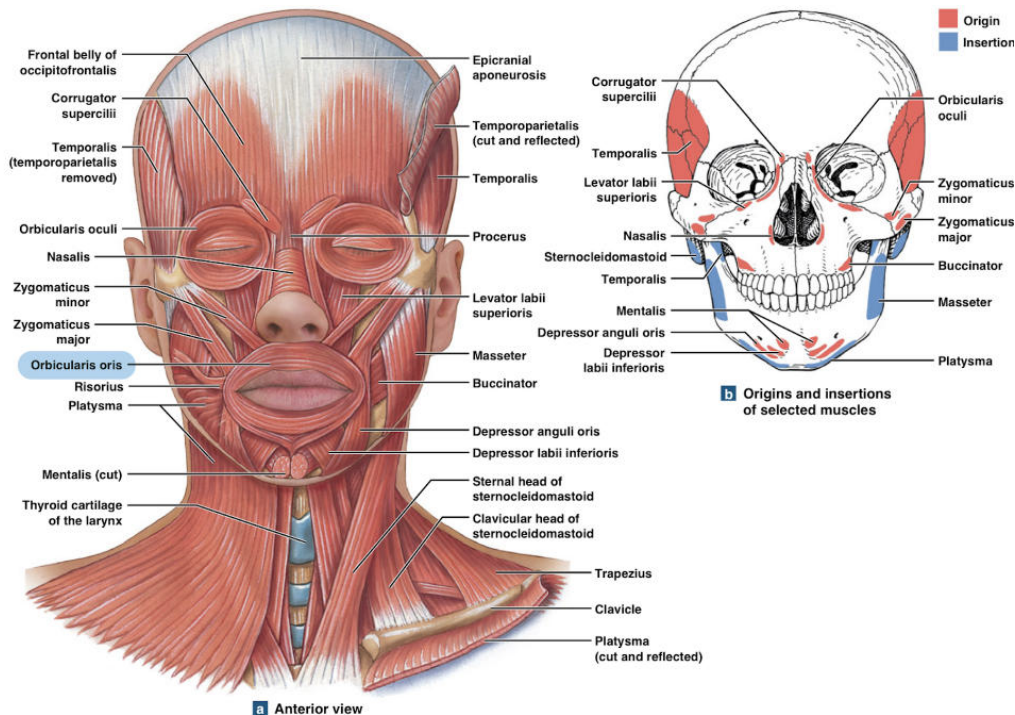
Compare & contrast histology of the three types of muscle tissue

	Main features	Location	Type of cells	Histology
Skeletal muscle	- Fibers: striated, tubular and multi nucleated - Voluntary - Usually attached to skeleton			
Smooth muscle	- Fibers: non-striated, spindle-shaped, and uninucleated. - Involuntary - Usually covering wall of internal organs.			
Cardiac muscle	- Fibers: striated, branched and uninucleated. - Involuntary - Only covering walls of the heart.			

Differentiate between axial and appendicular muscles

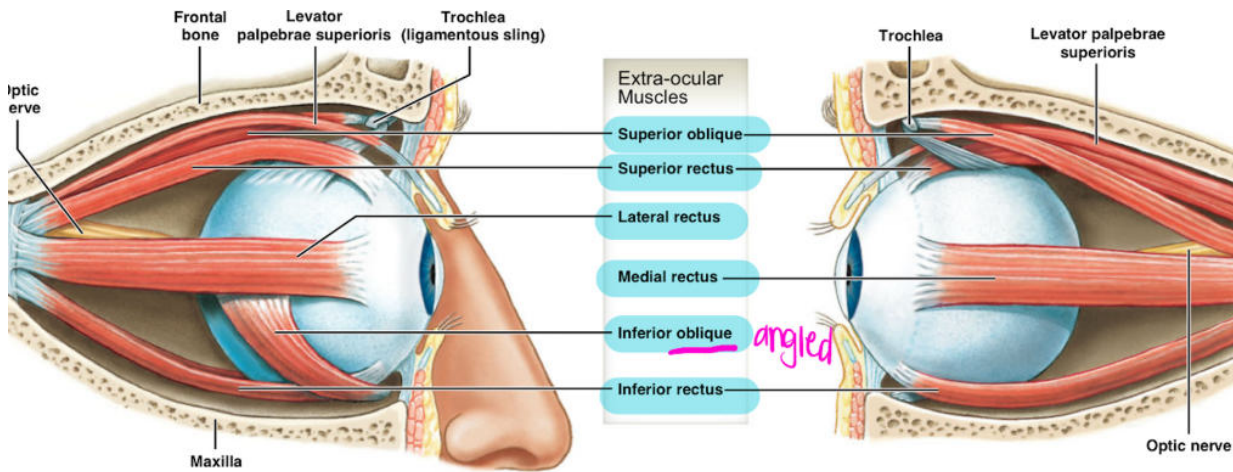
- **Axial musculature**
 - Muscles that position the head and vertebral column
 - Muscles that move the rib cage
 - Innervated by nerves
 - **Placed into four groups based on location/function**
 - Head/neck
 - Muscles of facial expression
 - Extraocular muscles
 - Muscles of mastication
 - Muscles of tongue
 - Muscles of pharynx
 - Anterior muscles of the neck
 - Vertebral column
 - Rib cage / abdominal and pelvic cavities
 - Pelvic floor
- **Muscles of facial expression (head & neck)**

- Innervated by **CN VII** *facial nerve*
- Divided into 5 groups
 - Mouth
 - Eyes
 - Scalp
 - Nose
 - neck
- **Orbicularis oris** *} mouth*
- **Buccinator**
- **Temporoparietalis**
- **Occipitofrontalis**
- **Platysma** - changes shape of neck skin when tenses



● Extraocular muscles (head & neck)

- Extrinsic eye muscles (control eye movement)
 - **Medial rectus** - rotates eye medially
 - **lateral rectus** - rotates eye laterally
 - **Superior rectus** - rotates eye upwards
 - **Inferior rectus** - rotates eye downwards
 - **Superior oblique** - rotates eye down and laterally
 - **Inferior oblique** - rotates eye up and laterally
- LR6, SO4, everything else 3

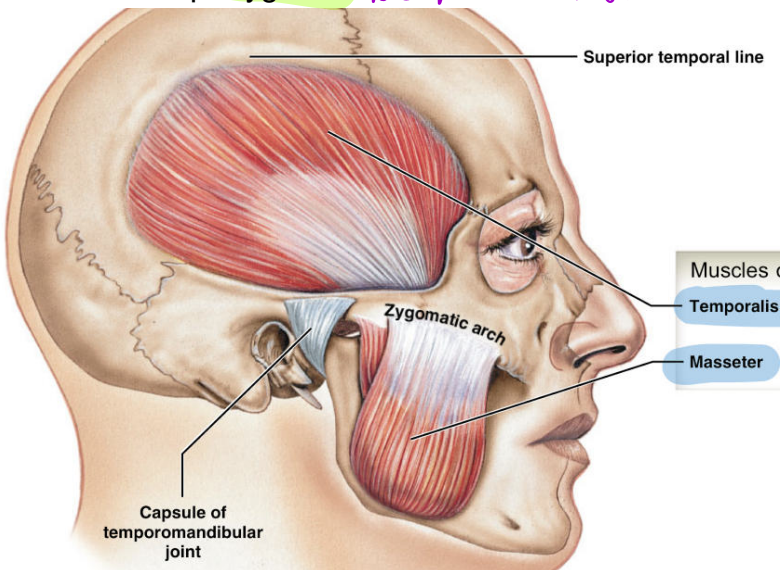


a Muscles on the lateral surface of the right eye.

b Muscles on the medial surface of the right eye.

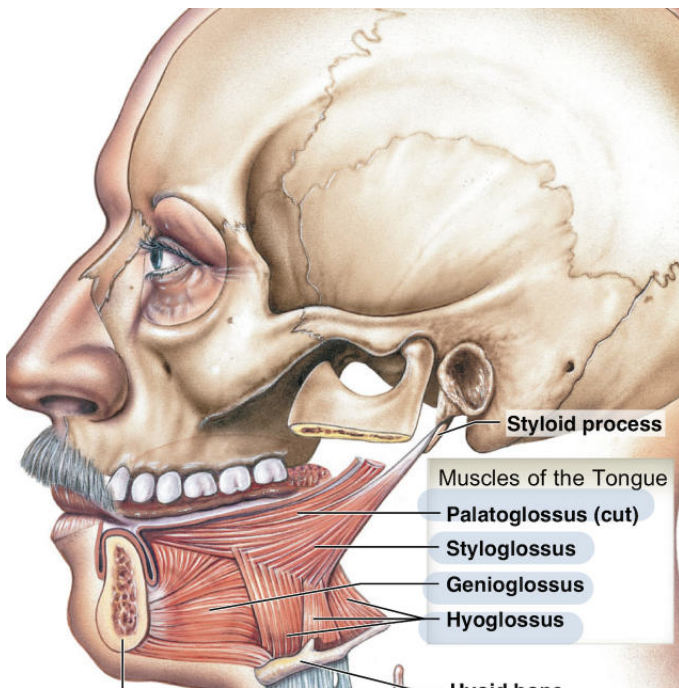
• Muscles of mastication (head and neck)

- Innervated by CN V
- **Masseter**
- **Temporalis**
- **pterygoids** *Deep to Masseter*



• Muscles of the tongue (head & neck)

- **Genioglossus**
 - **Hyo**glossus
 - **Palato**glossus
 - **stylo**glossus
- glossus means tongue*



- **Muscles of the pharynx (head and neck) *not super important*** → pharyngeus
 - **Pharyngeal constrictors:** superior/middle/inferior constrictors
 - **Laryngeal elevators:** palatopharyngeus / salpingopharyngeus / stylopharyngeus
 - **Palatal muscles:** tensor veli palatini / levator veli palatini
- **Anterior muscles of neck (head and neck)**
 - **Mylohyoid:** CN V
 - **Geniohyoid:** CN XII
 - **Stylohyoid:** CN VII
 - **Sternocleidomastoid: CN XI**
 - O - medial head - cranial part of manubrium sterni
 - O - lateral head - medial 1/3 of clavicle
 - I - lateral surface of mastoid process, lateral 1/2 of superior nuchal line of occipital bone

- **Muscles of the vertebral column**
 - Superficial layer (extrinsic back muscles): move the neck
 - Intermediate layer (extrinsic back muscles): move vertebral column
 - Deep layer (intrinsic back muscles): interconnect the vertebrae

• **Superficial layer of the ^{extrinsic} back muscles (vertebral column)**

- Innervated by cervical nerves
- Splenius capitis baseball cap - Head
- Splenius cervicis cervical spine

• **Intermediate layer of ^{extrinsic} back muscles (vertebral column)**

- **Erector spinae** (group of three muscles - have on each side) - help you stay upright
 - **Spinalis** (most medial of all three)
 - **Longissimus**
 - Long muscle
 - **Iliocostalis** (most lateral of all three)
 - Ilium to to costal ribs

I Love Spaghetti!
makes you erect.
1 L S
|
Spine

NECK
Vertebral column
vertebrae

NECK

Vertebral stay erect like a spine

- Muscle crosses over the joint that it moves

- **Deep layers of intrinsic back muscles (vertebral column)**

vertebrae

- Hold the vertebrae together
- Transversospinales (group of 5 muscles)

⑤

- S ■ Semispinalis
- M ■ Multifidus
- R ■ Rotatores
- I ■ Interspinales
- I ■ intertransversarii

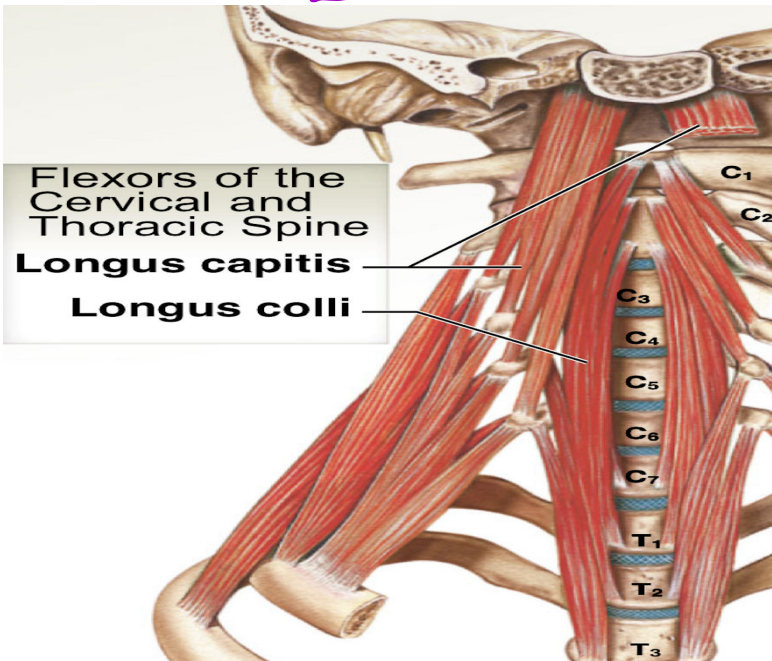
some
mighty
Rodents
infiltrate
interiors

- **Spinal flexors (vertebral column)**

- Longus capitis > neck
- Longus colli

③

flexion



O: iliac crest

I: last rib & lumbar

- **Quadratus lumborum** → laterally

- **Flexes vertebral column laterally**
- **O - iliac crest and iliolumbar ligament**
- **I - last rib and transverse processes of lumbar vertebrae**

- **Oblique and rectus muscles**

- **Cervical muscles flex**

- geniohyoid/omohyoid
- sternohyoid/sternothyroid/thyrohyoid
- Scalene muscles elevate ribs (inhalation)

} rectus muscles

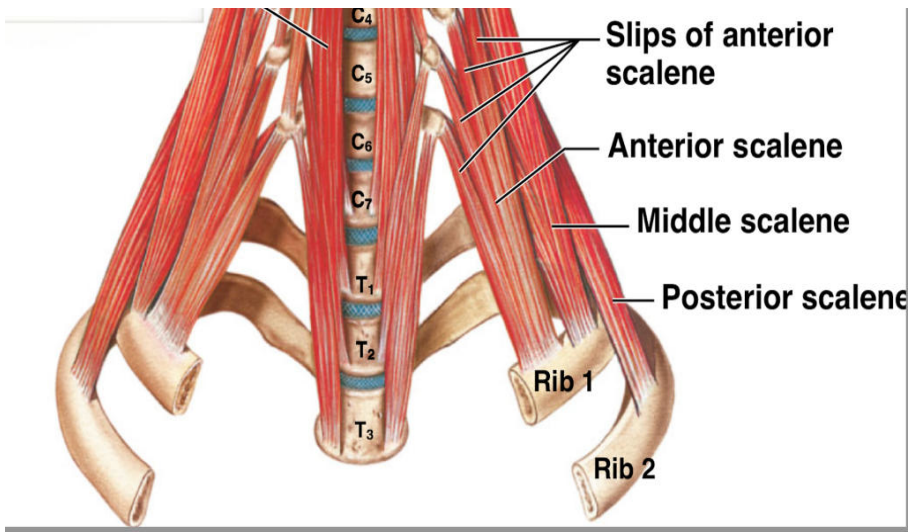
cervical

- Anterior
- Middle
- posterior

Scalenes

A
M
P

up the ribs (inhalation)



○ Thoracic muscles

■ Diaphragm

- Major breathing muscle
- Contracts → diaphragm lowers to increase the volume of thoracic cavity → inhalation
- Relaxes → diaphragm rises to lower the volume of thoracic cavity → exhalation

■ Intercostals (in between ribs)

- External intercostals: elevates ribs
- Internal intercostals: depresses ribs

■ Transversus thoracis muscles

- Depresses ribs

■ Serratus posterior muscles

- Superior: elevates ribs
- Inferior: depresses ribs

○ Abdominal muscles

■ Rectus abdominis - separated by linea alba

- O - superior surface of pubis around symphysis
- I - inferior surfaces of ribs 5-7 and xiphoid process

■ Oblique

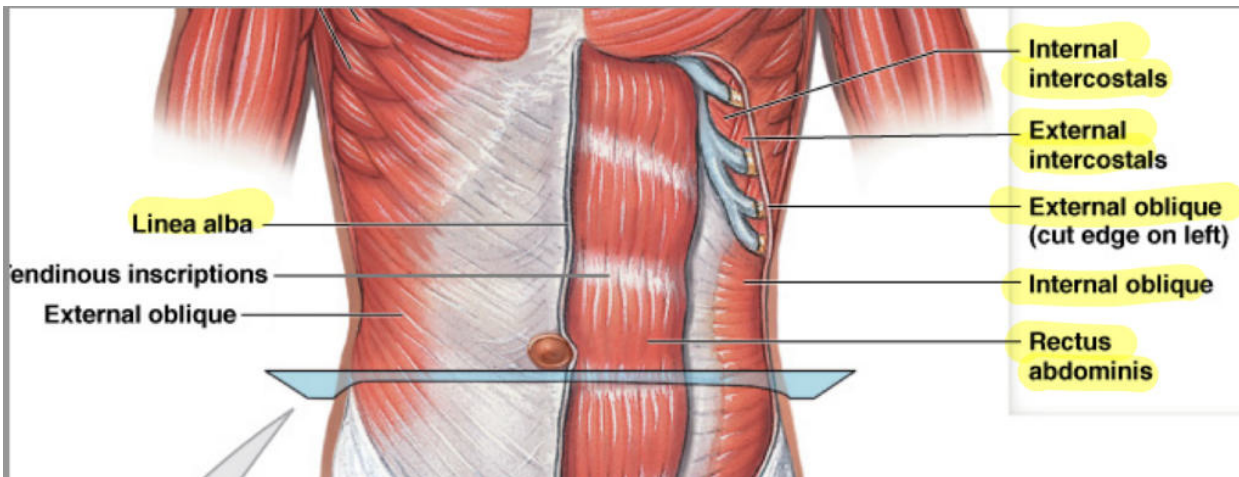
- External oblique: compresses abdomen/depresses ribs/ laterally flexes torso
- Internal oblique: compresses abdomen/depresses ribs/laterally flexes torso

■ Transversus abdominis

- Compresses abdomen

depressed
&
compressed
&
laterally

Skinny Queens



- **Muscles of perineal region & pelvic diaphragm**

- **The perineal region (pelvic floor)**

- Divided into two triangles:

- **Urogenital triangle** (anterior)

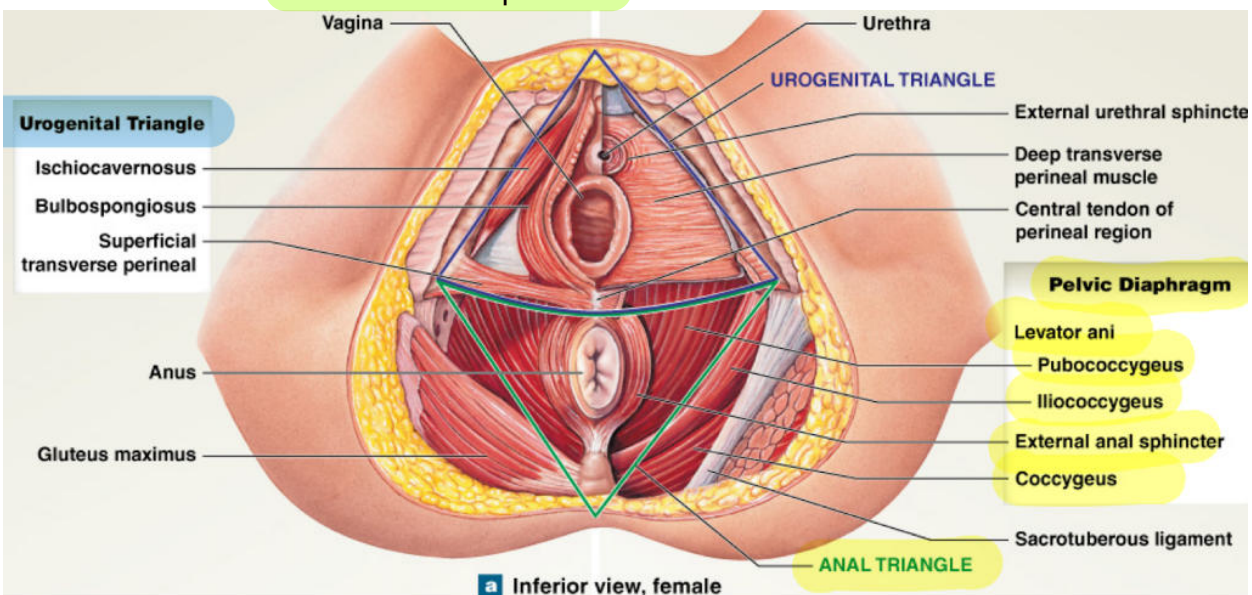
- Consists of **superficial** urogenital triangle muscles & **deep** urogenital triangle muscles)

- **Anal triangle** (posterior)

- Pelvic diaphragm (forms the foundation)

- **Pelvic diaphragm**

- Levator ani
 - Pubococcygeus
 - Iliococcygeus
 - External anal sphincter



Describe the structures that make up the neuromuscular junction

- **Neuromuscular junction:** specialized synapse where a motor neuron from the nervous system communicates with a skeletal muscle fiber to initiate muscle contraction

- **1) motor neuron terminal (presynaptic terminal)**
 - **Synaptic vesicles** contain the neurotransmitter ACh
 - **Voltage gated Ca²⁺ channels** open in response to action potential, allowing Ca²⁺ to enter and trigger ACh release
 - **Mitochondria**: provide energy for neurotransmitter synthesis and vesicle recycling
- **2) synaptic cleft**
 - A narrow space between the neuron and muscle fiber
 - Contains AChE, an enzyme that breaks down ACh to terminate the signal
- **3) Motor end plate (postsynaptic membrane)**
 - A specialized region of the **sarcolemma** (muscle cell membrane)
 - Contains nicotinic acetylcholine receptors (nAChRs) that bind ACh and initiate an end-plate potential
 - Has junctional folds to increase surface area & receptor density

LECTURE 22 APPENDICULAR MUSCLES

Discuss appendicular muscles

- **Appendicular musculature:**

- **Responsible for..**
 - Stabilize pectoral and pelvic girdles
 - Move the upper & lower limbs
 - Absorb shocks and jolts as you walk/run/jump
 - Aid in strengthening joint area
- **Two major groups:**
 - Muscles of pectoral girdle and upper limbs
 - Muscles of pelvic girdle and lower limbs
- **Factors affecting function**
 - **Action lines** of the shoulder joint are the **direction of pull** = direction that muscle fibers are moving
 - **Oblique** - twisting / angle of pull → muscle will take on more than one movement
 - **Rectus** - middle fibers of deltoid are straight → abduction of shoulder
 - **Action lines at the hip joint** → **movements of hip joint**
 - Flexion and extension of leg
 - Abduction and adduction of the leg
 - Medial rotation and lateral rotation of the leg

- **Muscles of pectoral girdle and upper limbs - muscles that position the pectoral girdle**

- **Trapezius:** rotates scapula and extends the neck
 - Goes from C spine to L spine
 - Big muscle! Does lots of pulling
 - Origin - occipital bone, ligamentum nuchae, and spinous process of T-spine

positions

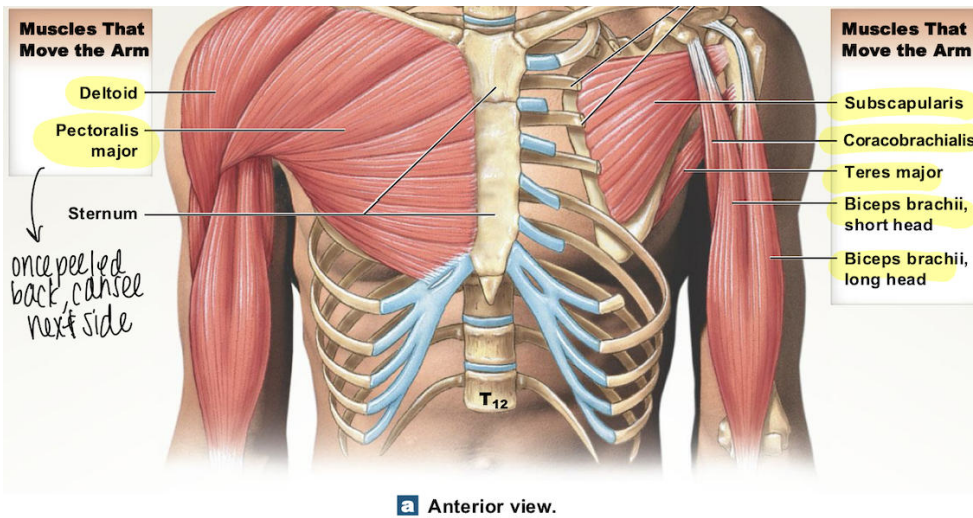
Two
Real
Ladies

Prep
SALad
Sometimes

- Insertion - clavicle and scapula
- **Rhomboid:** adducts scapula brings toward midline
- **Levator scapulae:** elevates scapula Levitate
 - Origin - transverse process of C1-C4
 - Insertion - vertebral border of scapula
- **Pectoralis minor:** protracts shoulder
 - Origin - anterior surface of ribs 3-5
 - Insertion - coracoid process of scapula
- **Serratus anterior:** protracts (moving body part forward) scapula
- **Subclavius:** protracts scapula
- **Muscles of pectoral girdle and upper limbs - muscles that move the arm**
 - **Deltoid:** abducts arm - main function - slight oblique pull
 - Origin - Clavicle and scapula
 - Insertion - deltoid tuberosity of humerus
 - **Supraspinatus:** abduction of shoulder
 - Origin - supraspinous fossa
 - Insertion - greater tubercle of humerus
 - **Infraspinatus:** lateral rotation at shoulder
 - Origin - infraspinous fossa
 - Insertion - greater tubercle of humerus
 - **Subscapularis:** medial rotation of shoulder
 - Origin - subspinous fossa
 - Insertion - lesser tubercle of humerus
 - **Teres major:** extension and medial rotation at shoulder
 - **Teres minor:** lateral rotation and adduction at shoulder pulls back down
 - **Coracobrachialis:** adduction and flexion at shoulder
 - Origin - coracoid process of scapula
 - Insertion - medial shaft of humerus
 - **Pectoralis major:** adducts, flexes, and medially rotates arms
 - Origin - ribs 2-6, sternum, inferior & medial clavicle
 - Insertion - greater tubercle of humerus
 - **Latissimus dorsi:** extension, adduction, medial rotation at shoulder
 - Multiple directions of pull add me ex - Lat
- **Rotator cuff muscles**
 - **Supraspinatus**
 - **Infraspinatus**
 - **Teres minor**
 - **Subscapularis**

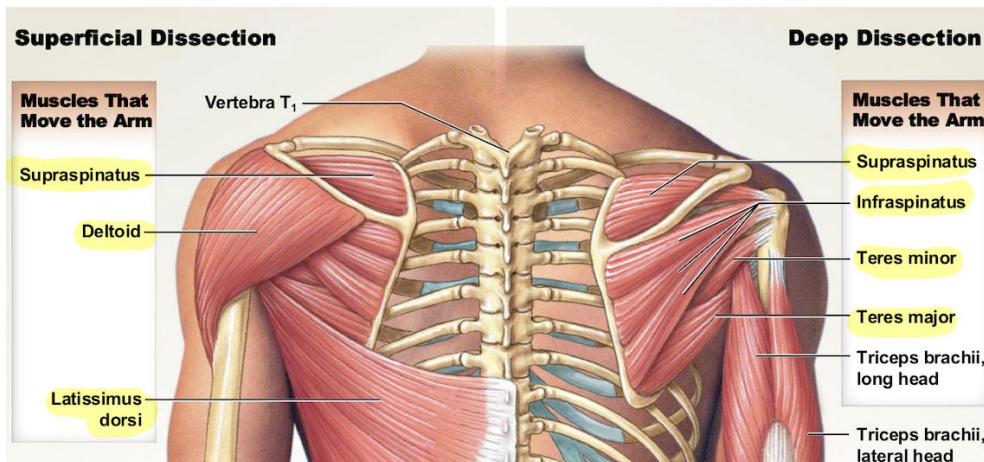
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Figure 11.7b Muscles That Move the Arm



• Muscles of pectoral girdle and upper limbs - muscles that move the forearm and hand

- **Triceps brachii:** extension at elbow
 - Insertion - olecranon of ulna
- **Biceps brachii:** flexion at elbow and supinates forearm
 - Origin - short head (coracoid), long head (scapula)
 - Insertion - radial tuberosity
- **Brachialis:** flexion at elbow → does most of your work!
 - Origin - distal ½ anterior humerus
 - Insertion - ulnar tuberosity and coronoid process
- **Brachioradialis:** flexion at elbow
- **Anconeus:** extension at elbow
- **Pronator teres:** pronates forearm
- **Supinator:** supinates forearm
- **Flexor carpi ulnaris:** flexion and adduction at wrist
- **Flexor carpi radialis:** flexion and abduction at wrist
- **Palmaris longus:** flexion at wrist
- **Extensor carpi ulnaris:** extension and adduction at wrist
 - Attaches to ulna

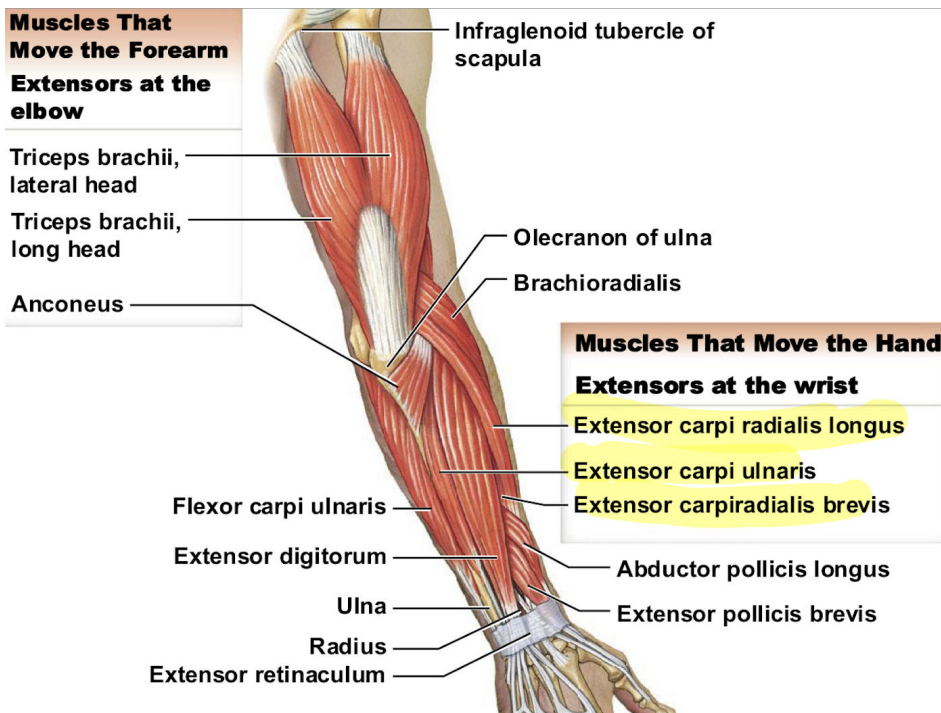
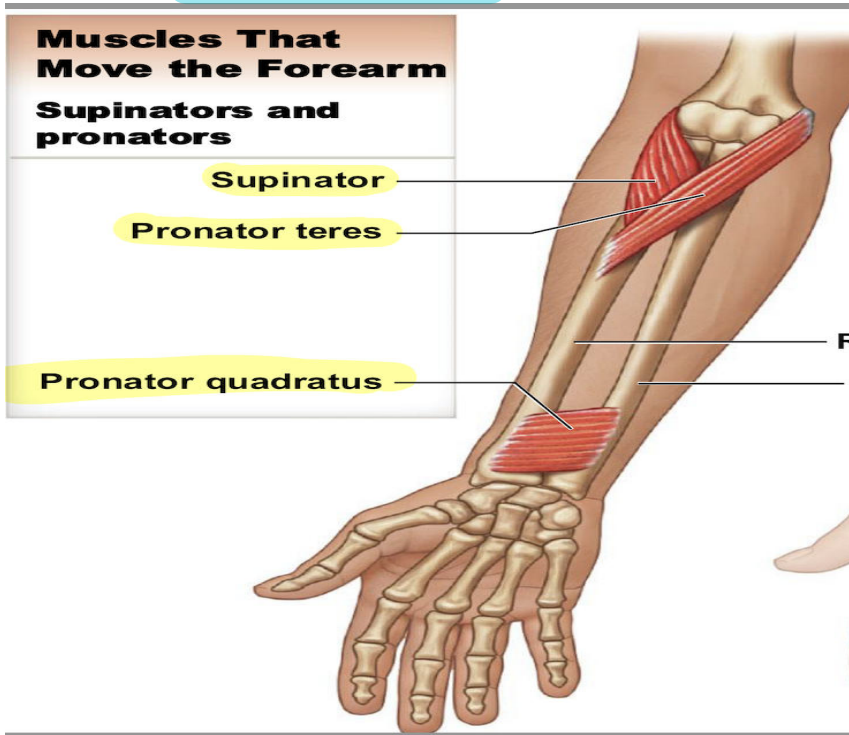
Triceps extensions
Brachii
1: radial

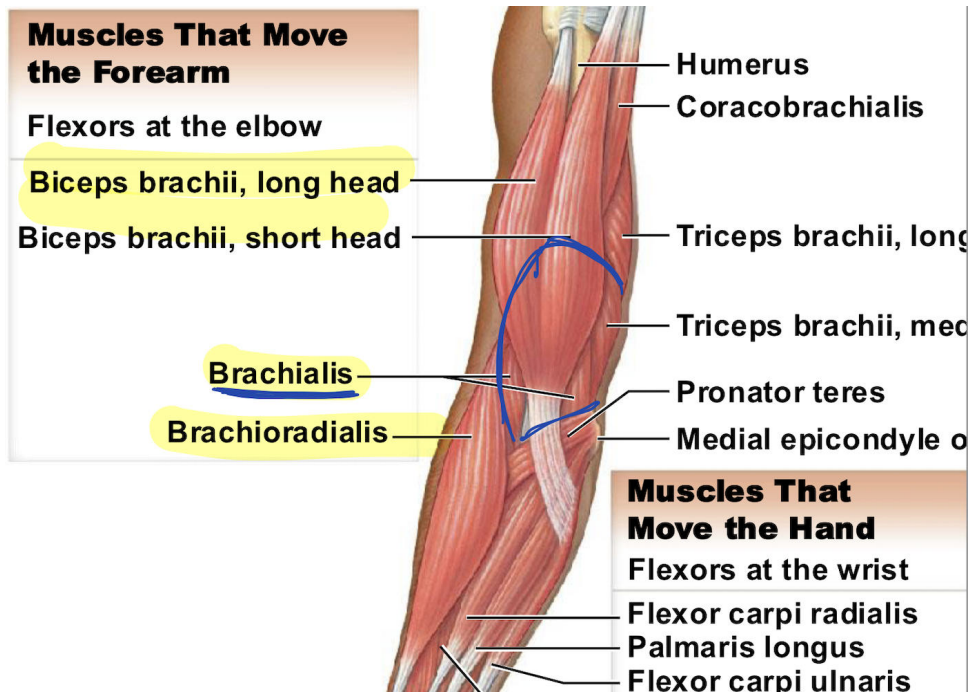
BBB flex

flexion

u-add
rad abduction

- **Extensor carpi radialis:** extension and abduction at wrist
- **Pronator quadratus:** pronates forearm and hand



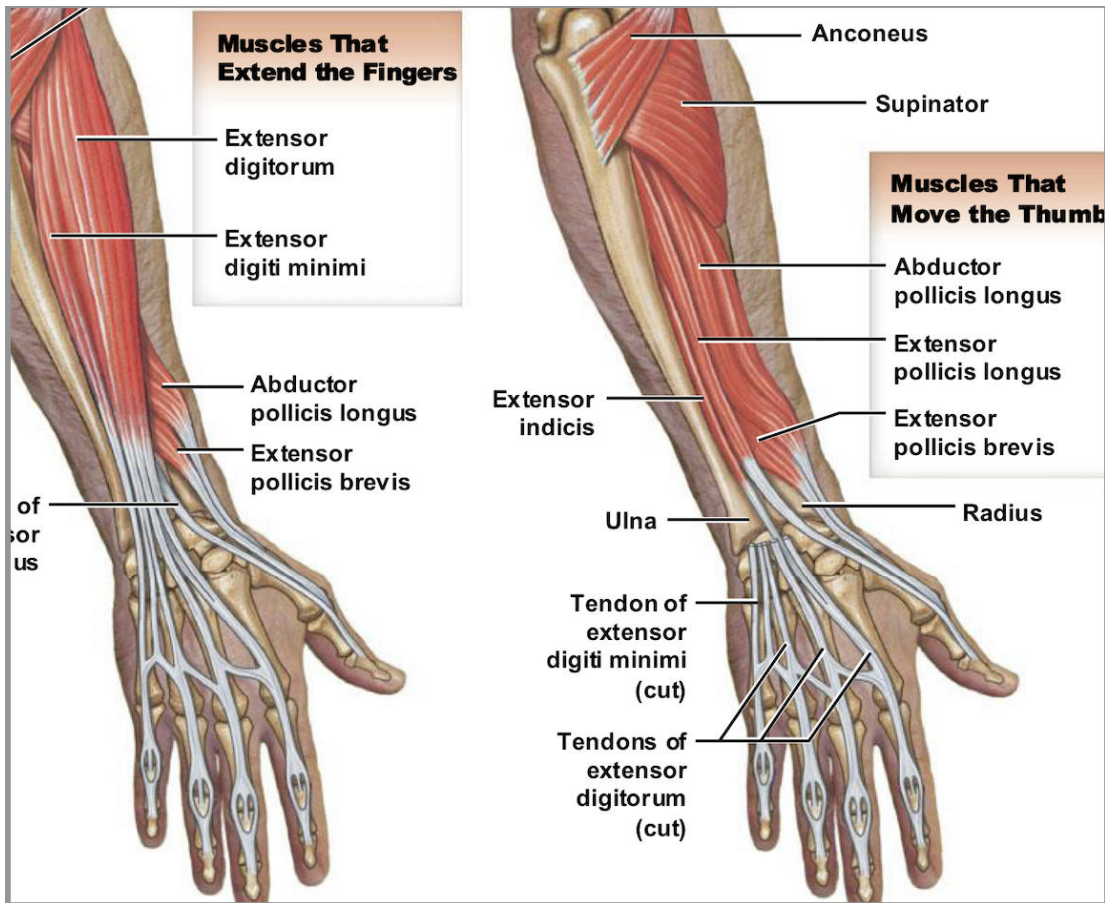


- **Muscles of pectoral girdle and upper limbs - muscles that move the hands and fingers**

- **Extrinsic muscles of hand**

- Abductor pollicis longus → thumb is only finger that does abduction
 - Extensor digitorum
 - Extensor pollicis brevis
 - Extensor pollicis longus
 - Extensor indicis
 - Extensor digiti minimi
 - Flexor digitorum superficialis
 - Flexor digitorum profundus
 - Flexor pollicis longus
 - Tendons of these muscles pass through **synovial tendon sheaths** (can lead to trigger finger, carpal tunnel), located within wrist area
 - Fibrous coverings: extensor retinaculum and flexor retinaculum → not muscles

pollicis
- Abduct
- Ext
- flex



- **Intrinsic muscles of hand**

- Found mostly in palm or dorsum of hand
 - Adductor pollicis
 - Opponens pollicis
 - Palmaris brevis
 - Abductor digiti minimi
 - Abductor pollicis brevis

- **Components and sectional anatomy of arm and forearm**

- Deep fascia (holds muscle together as one) extends between bones and superficial fascia & separates soft tissues of the limb into separate **compartments**
 - **Lateral intermuscular septum**
 - **Medial intermuscular septum**
 - Both septa create compartments within upper arm
 - **Anterior compartment**
 - **Posterior compartments**

Table 11.6 | Compartments of the Upper Limb

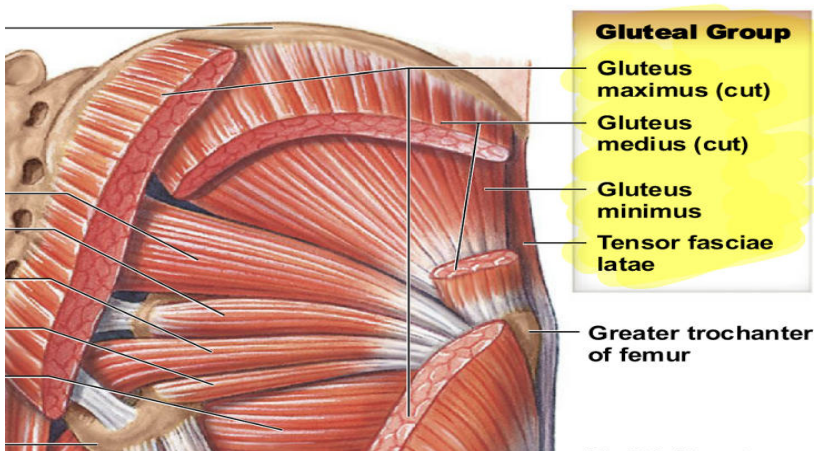
Compartment	Muscles
ARM	
Anterior compartment	Biceps brachii Brachialis Coracobrachialis
Posterior compartment	Triceps brachii

- The forearm is divided into four compartments
 - Superficial anterior compartment
 - Deep anterior compartment
 - Lateral compartment
 - Posterior compartment

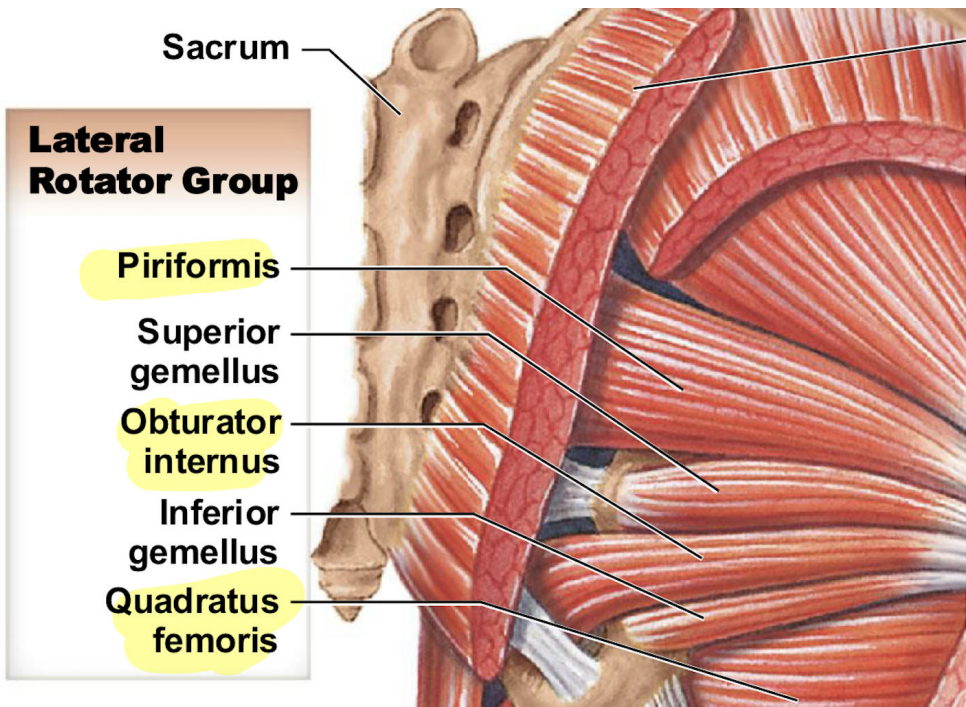
Compartment	Muscles
FOREARM	
Anterior compartment Superficial	Flexor carpi radialis Flexor carpi ulnaris Flexor digitorum superficialis Palmaris longus Pronator teres
Deep	Flexor digitorum profundus Flexor pollicis longus Pronator quadratus
Lateral compartment^s	Brachioradialis Extensor carpi radialis brevis Extensor carpi radialis longus
Posterior compartment	Abductor pollicis longus Anconeus Extensor carpi ulnaris Extensor digitorum Extensor digiti minimi Extensor indicis Extensor pollicis brevis Extensor pollicis longus Supinator

- **Muscles of the pelvic girdle and lower limb**
 - Three groups:
 - Muscles that move the thigh
 - Muscles that move the leg
 - Muscles that move the foot and toes
- **Muscles of the pelvic girdle and lower limb: Muscles that move the thigh/femur**
 - Origin - on the pelvis → many are large & powerful
 - Four groups:
 - Gluteal group
 - Lateral rotator group
 - Adductor group
 - Iliopsoas group
 - **Gluteal muscles**

- **Gluteus maximus:** mostly extension and lateral rotation at hip
 - Origin - iliac crest
 - Insertion - iliotibial tract and gluteal tuberosity of femur
- **Gluteus medius:** mostly abduction and medial rotation at hip
 - Origin - anterior iliac crest, lateral ileum
 - Insertion - greater trochanter of femur
- **Gluteus minimus:** abduction and medial rotation at hip
 - Origin - lateral ileum
 - Insertion - greater trochanter of femur
 - Medius and minimus are more lateral on pelvis than maximus
- **Tensor fasciae latae:** extension of knee and lateral rotation of leg
 - Crosses knee, involved with knee motion
 - Origin - iliac crest
 - Insertion - iliotibial tract



- **Lateral rotator group**
 - Oblique movements b/c they are rotating
 - Obturator muscles
 - Piriformis
 - Quadratus femoris



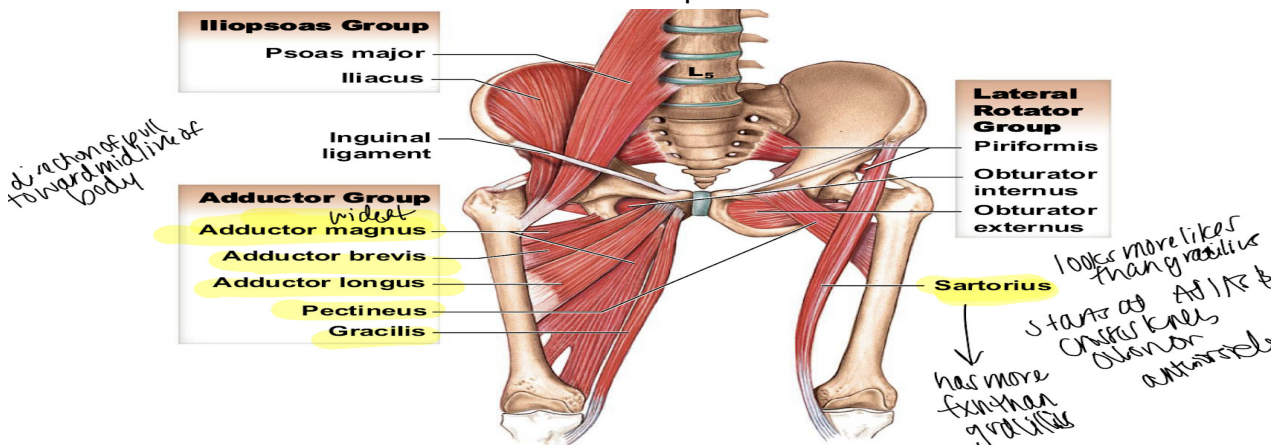
○ Adductor group

- Adductor brevis
- Adductor longus
- Adductor magnus
- Pectineus
- Gracilis: flexion and medial rotation at knee
 - Pubic symphysis to knee
 - Little bit of adduction and medial rotation at hip

O: inferior pubic rami
I: medial tibia inferior to medial condyle

○ Iliopsoas group

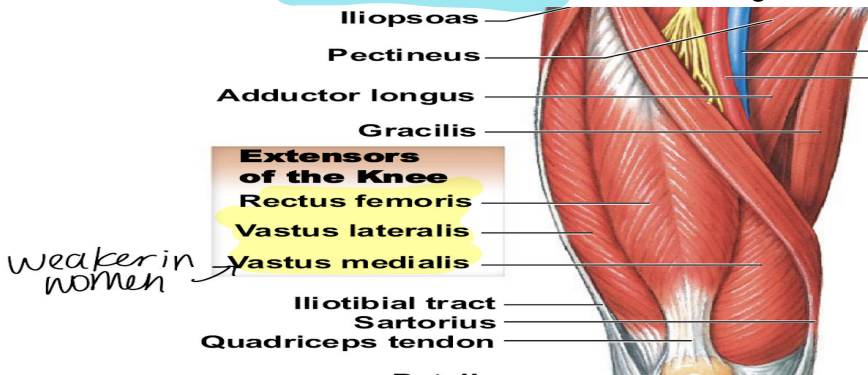
- Iliacus: flexion at hip
 - Sits sling ilium
- Psoas major: flexion at hip
 - Attached to lumbar spine



● Muscles of the pelvic girdle and lower limb: Muscles that move the leg

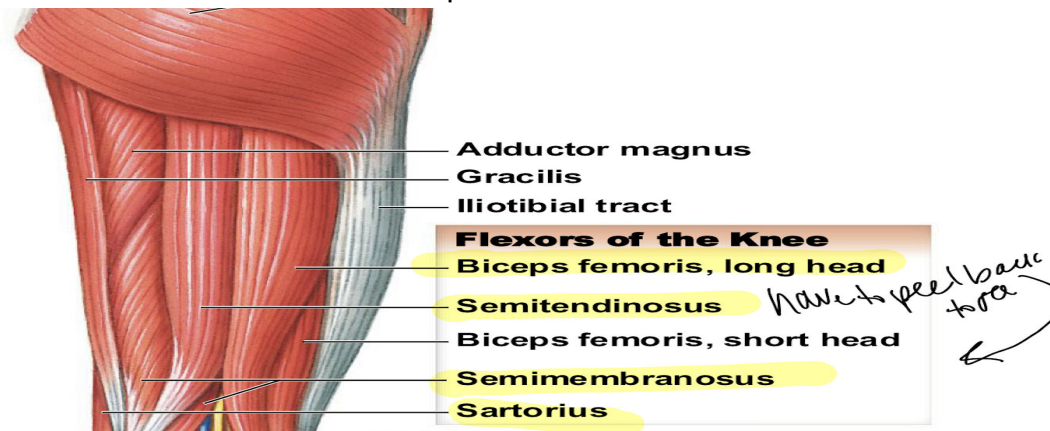
- Extensors of the knee → quadriceps femoris
- With extensors, you usually think about posterior side; this is not true with the legs
 - Rectus femoris: extends leg and flexion at hip

- **Vastus medialis**: extends leg
 - Weaker in women
- **Vastus lateralis**: extends leg
- **Vastus intermedius**: extends leg



○ **Flexors of the knee** → hamstrings

- **Biceps femoris**: Flexes legs at knee, extension at hip
- **Semimembranosus**: flexes the knee
- **Semitendinosus**: flexes the knee
- **Sartorius**: flexes hip *NOT part of Quad*
 - Origin - ASIS
 - Insertion - medial surface of tibia near tuberosity
- **Popliteus**: medially rotates knee (not part of hamstrings)!
 - Oblique muscle

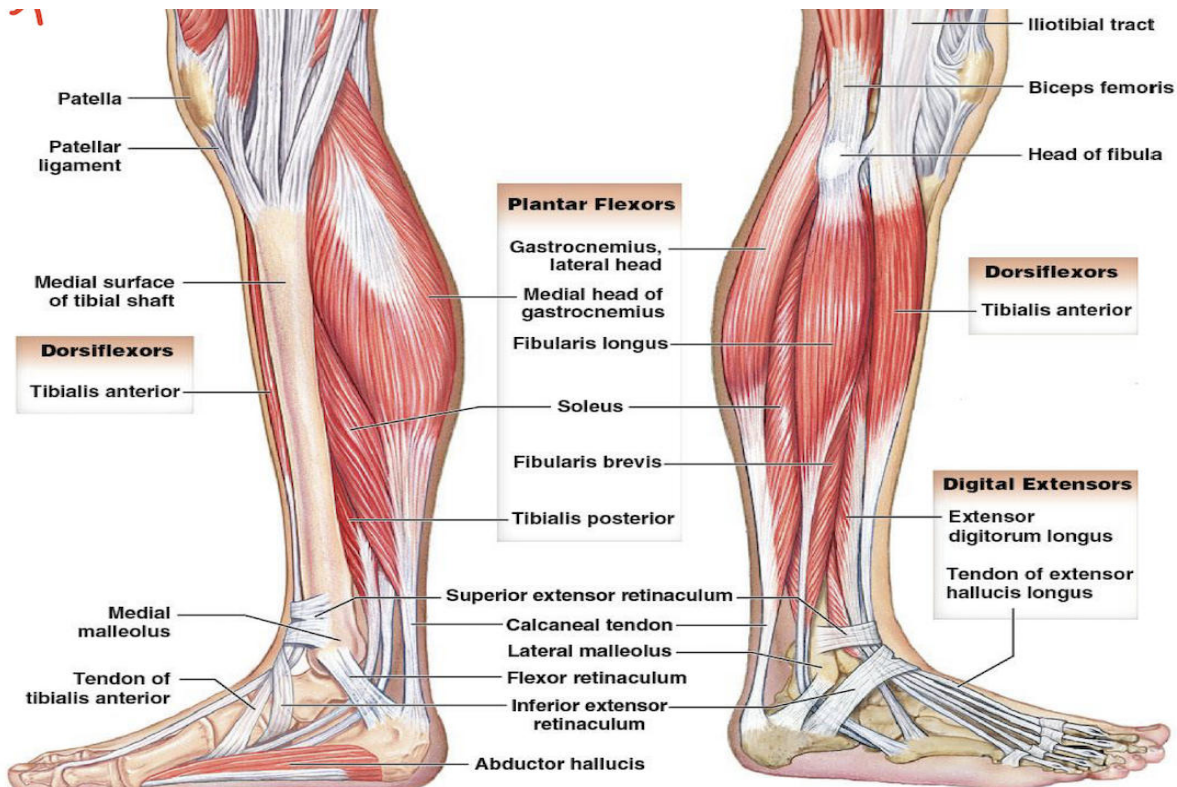


● **Muscles of the pelvic girdle and lower limb: Muscles that move the foot and toes**

○ **Extrinsic muscles** of foot: moves ankle

- O - distal end of the femur or on the tibia or fibula, but move the foot and toes
- Tibialis anterior
- **Gastrocnemius**
 - Origin - femoral condyles
 - Insertion - calcaneus tendon which inserts on calcaneus
- Fibularis brevis
- Fibularis longus
- Plantaris: plantar flexion
- **Soleus**
 - Directly under gastrocnemius, just doesn't start as high

- Origin - head and proximal shaft of fibula
- Insertion - calcaneus tendon which inserts on calcaneus
- Tibialis posterior
- Flexor digitorum longus: flexion of toes 2-5
- Flexor hallucis longus: flexion of hallux
- Extensor digitorum longus: extension of toes 2-5
- Extensor hallucis longus: extension of hallux
- Superior and inferior extensor retinaculum stabilize tendons in the tarsal area (CT pulley system) → not muscle



○ **Intrinsic muscles of foot moves the toes**

- Origin - some aspect of the foot but move the toes
- Abductor hallucis
- Flexor digitorum brevis
- Abductor digiti minimi
- Flexor hallucis brevis
- Flexor digiti minimi brevis
- plantar aponeurosis: superficial fascia layer
- Quadratus plantae
- Adductor hallucis
- Plantar interossei
- Extensor digitorum brevis
- Dorsal interossei
- Compartments and sectional anatomy of thigh and leg - musculoskeletal compartments

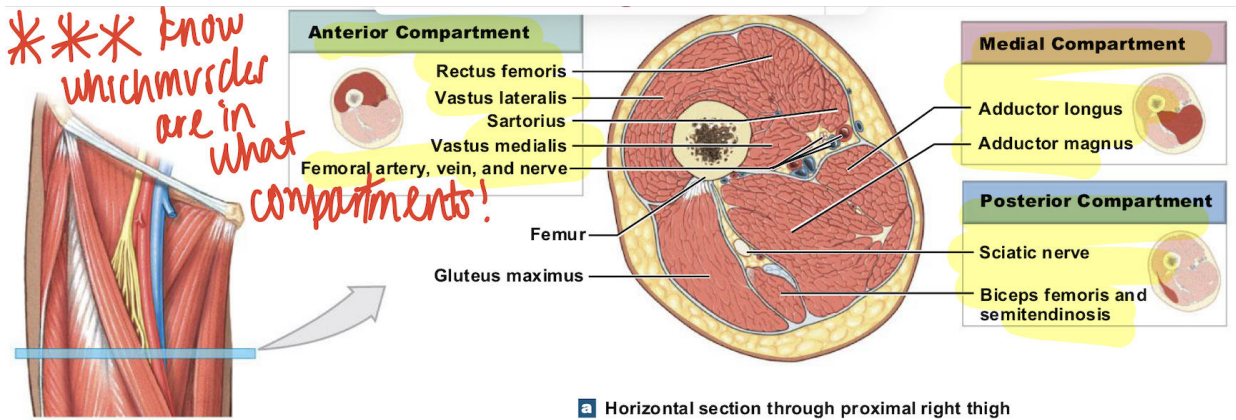


Figure 11.27 Musculoskeletal Compartments of the Leg, Anterior View

